Drug Discovery Sciences

Vrije Universiteit Amsterdam - Faculty of Science - Drug Discovery Sciences - 2025-2026

The Master's programme in Drug Discovery Sciences (DDS) offers a cutting-edge, interdisciplinary education in early-stage drug discovery from a molecular and chemical perspective. The programme prepares students to contribute to the development of new molecular therapeutics and studying disease mechanisms at the molecular level by equipping them with a strong foundation in both theory and research skills essential to pharmaceutical innovation. The DDS programme focusses on drug discovery and revolves around drug research from various complementary chemical angles. The curriculum is designed with focus on academic knowledge, practical laboratory experience, and critical thinking skills, allowing graduates to play pivotal roles in interdisciplinary research teams across academia, industry, and healthcare. Graduates also find career paths in policy, management, or education.

Students specialize in one of five specialisations:

• Molecular Pharmacology – Explore how compounds (including small molecules and proteins) interact with biological targets to modulate physiological and pathological processes.

• Synthesis of Drugs – Learn how to synthesize biologically active molecules with therapeutic potential using modern synthetic chemistry approaches and techniques.

• Molecular Toxicology – Understand how drugs are metabolized in the human body, how drugs impact cellular metabolism, and investigate mechanisms of toxicity and adverse effects.

• Bioanalytics – Develop expertise in detecting, identifying, and quantifying drugs, metabolites, and biomarkers, and develop expertise in studying disease-mediated molecular pathways, using state-of-the-art analytical methods.

• Computer-Aided Drug Design (CADD) – Apply computational methods to model and predict interactions of small-molecule compounds with both drug targets as well as off-targets.

Year 1: Foundation and specialisation

In the first year, students take specialisation-specific core courses, offering in-depth scientific training in their chosen specialisation. These are complemented by elective courses, enabling students to build a deeper and/or broader understanding of pharmaceutical sciences and connect with other areas relevant to drug discovery. This combination supports collaboration in an interdisciplinary research environment.

Throughout the programme, attention is paid to the development of academic and professional skills. The compulsory course Research Skills and Career Perspectives helps students reflect on their scientific identity, develops essential transferable skills and critical thinking, and prepares them for internships and future career paths.

By the end of Year 1, many students start their major research internship at a university, institute or company. Students can also start with their major internship in Year 2. Research internships are available within the VU Amsterdam in research groups either within the department of Chemistry & Pharmaceutical Sciences or within other VU departments. External internships in the Netherlands and abroad are also possible.

Year 2: Focus on the Research (R) variant internship or on the Education (E) variant

The second year offers tailored pathways that match students' goals and ambitions:

• In the Research profile, many students continue conducting their major internship in year 2. In this year they explore scientific topics in greater depth depending on their interests and career goals. Additional elective courses and/or an optional second internship can be done, in a company, research institute, academic hospital, or university. The Research variant of the programme is designed to offer flexibility in internship environment and focus, allowing students to shape their learning experience according to their professional aspirations. Students can also choose to perform an in-depth literature review in year 2.

• In the Education profile, students in the second year take courses and an internship, all with an educational focus. This enables them to become a first-degree Dutch high school teacher in chemistry. This profile (i.e., the second year) is only provided in the Dutch language.

Info

Level	Master
Language	English
Duration	2 years

Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Drug Discovery Sciences specialisation Bioanalytics

Programme Sections

- DDS elective courses
- DDS specialisation Bioanalytics R variant
- Drug Discovery and Safety CSE variants
- DDS specialisation Bioanalytics compulsory courses
- DDS core courses

DDS elective courses

Courses

Course Name	Period	Credits	Code
Minor Research Project Bioanalytical Chemistry	Ac. Year (September)	18EC	XM_0044
Minor Research Project Environmental Bioanalysis	Ac. Year (September)	18EC	XM_0045
Minor Research Project Medicinal Chemistry	Ac. Year (September)	18EC	XM_0046
Minor Research Project Molecular & Computational Toxicology	Ac. Year (September)	18EC	XM_0047
Minor Research Project Organic Chemistry	Ac. Year (September)	18EC	XM_0048
Minor Research Project Radiopharmaceutical Chemistry	Ac. Year (September)	18EC	XM_0049
Minor Research Project Theoretical Chemistry	Ac. Year (September)	18EC	XM_0050
ADME	P1	6EC	XM_0136
Data Management in Drug Discovery	P1	6EC	XM_0117
Fundamentals of Bioinformatics	P1	6EC	X_405052
Oncogenesis	P1	6EC	M_OONC03
Physical Organic Chemistry	P1	6EC	X_435663
Advanced Bioanalytical Approaches	P2	6EC	XM_0107
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Molecular and Cellular Toxicology	P2	6EC	XM_0113
Spectroscopic Approaches in Medicinal Chemistry	P2	6EC	XM_0109
Advanced Course in Molecular Toxicology	P4	6EC	XM_0114
Advanced Molecular Orbital Theory	P4	6EC	XMU_0028
Advanced Pharmacology	P4	6EC	XM_0112
Biopharmaceuticals & Biopharma Proteomics	P4	6EC	XM_0108
Concepts in Chemical Biology	P4	6EC	XM_0080
Molecular Modelling	P4	6EC	XMU_0059
Synthetic Approaches in Medicinal Chemistry	P4	6EC	XM_0110
Translational Radiopharmaceutical Sciences	P4	6EC	XM_0067
Advanced Radiopharmaceutical Chemistry	P5	6EC	XM_0069
Protein Analysis	P5	6EC	X_435045
Replacement, Refinement and Reduction of animal experiments applying human cells in culture	P5	6EC	XM_0148

DDS specialisation Bioanalytics R variant

Programme Sections

- DDS ethics and academic skills
- DDS specialisation Bioanalytics research project (mj)
- DDS specialisation Bioanalytics colloq. & lit. thes.

DDS ethics and academic skills

Description

In consultation with the master coordinator and depending on the chosen specialisation, 6 EC have to be chosen from the following list for students participating in the research-variant.

Courses

Course Name	Period	Credits	Code
Teaching Assistant	Ac. Year (September)	3EC	XM_432741
Teaching Assistant	Ac. Year (September)	6EC	XM_432742
Managing Science and Technology in Society	P1	6EC	AM_470586
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587
Societal Entrepreneurship in Health and Life Sciences	P1	6EC	AM_470575
Business Management in Health and Life Sciences	P2	6EC	AM_470584
Communication, Organization and Management	P2	6EC	AM_470572
Science Journalism	P2	6EC	AM_471014
Clinical Development and Clinical Trials	P3	3EC	AM_1180
Epidemiology	P3	3EC	AM_1179
Ethics in Life Sciences	P3	3EC	AM_470707
Science in Dialogue	P3	6EC	AM_1002
Tutoring Students	P3	3EC	X_432625
Scientific Writing in English	P3, P6	3EC	X_400592

DDS specialisation Bioanalytics research project (mj)

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry	Ac. Year (September)	42EC	XM_0037
Major Research Project Environmental Bioanalysis	Ac. Year (September)	42EC	XM_0038
Major Research Project Molecular & Computational Toxicology	Ac. Year (September)	42EC	XM_0040

DDS specialisation Bioanalytics colloq. & lit. thes.

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry	Ac. Year (September)	12EC	XM_0030
Colloquium & Literature Thesis Environmental Bioanalysis	Ac. Year (September)	12EC	XM_0031

Drug Discovery and Safety CSE variants

Programme Sections

- DDS Colloquium & Literature Thesis
- DDS research project
- DDS specialisation Education

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description
Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Education

Description

The teaching in these specialisation is mainly in Dutch. Therefore we also give the requirements in Dutch.

Programma

De opleiding voor het behalen van de eerstegraads lesbevoegdheid start twee keer per jaar, in september en in februari. De opleiding wordt aangeboden in twee semesters. Uitgaande van de start in september duurt semester 1 tot en met januari en semester 2 tot juli. De opleiding is sterk praktijkgericht. De helft van de opleiding bestaat uit praktijk door werkervaring of stage (ook wel schoolpracticum genoemd) op een school voor voortgezet onderwijs. Daarnaast kent de opleiding vier componenten: vakdidactiek, algemene didactiek/pedagogiek, praktijkonderzoek en verdiepingsmodulen.

Naast de educatievakken volgt de student 60 sp Pharmaceutical Sciences / Chemistry vakken, in overleg met de mastercoördinator van de gekozen specialisatie. Hierbij zijn de twee vakken Literature thesis and Colloquium Chemistry Education en Master Research Project Chemistry-Education verplicht.

Studenten die bij de Communicatie variant de vakken 'interpersoonlijke communicatie' en 'museologie en buitenschoolse educatie' volgen, krijgen bij de lerarenopleiding een vrijstelling voor verdiepingsmodulen, een deel van het praktijkonderzoek en een deel van algemene didactiek.

Programme Sections

- Teaching Chemistry
- DDS Courses

Teaching Chemistry

Description

Scheikunde is overal om ons heen te vinden. Neem jij leerlingen stap voor stap mee in de wereld van atomen,

moleculen en chemische reacties? Speel jij graag een rol in hun ontwikkeling en wil je jouw passie voor scheikunde met hen delen? Als docent Scheikunde behandel je niet alleen (bio)chemie en de transformatie van materialen, maar je kijkt ook naar ook actuele maatschappelijke thema's vanuit een scheikundig oogpunt. Het visueel maken van abstracte ideeën speelt bij het schoolvak Scheikunde een belangrijke rol. Tijdens de lerarenopleiding Scheikunde leer je hoe je visualisaties, analogieën en simulaties effectief kunt inzetten om complexe onderwerpen begrijpelijk te maken voor je leerlingen. Je ontdekt ook hoe je leerlingen kunt begeleiden in het onderzoeken en verklaren van de wereld om hen heen en hoe je hen actief aan het denken kunt zetten. Ook leer je hoe je de effectiviteit van practica en demonstratieproeven kunt vergroten. Welke proef zorgt ervoor dat een ingewikkeld concept écht blijft hangen bij je leerlingen?

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1

Course Name	Period	Credits	Code
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2
Practice 2	P2+3	9EC	O_MLPRAK_2
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

DDS Courses

Description

This specialisation is intended for students with a BSc degree in any of the bèta-studies who want to specialise in communication. The programme focuses on science communication theory, research and practice. The programme of the communication (C) specialisation is 1 year (60 credits). This specialisation may not be combined with te Societal specialisation (M) or the Education specialisation (E). C-courses are shared with master students from the Faculty of Earth and Life Sciences.

To complete the entire Master programme (120 EC) of the Communication, education or social variant, the student has to choose 60 EC in DDS courses in consultation with the master track coordinator.

Programme Sections

- Specialisation Courses
- DDS Colloquium & Literature Thesis
- DDS research project

Specialisation Courses

Description

In consultation with the master coordinator and depending on the chosen specialization, 6 EC have to be chosen from the following list.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Physical Organic Chemistry	P1	6EC	X_435663
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description

Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023

Course Name	Period	Credits	Code
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Bioanalytics compulsory courses

Courses

Course Name	Period	Credits	Code
ADME	P1	6EC	XM_0136
Advanced Bioanalytical Approaches	P2	6EC	XM_0107
Biopharmaceuticals & Biopharma Proteomics	P4	6EC	XM_0108

DDS core courses

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Principles of Drug Targets	P1	6EC	XM_0105
Principles of Drug Discovery	P2	6EC	XM_0106

Drug Discovery Sciences specialisation Synthesis of Drugs

Programme Sections

- DDS elective courses
- DDS specialisation Synthesis of Drugs R variant
- Drug Discovery and Safety CSE variants
- DDS specialisation Synthesis of Drugs specific courses
- DDS core courses

DDS elective courses

Courses

Course Name	Period	Credits	Code
Minor Research Project Bioanalytical Chemistry	Ac. Year (September)	18EC	XM_0044
Minor Research Project Environmental Bioanalysis	Ac. Year (September)	18EC	XM_0045
Minor Research Project Medicinal Chemistry	Ac. Year (September)	18EC	XM_0046
Minor Research Project Molecular & Computational Toxicology	Ac. Year (September)	18EC	XM_0047
Minor Research Project Organic Chemistry	Ac. Year (September)	18EC	XM_0048
Minor Research Project Radiopharmaceutical Chemistry	Ac. Year (September)	18EC	XM_0049
Minor Research Project Theoretical Chemistry	Ac. Year (September)	18EC	XM_0050
ADME	P1	6EC	XM_0136
Data Management in Drug Discovery	P1	6EC	XM_0117
Fundamentals of Bioinformatics	P1	6EC	X_405052
Oncogenesis	P1	6EC	M_OONC03
Physical Organic Chemistry	P1	6EC	X_435663

Course Name	Period	Credits	Code
Advanced Bioanalytical Approaches	P2	6EC	XM_0107
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Molecular and Cellular Toxicology	P2	6EC	XM_0113
Spectroscopic Approaches in Medicinal Chemistry	P2	6EC	XM_0109
Advanced Course in Molecular Toxicology	P4	6EC	XM_0114
Advanced Molecular Orbital Theory	P4	6EC	XMU_0028
Advanced Pharmacology	P4	6EC	XM_0112
Biopharmaceuticals & Biopharma Proteomics	P4	6EC	XM_0108
Concepts in Chemical Biology	P4	6EC	XM_0080
Molecular Modelling	P4	6EC	XMU_0059
Synthetic Approaches in Medicinal Chemistry	P4	6EC	XM_0110
Translational Radiopharmaceutical Sciences	P4	6EC	XM_0067
Advanced Radiopharmaceutical Chemistry	P5	6EC	XM_0069
Protein Analysis	P5	6EC	X_435045
Replacement, Refinement and Reduction of animal experiments applying human cells in culture	P5	6EC	XM_0148

DDS specialisation Synthesis of Drugs R variant

Programme Sections

- DDS ethics and academic skills
- DDS specialisation Synthesis of Drugs research project (mj)
- DDS specialisation Synthesis of Drugs colloq. & lit. thes.

DDS ethics and academic skills

Description

In consultation with the master coordinator and depending on the chosen specialisation, 6 EC have to be chosen from the following list for students participating in the research-variant.

Courses

Course Name	Period	Credits	Code
Teaching Assistant	Ac. Year (September)	3EC	XM_432741
Teaching Assistant	Ac. Year (September)	6EC	XM_432742
Managing Science and Technology in Society	P1	6EC	AM_470586
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587
Societal Entrepreneurship in Health and Life Sciences	P1	6EC	AM_470575
Business Management in Health and Life Sciences	P2	6EC	AM_470584
Communication, Organization and Management	P2	6EC	AM_470572
Science Journalism	P2	6EC	AM_471014
Clinical Development and Clinical Trials	P3	3EC	AM_1180
Epidemiology	P3	3EC	AM_1179
Ethics in Life Sciences	P3	3EC	AM_470707
Science in Dialogue	P3	6EC	AM_1002
Tutoring Students	P3	3EC	X_432625
Scientific Writing in English	P3, P6	3EC	X_400592

DDS specialisation Synthesis of Drugs research project (mj)

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry	Ac. Year (September)	42EC	XM_0039
Major Research Project Molecular & Computational Toxicology	Ac. Year (September)	42EC	XM_0040
Major Research Project Organic Chemistry	Ac. Year (September)	42EC	XM_0041
Major Research Project Radiopharmaceutical Chemistry	Ac. Year (September)	42EC	XM_0042

DDS specialisation Synthesis of Drugs colloq. & lit. thes.

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry	Ac. Year (September)	12EC	XM_0032
Colloquium & Literature Thesis Molecular & Computational Toxicology	Ac. Year (September)	12EC	XM_0033
Colloquium & Literature Thesis Organic Chemistry	Ac. Year (September)	12EC	XM_0034
Colloquium & Literature Thesis Radiopharmaceutical Chemistry	Ac. Year (September)	12EC	XM_0035

Drug Discovery and Safety CSE variants

Programme Sections

- DDS Colloquium & Literature Thesis
- DDS research project
- DDS specialisation Education

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description
Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description
Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Education

Description

The teaching in these specialisation is mainly in Dutch. Therefore we also give the requirements in Dutch.

Programma

De opleiding voor het behalen van de eerstegraads lesbevoegdheid start twee keer per jaar, in september en in februari. De opleiding wordt aangeboden in twee semesters. Uitgaande van de start in september duurt semester 1 tot en met januari en semester 2 tot juli. De opleiding is sterk praktijkgericht. De helft van de opleiding bestaat uit praktijk door werkervaring of stage (ook wel schoolpracticum genoemd) op een school voor voortgezet onderwijs. Daarnaast kent de opleiding vier componenten: vakdidactiek, algemene didactiek/pedagogiek, praktijkonderzoek en verdiepingsmodulen.

Naast de educatievakken volgt de student 60 sp Pharmaceutical Sciences / Chemistry vakken, in overleg met de mastercoördinator van de gekozen specialisatie. Hierbij zijn de twee vakken Literature thesis and Colloquium Chemistry Education en Master Research Project Chemistry-Education verplicht.

Studenten die bij de Communicatie variant de vakken 'interpersoonlijke communicatie' en 'museologie en buitenschoolse educatie' volgen, krijgen bij de lerarenopleiding een vrijstelling voor verdiepingsmodulen, een deel van het praktijkonderzoek en een deel van algemene didactiek.

Programme Sections

- Teaching Chemistry
- DDS Courses

Teaching Chemistry

Description

Scheikunde is overal om ons heen te vinden. Neem jij leerlingen stap voor stap mee in de wereld van atomen, moleculen en chemische reacties? Speel jij graag een rol in hun ontwikkeling en wil je jouw passie voor scheikunde met hen delen? Als docent Scheikunde behandel je niet alleen (bio)chemie en de transformatie van materialen, maar je kijkt ook naar ook actuele maatschappelijke thema's vanuit een scheikundig oogpunt. Het visueel maken van abstracte ideeën speelt bij het schoolvak Scheikunde een belangrijke rol. Tijdens de lerarenopleiding Scheikunde leer je hoe je visualisaties, analogieën en simulaties effectief kunt inzetten om complexe onderwerpen begrijpelijk te maken voor je leerlingen. Je ontdekt ook hoe je leerlingen kunt begeleiden in het onderzoeken en verklaren van de wereld om hen heen en hoe je hen actief aan het denken kunt zetten. Ook leer je hoe je de effectiviteit van practica en demonstratieproeven kunt vergroten. Welke proef zorgt ervoor dat een ingewikkeld concept écht blijft hangen bij je leerlingen?

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F

Course Name	Period	Credits	Code
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2
Practice 2	P2+3	9EC	O_MLPRAK_2
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

DDS Courses

Description

This specialisation is intended for students with a BSc degree in any of the bèta-studies who want to specialise in communication. The programme focuses on science communication theory, research and practice. The programme of the communication (C) specialisation is 1 year (60 credits). This specialisation may not be combined with te Societal specialisation (M) or the Education specialisation (E). C-courses are shared with master students from the Faculty of Earth and Life Sciences.

To complete the entire Master programme (120 EC) of the Communication, education or social variant, the student has to choose 60 EC in DDS courses in consultation with the master track coordinator.

Programme Sections

- Specialisation Courses
- DDS Colloquium & Literature Thesis
- DDS research project

Specialisation Courses

Description

In consultation with the master coordinator and depending on the chosen specialization, 6 EC have to be chosen from the following list.

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Physical Organic Chemistry	P1	6EC	X_435663
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description

Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Synthesis of Drugs specific courses

Courses

Course Name	Period	Credits	Code
Physical Organic Chemistry	P1	6EC	X_435663
Spectroscopic Approaches in Medicinal Chemistry	P2	6EC	XM_0109
Synthetic Approaches in Medicinal Chemistry	P4	6EC	XM_0110

DDS core courses

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Principles of Drug Targets	P1	6EC	XM_0105
Principles of Drug Discovery	P2	6EC	XM_0106

Drug Discovery Sciences specialisation Molecular Pharmacology

Programme Sections

- DDS elective courses
- DDS specialisation Molecular Pharmacology R variant
- Drug Discovery and Safety CSE variants
- DDS specialisation Molecular Pharmacology specific courses
- DDS core courses

DDS elective courses

Courses

Course Name	Period	Credits	Code
Minor Research Project Bioanalytical Chemistry	Ac. Year (September)	18EC	XM_0044
Minor Research Project Environmental Bioanalysis	Ac. Year (September)	18EC	XM_0045
Minor Research Project Medicinal Chemistry	Ac. Year (September)	18EC	XM_0046
Minor Research Project Molecular & Computational Toxicology	Ac. Year (September)	18EC	XM_0047
Minor Research Project Organic Chemistry	Ac. Year (September)	18EC	XM_0048
Minor Research Project Radiopharmaceutical Chemistry	Ac. Year (September)	18EC	XM_0049
Minor Research Project Theoretical Chemistry	Ac. Year (September)	18EC	XM_0050
ADME	P1	6EC	XM_0136
Data Management in Drug Discovery	P1	6EC	XM_0117
Fundamentals of Bioinformatics	P1	6EC	X_405052
Oncogenesis	P1	6EC	M_OONC03
Physical Organic Chemistry	P1	6EC	X_435663
Advanced Bioanalytical Approaches	P2	6EC	XM_0107
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Molecular and Cellular Toxicology	P2	6EC	XM_0113
Spectroscopic Approaches in Medicinal Chemistry	P2	6EC	XM_0109
Advanced Course in Molecular Toxicology	P4	6EC	XM_0114
Advanced Molecular Orbital Theory	P4	6EC	XMU_0028
Advanced Pharmacology	P4	6EC	XM_0112
Biopharmaceuticals & Biopharma Proteomics	P4	6EC	XM_0108
Concepts in Chemical Biology	P4	6EC	XM_0080
Molecular Modelling	P4	6EC	XMU_0059
Synthetic Approaches in Medicinal Chemistry	P4	6EC	XM_0110
Translational Radiopharmaceutical Sciences	P4	6EC	XM_0067
Advanced Radiopharmaceutical Chemistry	P5	6EC	XM_0069
Protein Analysis	P5	6EC	X_435045
Replacement, Refinement and Reduction of animal experiments applying human cells in culture	P5	6EC	XM_0148

DDS specialisation Molecular Pharmacology R variant

Programme Sections

- DDS ethics and academic skills
- DDS specialisation Molecular Pharmacology research project (mj)
- DDS specialisation Molecular Pharmacology colloq. & lit. thes.

DDS ethics and academic skills

Description

In consultation with the master coordinator and depending on the chosen specialisation, 6 EC have to be chosen

from the following list for students participating in the research-variant.

Courses

Course Name	Period	Credits	Code
Teaching Assistant	Ac. Year (September)	3EC	XM_432741
Teaching Assistant	Ac. Year (September)	6EC	XM_432742
Managing Science and Technology in Society	P1	6EC	AM_470586
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587
Societal Entrepreneurship in Health and Life Sciences	P1	6EC	AM_470575
Business Management in Health and Life Sciences	P2	6EC	AM_470584
Communication, Organization and Management	P2	6EC	AM_470572
Science Journalism	P2	6EC	AM_471014
Clinical Development and Clinical Trials	P3	3EC	AM_1180
Epidemiology	P3	3EC	AM_1179
Ethics in Life Sciences	P3	3EC	AM_470707
Science in Dialogue	P3	6EC	AM_1002
Tutoring Students	P3	3EC	X_432625
Scientific Writing in English	P3, P6	3EC	X_400592

DDS specialisation Molecular Pharmacology research project (mj)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry	Ac. Year (September)	42EC	XM_0039
Major Research Project Organic Chemistry	Ac. Year (September)	42EC	XM_0041

DDS specialisation Molecular Pharmacology colloq. & lit. thes.

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry	Ac. Year (September)	12EC	XM_0032
Colloquium & Literature Thesis Organic Chemistry	Ac. Year (September)	12EC	XM_0034

Drug Discovery and Safety CSE variants

Programme Sections

- DDS Colloquium & Literature Thesis
- DDS research project
- DDS specialisation Education

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description
Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description
Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Education

Description

The teaching in these specialisation is mainly in Dutch. Therefore we also give the requirements in Dutch.

Programma

De opleiding voor het behalen van de eerstegraads lesbevoegdheid start twee keer per jaar, in september en in februari. De opleiding wordt aangeboden in twee semesters. Uitgaande van de start in september duurt semester 1 tot en met januari en semester 2 tot juli. De opleiding is sterk praktijkgericht. De helft van de opleiding bestaat uit praktijk door werkervaring of stage (ook wel schoolpracticum genoemd) op een school voor voortgezet onderwijs. Daarnaast kent de opleiding vier componenten: vakdidactiek, algemene didactiek/pedagogiek, praktijkonderzoek en verdiepingsmodulen.

Naast de educatievakken volgt de student 60 sp Pharmaceutical Sciences / Chemistry vakken, in overleg met de mastercoördinator van de gekozen specialisatie. Hierbij zijn de twee vakken Literature thesis and Colloquium Chemistry Education en Master Research Project Chemistry-Education verplicht.

Studenten die bij de Communicatie variant de vakken 'interpersoonlijke communicatie' en 'museologie en buitenschoolse educatie' volgen, krijgen bij de lerarenopleiding een vrijstelling voor verdiepingsmodulen, een deel van het praktijkonderzoek en een deel van algemene didactiek.

Programme Sections

- Teaching Chemistry
- DDS Courses

Teaching Chemistry

Description

Scheikunde is overal om ons heen te vinden. Neem jij leerlingen stap voor stap mee in de wereld van atomen, moleculen en chemische reacties? Speel jij graag een rol in hun ontwikkeling en wil je jouw passie voor scheikunde met hen delen? Als docent Scheikunde behandel je niet alleen (bio)chemie en de transformatie van materialen,

maar je kijkt ook naar ook actuele maatschappelijke thema's vanuit een scheikundig oogpunt. Het visueel maken van abstracte ideeën speelt bij het schoolvak Scheikunde een belangrijke rol. Tijdens de lerarenopleiding Scheikunde leer je hoe je visualisaties, analogieën en simulaties effectief kunt inzetten om complexe onderwerpen begrijpelijk te maken voor je leerlingen. Je ontdekt ook hoe je leerlingen kunt begeleiden in het onderzoeken en verklaren van de wereld om hen heen en hoe je hen actief aan het denken kunt zetten. Ook leer je hoe je de effectiviteit van practica en demonstratieproeven kunt vergroten. Welke proef zorgt ervoor dat een ingewikkeld concept écht blijft hangen bij je leerlingen?

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2
Practice 2	P2+3	9EC	O_MLPRAK_2

Course Name	Period	Credits	Code
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

DDS Courses

Description

This specialisation is intended for students with a BSc degree in any of the bèta-studies who want to specialise in communication. The programme focuses on science communication theory, research and practice. The programme of the communication (C) specialisation is 1 year (60 credits). This specialisation may not be combined with te Societal specialisation (M) or the Education specialisation (E). C-courses are shared with master students from the Faculty of Earth and Life Sciences.

To complete the entire Master programme (120 EC) of the Communication, education or social variant, the student has to choose 60 EC in DDS courses in consultation with the master track coordinator.

Programme Sections

- Specialisation Courses
- DDS Colloquium & Literature Thesis
- DDS research project

Specialisation Courses

Description

In consultation with the master coordinator and depending on the chosen specialization, 6 EC have to be chosen from the following list.

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Physical Organic Chemistry	P1	6EC	X_435663
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description

Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Molecular Pharmacology specific courses

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
ADME	P1	6EC	XM_0136
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Advanced Pharmacology	P4	6EC	XM_0112

DDS core courses

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Principles of Drug Targets	P1	6EC	XM_0105
Principles of Drug Discovery	P2	6EC	XM_0106

Drug Discovery Sciences specialisation Molecular Toxicology

Programme Sections

- DDS elective courses
- DDS specialisation Molecular Toxicology R variant
- Drug Discovery and Safety CSE variants
- DDS specialisation Molecular Toxicology specific courses
- DDS core courses

DDS elective courses

Courses

Course Name	Period	Credits	Code
Minor Research Project Bioanalytical Chemistry	Ac. Year (September)	18EC	XM_0044
Minor Research Project Environmental Bioanalysis	Ac. Year (September)	18EC	XM_0045
Minor Research Project Medicinal Chemistry	Ac. Year (September)	18EC	XM_0046
Minor Research Project Molecular & Computational Toxicology	Ac. Year (September)	18EC	XM_0047
Minor Research Project Organic Chemistry	Ac. Year (September)	18EC	XM_0048
Minor Research Project Radiopharmaceutical Chemistry	Ac. Year (September)	18EC	XM_0049
Minor Research Project Theoretical Chemistry	Ac. Year (September)	18EC	XM_0050
ADME	P1	6EC	XM_0136
Data Management in Drug Discovery	P1	6EC	XM_0117
Fundamentals of Bioinformatics	P1	6EC	X_405052
Oncogenesis	P1	6EC	M_OONC03
Physical Organic Chemistry	P1	6EC	X_435663
Advanced Bioanalytical Approaches	P2	6EC	XM_0107
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Molecular and Cellular Toxicology	P2	6EC	XM_0113
Spectroscopic Approaches in Medicinal Chemistry	P2	6EC	XM_0109
Advanced Course in Molecular Toxicology	P4	6EC	XM_0114
Advanced Molecular Orbital Theory	P4	6EC	XMU_0028
Advanced Pharmacology	P4	6EC	XM_0112
Biopharmaceuticals & Biopharma Proteomics	P4	6EC	XM_0108

Course Name	Period	Credits	Code
Concepts in Chemical Biology	P4	6EC	XM_0080
Molecular Modelling	P4	6EC	XMU_0059
Synthetic Approaches in Medicinal Chemistry	P4	6EC	XM_0110
Translational Radiopharmaceutical Sciences	P4	6EC	XM_0067
Advanced Radiopharmaceutical Chemistry	P5	6EC	XM_0069
Protein Analysis	P5	6EC	X_435045
Replacement, Refinement and Reduction of animal experiments applying human cells in culture	P5	6EC	XM_0148

DDS specialisation Molecular Toxicology R variant

Programme Sections

- DDS ethics and academic skills
- DDS specialisation Molecular Toxicology research project (mj)
- DDS specialisation Molecular Toxicology colloq. & lit. thes.

DDS ethics and academic skills

Description

In consultation with the master coordinator and depending on the chosen specialisation, 6 EC have to be chosen from the following list for students participating in the research-variant.

Courses

Course Name	Period	Credits	Code
Teaching Assistant	Ac. Year (September)	3EC	XM_432741
Teaching Assistant	Ac. Year (September)	6EC	XM_432742
Managing Science and Technology in Society	P1	6EC	AM_470586
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587
Societal Entrepreneurship in Health and Life Sciences	P1	6EC	AM_470575
Business Management in Health and Life Sciences	P2	6EC	AM_470584
Communication, Organization and Management	P2	6EC	AM_470572
Science Journalism	P2	6EC	AM_471014
Clinical Development and Clinical Trials	P3	3EC	AM_1180
Epidemiology	P3	3EC	AM_1179
Ethics in Life Sciences	P3	3EC	AM_470707
Science in Dialogue	P3	6EC	AM_1002
Tutoring Students	P3	3EC	X_432625
Scientific Writing in English	P3, P6	3EC	X_400592

DDS specialisation Molecular Toxicology research project (mj)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology	Ac. Year (September)	42EC	XM_0040

DDS specialisation Molecular Toxicology colloq. & lit. thes.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology	Ac. Year (September)	12EC	XM_0033

Drug Discovery and Safety CSE variants

Programme Sections

- DDS Colloquium & Literature Thesis
- DDS research project
- DDS specialisation Education

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description

Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Education

Description

The teaching in these specialisation is mainly in Dutch. Therefore we also give the requirements in Dutch.

Programma

De opleiding voor het behalen van de eerstegraads lesbevoegdheid start twee keer per jaar, in september en in februari. De opleiding wordt aangeboden in twee semesters. Uitgaande van de start in september duurt semester 1 tot en met januari en semester 2 tot juli. De opleiding is sterk praktijkgericht. De helft van de opleiding bestaat uit praktijk door werkervaring of stage (ook wel schoolpracticum genoemd) op een school voor voortgezet onderwijs. Daarnaast kent de opleiding vier componenten: vakdidactiek, algemene didactiek/pedagogiek, praktijkonderzoek en verdiepingsmodulen.

Naast de educatievakken volgt de student 60 sp Pharmaceutical Sciences / Chemistry vakken, in overleg met de mastercoördinator van de gekozen specialisatie. Hierbij zijn de twee vakken Literature thesis and Colloquium Chemistry Education en Master Research Project Chemistry-Education verplicht.

Studenten die bij de Communicatie variant de vakken 'interpersoonlijke communicatie' en 'museologie en buitenschoolse educatie' volgen, krijgen bij de lerarenopleiding een vrijstelling voor verdiepingsmodulen, een deel van het praktijkonderzoek en een deel van algemene didactiek.

Programme Sections

- Teaching Chemistry
- DDS Courses

Teaching Chemistry

Description

Scheikunde is overal om ons heen te vinden. Neem jij leerlingen stap voor stap mee in de wereld van atomen, moleculen en chemische reacties? Speel jij graag een rol in hun ontwikkeling en wil je jouw passie voor scheikunde met hen delen? Als docent Scheikunde behandel je niet alleen (bio)chemie en de transformatie van materialen, maar je kijkt ook naar ook actuele maatschappelijke thema's vanuit een scheikundig oogpunt. Het visueel maken van abstracte ideeën speelt bij het schoolvak Scheikunde een belangrijke rol. Tijdens de lerarenopleiding Scheikunde leer je hoe je visualisaties, analogieën en simulaties effectief kunt inzetten om complexe onderwerpen begrijpelijk te maken voor je leerlingen. Je ontdekt ook hoe je leerlingen kunt begeleiden in het onderzoeken en verklaren van de wereld om hen heen en hoe je hen actief aan het denken kunt zetten. Ook leer je hoe je de effectiviteit van practica en demonstratieproeven kunt vergroten. Welke proef zorgt ervoor dat een ingewikkeld concept écht blijft hangen bij je leerlingen?

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2
Practice 2	P2+3	9EC	O_MLPRAK_2
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

DDS Courses

Description

This specialisation is intended for students with a BSc degree in any of the bèta-studies who want to specialise in communication. The programme focuses on science communication theory, research and practice. The programme of the communication (C) specialisation is 1 year (60 credits). This specialisation may not be combined with te Societal specialisation (M) or the Education specialisation (E). C-courses are shared with master students

from the Faculty of Earth and Life Sciences.

To complete the entire Master programme (120 EC) of the Communication, education or social variant, the student has to choose 60 EC in DDS courses in consultation with the master track coordinator.

Programme Sections

- Specialisation Courses
- DDS Colloquium & Literature Thesis
- DDS research project

Specialisation Courses

Description

In consultation with the master coordinator and depending on the chosen specialization, 6 EC have to be chosen from the following list.

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Physical Organic Chemistry	P1	6EC	X_435663
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description

Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Course Name	Period	Credits	Code
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description
Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Molecular Toxicology specific courses

Courses

Course Name	Period	Credits	Code
ADME	P1	6EC	XM_0136
Molecular and Cellular Toxicology	P2	6EC	XM_0113
Advanced Course in Molecular Toxicology	P4	6EC	XM_0114

DDS core courses

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Principles of Drug Targets	P1	6EC	XM_0105
Principles of Drug Discovery	P2	6EC	XM_0106

Drug Discovery Sciences specialisation Computer-aided Drug Design

Programme Sections

- DDS elective courses
- DDS specialisation Computer-aided Drug Design R variant
- Drug Discovery and Safety CSE variants
- DDS specialisation Computer-aided Drug Design specific courses
- DDS core courses

DDS elective courses

Courses

Course Name	Period	Credits	Code
Minor Research Project Bioanalytical Chemistry	Ac. Year (September)	18EC	XM_0044
Minor Research Project Environmental Bioanalysis	Ac. Year (September)	18EC	XM_0045

Course Name	Period	Credits	Code
Minor Research Project Medicinal Chemistry	Ac. Year (September)	18EC	XM_0046
Minor Research Project Molecular & Computational Toxicology	Ac. Year (September)	18EC	XM_0047
Minor Research Project Organic Chemistry	Ac. Year (September)	18EC	XM_0048
Minor Research Project Radiopharmaceutical Chemistry	Ac. Year (September)	18EC	XM_0049
Minor Research Project Theoretical Chemistry	Ac. Year (September)	18EC	XM_0050
ADME	P1	6EC	XM_0136
Data Management in Drug Discovery	P1	6EC	XM_0117
Fundamentals of Bioinformatics	P1	6EC	X_405052
Oncogenesis	P1	6EC	M_OONC03
Physical Organic Chemistry	P1	6EC	X_435663
Advanced Bioanalytical Approaches	P2	6EC	XM_0107
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Molecular and Cellular Toxicology	P2	6EC	XM_0113
Spectroscopic Approaches in Medicinal Chemistry	P2	6EC	XM_0109
Advanced Course in Molecular Toxicology	P4	6EC	XM_0114
Advanced Molecular Orbital Theory	P4	6EC	XMU_0028
Advanced Pharmacology	P4	6EC	XM_0112
Biopharmaceuticals & Biopharma Proteomics	P4	6EC	XM_0108
Concepts in Chemical Biology	P4	6EC	XM_0080
Molecular Modelling	P4	6EC	XMU_0059
Synthetic Approaches in Medicinal Chemistry	P4	6EC	XM_0110
Translational Radiopharmaceutical Sciences	P4	6EC	XM_0067
Advanced Radiopharmaceutical Chemistry	P5	6EC	XM_0069
Protein Analysis	P5	6EC	X_435045
Replacement, Refinement and Reduction of animal experiments applying human cells in culture	P5	6EC	XM_0148

DDS specialisation Computer-aided Drug Design R variant

Programme Sections

- DDS ethics and academic skills
- DDS specialisation Computer-aided Drug Design research project (mj)
- DDS specialisation Computer-Aided Drug Design colloq. & lit. thes.

DDS ethics and academic skills

Description

In consultation with the master coordinator and depending on the chosen specialisation, 6 EC have to be chosen from the following list for students participating in the research-variant.

Courses

Course Name	Period	Credits	Code
Teaching Assistant	Ac. Year (September)	3EC	XM_432741
Teaching Assistant	Ac. Year (September)	6EC	XM_432742
Managing Science and Technology in Society	P1	6EC	AM_470586
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587
Societal Entrepreneurship in Health and Life Sciences	P1	6EC	AM_470575

Course Name	Period	Credits	Code
Business Management in Health and Life Sciences	P2	6EC	AM_470584
Communication, Organization and Management	P2	6EC	AM_470572
Science Journalism	P2	6EC	AM_471014
Clinical Development and Clinical Trials	P3	3EC	AM_1180
Epidemiology	P3	3EC	AM_1179
Ethics in Life Sciences	P3	3EC	AM_470707
Science in Dialogue	P3	6EC	AM_1002
Tutoring Students	P3	3EC	X_432625
Scientific Writing in English	P3, P6	3EC	X_400592

DDS specialisation Computer-aided Drug Design research project (mj)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry	Ac. Year (September)	42EC	XM_0039
Major Research Project Molecular & Computational Toxicology	Ac. Year (September)	42EC	XM_0040
Major Research Project Theoretical Chemistry	Ac. Year (September)	42EC	XM_0043

DDS specialisation Computer-Aided Drug Design colloq. & lit. thes.

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry	Ac. Year (September)	12EC	XM_0032
Colloquium & Literature Thesis Molecular & Computational Toxicology	Ac. Year (September)	12EC	XM_0033
Colloquium & Literature Thesis Theoretical Chemistry	Ac. Year (September)	12EC	XM_0036

Drug Discovery and Safety CSE variants

Programme Sections

- DDS Colloquium & Literature Thesis
- DDS research project
- DDS specialisation Education

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description

Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Course Name	Period	Credits	Code
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description
Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Education

Description

The teaching in these specialisation is mainly in Dutch. Therefore we also give the requirements in Dutch.

Programma

De opleiding voor het behalen van de eerstegraads lesbevoegdheid start twee keer per jaar, in september en in februari. De opleiding wordt aangeboden in twee semesters. Uitgaande van de start in september duurt semester 1 tot en met januari en semester 2 tot juli. De opleiding is sterk praktijkgericht. De helft van de opleiding bestaat uit praktijk door werkervaring of stage (ook wel schoolpracticum genoemd) op een school voor voortgezet onderwijs. Daarnaast kent de opleiding vier componenten: vakdidactiek, algemene didactiek/pedagogiek, praktijkonderzoek en verdiepingsmodulen.

Naast de educatievakken volgt de student 60 sp Pharmaceutical Sciences / Chemistry vakken, in overleg met de mastercoördinator van de gekozen specialisatie. Hierbij zijn de twee vakken Literature thesis and Colloquium Chemistry Education en Master Research Project Chemistry-Education verplicht.

Studenten die bij de Communicatie variant de vakken 'interpersoonlijke communicatie' en 'museologie en buitenschoolse educatie' volgen, krijgen bij de lerarenopleiding een vrijstelling voor verdiepingsmodulen, een deel van het praktijkonderzoek en een deel van algemene didactiek.

Programme Sections

- Teaching Chemistry
- DDS Courses

Teaching Chemistry

Description

Scheikunde is overal om ons heen te vinden. Neem jij leerlingen stap voor stap mee in de wereld van atomen, moleculen en chemische reacties? Speel jij graag een rol in hun ontwikkeling en wil je jouw passie voor scheikunde met hen delen? Als docent Scheikunde behandel je niet alleen (bio)chemie en de transformatie van materialen, maar je kijkt ook naar ook actuele maatschappelijke thema's vanuit een scheikundig oogpunt. Het visueel maken van abstracte ideeën speelt bij het schoolvak Scheikunde een belangrijke rol. Tijdens de lerarenopleiding Scheikunde leer je hoe je visualisaties, analogieën en simulaties effectief kunt inzetten om complexe onderwerpen begrijpelijk te maken voor je leerlingen. Je ontdekt ook hoe je leerlingen kunt begeleiden in het onderzoeken en verklaren van de wereld om hen heen en hoe je hen actief aan het denken kunt zetten. Ook leer je hoe je de effectiviteit van practica en demonstratieproeven kunt vergroten. Welke proef zorgt ervoor dat een ingewikkeld concept écht blijft hangen bij je leerlingen?

Programme Sections

- Leraar VHO februari start
- Leraar VHO September start

Leraar VHO februari start

Courses

Course Name	Period	Credits	Code
Pedagogische en Didactische verdieping Semester 1	P1+2+3	9EC	O_MPEDIVER_F
Peergroup 2 Semester 1	P1+2+3	0EC	O_MLPEERG_2F
Practice 3	P1+2+3	15EC	O_MLPRAK_3F
Didactics 1	P4	6EC	O_MLDIDAC_1F
Practice 1	P4	6EC	O_MLPRAK_1F
Peergroup 1 Semester 2	P4+5+6	0EC	O_MLPEERG_1F
Didactiek 2 Semester 2	P5+6	9EC	O_MLDIDAC2F
Practice 2	P5+6	9EC	O_MLPRAK_2F

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

Leraar VHO September start

Courses

Course Name	Period	Credits	Code
Didactics 1	P1	6EC	O_MLDIDAC_1
Practice 1	P1	6EC	O_MLPRAK_1
Peergroup 1 Semester 1	P1+2+3	0EC	O_MLPEERGR_1
Didactiek 2 Semester 1	P2+3	9EC	O_MLDIDAC2
Practice 2	P2+3	9EC	O_MLPRAK_2
Pedagogische en Didactische Verdieping Semester 2	P4+5+6	9EC	O_MPEDIVER
Peergroup 2 Semester 2	P4+5+6	0EC	O_MLPEERGR_2
Practice 3	P4+5+6	15EC	O_MLPRAK_3

Programme Sections

- Leraar VHO Keuzevakken

Leraar VHO Keuzevakken

Description

Kies minimaal voor minimaal 6 EC aan keuzevakken.

Courses

Course Name	Period	Credits	Code
AI in het Onderwijs	P1, P4	3EC	O_MAIIHO
Wereldburgerschap deel 1	P1, P4	3EC	O_MWRLDBS_1
Tweetalig Onderwijs	P2	3EC	O_MTTO
Inclusieve Digitale Didactiek	P2, P5	3EC	O_MIDD
Vernieuwende onderwijsconcepten	P2, P5	3EC	O_MLVNON
Wereldburgerschap Deel 2	P2, P5	3EC	O_MWRLDBS_2
Praktijkonderzoek	P4+5+6	6EC	O_MLPROZ

DDS Courses

Description

This specialisation is intended for students with a BSc degree in any of the bèta-studies who want to specialise in communication. The programme focuses on science communication theory, research and practice. The programme of the communication (C) specialisation is 1 year (60 credits). This specialisation may not be combined with te Societal specialisation (M) or the Education specialisation (E). C-courses are shared with master students from the Faculty of Earth and Life Sciences.

To complete the entire Master programme (120 EC) of the Communication, education or social variant, the student has to choose 60 EC in DDS courses in consultation with the master track coordinator.

Programme Sections

- Specialisation Courses
- DDS Colloquium & Literature Thesis
- DDS research project

Specialisation Courses

Description

In consultation with the master coordinator and depending on the chosen specialization, 6 EC have to be chosen from the following list.

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002
Physical Organic Chemistry	P1	6EC	X_435663
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673

DDS Colloquium & Literature Thesis

Description

Students need to select a total of 6 EC or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)
- DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis
- Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)
- Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)
- DDS research variant Diagnostics and Imaging colloquium & literature thesis

Colloquium & Literature Thesis DDTF (Choose 1 of 2 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020

DDS research variant Drug Disposition and Safety Assessment colloquium & literature thesis

Description

Choose 1 (minimal 6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019

Colloquium & Literature Thesis CMCT (Choose 1 of 3 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

Colloquium & Literature Thesis DD&S (Choose 1 of 4 – 6 EC required)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0018
Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	6EC	XM_0019
Colloquium & Literature Thesis Organic Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0020
Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0021

DDS research variant Diagnostics and Imaging colloquium & literature thesis

Description
Choose 1 out of 2 (6EC)

Courses

Course Name	Period	Credits	Code
Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0016
Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)	Ac. Year (September)	6EC	XM_0017
Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)	Ac. Year (September)	6EC	XM_0022

DDS research project

Description

Students need to select at least 24 credits or more from the following list.

Note: Every programme, including the choice of optional courses, has to be discussed and agreed upon with the master coordinator or a personal mentor and approved by the Examination Board.

Programme Sections

- Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)
- Research project DD&SA (Compulsory for DD&SA, 24 EC required)
- Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)
- DDS research variant Drug Design and Synthesis research project (major)
- Research Project D&I (Major)

Research project DDTF (Compulsory for DDTF, choose 1 of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027

Research project DD&SA (Compulsory for DD&SA, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026

Research project CMCT (Compulsory for CMCT, choose 1 out of 3, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Theoretical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0029

DDS research variant Drug Design and Synthesis research project (major)

Description

Research project DD&S (Compulsory for DD&S, choose 1 out of 2, 24 EC required)

Courses

Course Name	Period	Credits	Code
Major Research Project Medicinal Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0025
Major Research Project Molecular & Computational Toxicology (C, E, S)	Ac. Year (September)	24EC	XM_0026
Major Research Project Organic Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0027
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

Research Project D&I (Major)

Description

Choose 1 out of 3. Minimal 24EC.

Courses

Course Name	Period	Credits	Code
Major Research Project Bioanalytical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0023
Major Research Project Environmental Bioanalysis (C, E, S)	Ac. Year (September)	24EC	XM_0024
Major Research Project Radiopharmaceutical Chemistry (C, E, S)	Ac. Year (September)	24EC	XM_0028

DDS specialisation Computer-aided Drug Design specific courses

Courses

Course Name	Period	Credits	Code
Data Management in Drug Discovery	P1	6EC	XM_0117
Computer-Aided Drug Design and Virtual Screening	P2	6EC	X_432673
Biomolecular Simulation in Drug Discovery	P4	6EC	XM_0118

DDS core courses

Courses

Course Name	Period	Credits	Code
Research Skills and Career Perspectives	Ac. Year (September)	0EC	XM_0002

Course Name	Period	Credits	Code
Principles of Drug Targets	P1	6EC	XM_0105
Principles of Drug Discovery	P2	6EC	XM_0106

Courses

ADME

Course Code	XM_0136
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Moco
Examiner	dr. S. Moco
Teaching Staff	dr. S. Moco, dr. J. Kool
Teaching method(s)	Study Group, Other, Lecture

Course Objective

“What happens to a drug inside the human body?”

This course will dive into the biochemical processes underlying the absorption (A), distribution (D), metabolism (M) and excretion (E) of drugs (or xenobiotics, in general): the ADME processes, an essential discipline in pharmaceutical sciences.

Course Content

Of the thousands of novel compounds that are developed in drug discovery programs, only a fraction has the appropriate pharmacokinetic properties to become a drug product. Pharmacokinetics is determined by the complex processes involved in absorption (A), distribution (D), metabolism (M) and excretion (E) of the drug in the human body, the so-called ADME processes. ADME processes directly impact drug effectiveness (pharmacodynamics), safety and dosage. ADME processes do not only apply to pharmaceutical drugs, but environmental contaminants, nutraceuticals or phytochemicals may follow analogous processes. In this course, the student will learn: the fundamentals of ADME, state-of-the-art analytical chemistry approaches to study ADME, and inter-individual variability impacting ADME.

In this course, the students will be familiarised with:

Drug **A**bsorption

- routes of administration, with emphasis on the gastro-intestinal tract (GIT);
- drug transporters involved in absorption;
- how much drug reaches systemic circulation

Drug **D**istribution

- how does the drug spread through the body’s tissues and fluids;
- blood brain barrier;
- drug protein binding and volume of distribution

Drug **M**etabolism

- how does the body chemically change the drug;
- xenobiotic metabolism mostly in the liver and gut, mediated by major enzymes classes and isoforms: CYPs, UGTs, SULTs, etc, and integrated metabolic pathway maps;
- (re)active metabolites and drug-drug interactions

Drug **E**xcretion

- how does the body eliminate the drug (often via urine or bile);
- drug transporters involved in excretion;

- drug clearance and entero-hepatic recirculation

Methods and approaches to study ADME processes

- biochemical, cellular, pre-clinical and clinical studies;
- analytical chemistry tools to monitor drugs and metabolites thereof in biological samples: LC-MS and NMR metabolomics;
- strategies to perform drug metabolite identification (metID) from LC-MS/MS;
- bioanalytical strategies for bio-affinity profiling of metabolic mixtures;
- use of (stable) isotopically labelled drugs to study ADME

Factors impacting ADME processes

- physico-chemical properties of drugs;
- inter-individual variability in: environmental factors, genetic variants (metabolic enzymes, transporters), sub-populations;
- personalised medicine

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Written exam and assignments

Literature

Consulting books:

1. ‘ADME Processes in Pharmaceutical Sciences’. Eds. Alan Talevi, Pablo A. M. Quiroga. Springer 2018, ISBN 978- 3-319-99593-9 2.
2. 'Rang & Dale's Pharmacology’. Eds. J.M. Ritter and R.J. Flower. Elsevier 2023, ISBN: 978-0-3238-7395-6.
3. ‘Drug-like properties: concepts, structure design and methods. From ADME to Toxicity optimization.’ Eds. E.H. Kerns and L. Di, Academic Press, 2008, ISBN: 978-0-1236-9520-8.

Additional course material will be provided in Canvas.

Additional Information Target Audience

All MSc DDS specialisations, MSc Biomolecular Sciences and MSc Chemistry (tracks Chemistry of Life and Analytical Sciences)

Recommended background knowledge

Background knowledge in chemistry is recommended.

Advanced Bioanalytical Approaches

Course Code	XM_0107
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. Kohler
Examiner	dr. I. Kohler
Teaching Staff	dr. I. Kohler
Teaching method(s)	Seminar, Lecture

Course Objective

The objectives of this course are:

- Get familiar with the bioanalytical approaches used in drug development and discovery, as well as in clinical,

forensic and toxicological fields.
- Learn how to set up, develop, validate and apply bioanalytical strategies based on a clinical or toxicological question - from the initial sampling to the final data analysis and interpretation.
- Increase theoretical knowledge in advanced sample preparation strategies and state-of-the-art MS-based chromatography and electrophoresis.
- Learn to use advanced analytical techniques to study both endogenous (i.e., metabolites) and exogenous (i.e., illicit drugs, pharmaceutical compounds, drug candidates) small molecules in complex human biological samples, such as serum, urine, saliva, and cerebrospinal fluid.
- Learn to work with targeted and untargeted clinical metabolomics in the context of biomarker and drug discovery.

Course Content

In the Advanced Bioanalytical Approaches course, you will get familiar with state-of-the-art bioanalytical approaches used in drug discovery.

You will learn the important aspects associated with the whole bioanalytical workflow, i.e., from sample preparation to data processing, analysis, and interpretation. Among others, the following techniques will be presented, discussed, and illustrated: advanced solid-phase extraction, liquid-liquid extractions (including microextractions), ultra high-pressure liquid chromatography, hydrophilic interaction chromatography, as well as advanced mass spectrometry approaches (for qualitative and quantitative analysis). These bioanalytical techniques will be discussed in the context of clinical toxicology, forensic toxicology, biomarker discovery, and therapeutic drug monitoring, focusing on endogenous (metabolites) and exogenous (e.g., drugs, pharmaceutical compounds) analytes. A special emphasis will be paid on sample collection, handling and storage.

The course will also provide you with an introduction to metabolomics, including targeted and untargeted strategies. You will explore how metabolomics can be used in drug discovery and in clinical settings, for instance enabling earlier diagnosis, more accurate prognosis, treatment staging, treatment individualization, and a better understanding of pathological (bio)molecular mechanisms in disease. The challenges associated with clinical metabolomics and future perspectives will also be discussed.

Finally, you will apply the knowledge gathered during the course during group assignments, which will also help you to develop academic skills (critical thinking, communication, etc.).

Additional Information Teaching Methods

This course follows a blended learning approach, combining online and in-person activities. The teaching methods include:

- Pre-class video lectures (1x/week)
- Live lectures with discussion of case studies (1x/week)
- Online quizzes (1x/week)
- Group assignments: self-study resulting in a poster and an oral presentation.

Method of Assessment

Written exam (individual grade)
Online quizzes (individual grade)
Group assignment (group grade), composed of a poster and a presentation.

Entry Requirements

Background knowledge in analytical chemistry, notably chromatography (reversed-phase liquid chromatography), electrophoresis, mass spectrometry, and sample preparation (liquid-liquid extraction and solid-phase extraction).

Literature

Recorded videos, PowerPoint slides, and additional study material (provided via web-links and/or via PDFs of papers, book chapters and/or other relevant material), which are made available through Canvas.

Additional Information Target Audience

MSc students of the Bioanalytics track; other DDS students interested in Bioanalysis and Metabolomics; students with a bioanalytical, biomedical and/or pharmacochemical background.

This course cannot be followed by students who have followed the course "Bioanalytical Sciences" from the Master Chemistry, track Analytical Sciences.

Recommended background knowledge

Background knowledge in analytical chemistry, notably chromatography (reversed-phase liquid chromatography), electrophoresis, mass spectrometry, and sample preparation (liquid-liquid extraction and solid-phase extraction).

Advanced Course in Molecular Toxicology

Course Code	XM_0114
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. P. Jennings
Examiner	prof. dr. P. Jennings
Teaching Staff	prof. dr. P. Jennings, dr. A. Wilmes
Teaching method(s)	Seminar, Lecture

Course Objective

Utilisation of knowledge of chemical properties and biological interactions for application to drug development, hazard identification and risk assessment.

Course Content

The course looks at pharmacology and toxicology from different view points, chemical-centric, tissue-centric and pathway centric. The course is designed to facilitate application of knowledge utilising a series of case study assignments.

These will include:

- Chemical properties and descriptors for determination of risk of toxicity.
- Tissue specific properties for determination of pharmacology, metabolism and sensitivity to chemical induced toxicity.
- Analysis of transcriptomic data for determining biological effects of selected chemicals.
- Principles required for Adverse Outcome Pathway building and deployment.

Additional Information Teaching Methods

Case study based assignments, presentations, data analysis and discussions.

Attendance is mandatory.

Students are expected to put in significant own time (approx. 20 hours a week).

Method of Assessment

Continuous assessment, assignments and examination.

Literature

USERS' HANDBOOK SUPPLEMENT TO THE GUIDANCE DOCUMENT FOR DEVELOPING AND ASSESSING AOPs.
http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env

Additional Information Target Audience

Anyone following the toxicological or pharmacological tracks at the VU or elsewhere.

Recommended background knowledge

A good working knowledge of toxicology.
The topics taught in 'ADME' and/or 'Drug-induced stress and Cellular Responses', and/or 'Molecular and Cellular Toxicology' or equivalent courses are regarded as known.

Advanced Molecular Orbital Theory

Course Code	XMU_0028
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.M. Bickelhaupt
Examiner	prof. dr. F.M. Bickelhaupt
Teaching Staff	dr. P. Vermeeren, prof. dr. C. Fonseca Guerra, prof. dr. F.M. Bickelhaupt
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130551

Additional Information Teaching Methods

-

Method of Assessment

-

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Advanced Pharmacology

Course Code	XM_0112
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. H.F. Vischer
Examiner	dr. H.F. Vischer
Teaching Staff	prof. dr. R. Leurs, dr. H.F. Vischer
Teaching method(s)	Seminar, Lecture

Course Objective

To understand the biological action of drugs and design the drugs of the future, the latest concepts of drug action will be discussed at the molecular level in this course.

Course Content

This course will discuss the latest modalities used for therapeutic targeting next to modern concepts like biased signalling, allosterism
and drug binding kinetics. Moreover, advanced analytical pharmacological approaches to dissect such concepts will be presented, next to the use of new assay technologies for efficient drug target characterization.

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Method of Assessment

written examination and assignments

Literature

Primary literature and handouts via Canvas.

Additional Information Target Audience

500-level, 1DDS, 1BMS

Additional Information

Students that did not attend Pharmaceutical Sciences (BSc) course Molecular Pharmacology (XB_43001) or in DDS (MSc) course Principles of drug targets (XM_0105) should contact the teacher(s) before signing-up.

Custom Course Registration

Students that did not attend Pharmaceutical Sciences (BSc) course Molecular Pharmacology (XB_43001) or in DDS (MSc) course Principles of drug targets (XM_0105) should contact the teacher(s) before signing-up.

Recommended background knowledge

Drug Target Biochemistry and Signaling (400).
In addition, students should be familiar at the start of this course with the topics saturation and competition binding ligand-receptor
binding, agonist, and competitive antagonism as learned in the Pharmaceutical Sciences (BSc) course Molecular Pharmacology (XB_43001) or in DDS (MSc) course Principles of drug targets (XM_0105). These topics are also examined in the Advanced Pharmacology course.

Advanced Radiopharmaceutical Chemistry

Course Code	XM_0069
Credits	6
Period	P5
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.J.L. Steen
Examiner	dr. E.J.L. Steen
Teaching Staff	prof. dr. A.D. Windhorst, prof. dr. D.J. Vugts
Teaching method(s)	Seminar, Practical, Lecture

Course Objective

To obtain detailed knowledge of radiopharmaceutical chemistry

Course Content

Radiopharmaceutical chemistry is the science of synthesising radiopharmaceuticals for imaging techniques like Positron Emission Tomography (PET) and Single Photon Emission Computed Tomography (SPECT) or for therapeutic applications. The great clinical successes of nuclear imaging and targeted radionuclide therapy have resulted in rapid growth of the field of radiopharmaceutical chemistry, an essential component of nuclear medicine. This course focusses on advanced radiopharmaceutical chemistry, with key coverage of the chemistry that is required to radiolabel compounds. The course will cover advanced radiochemistry of 11C, 13N, 15O, 18F, Iodine radionuclides as well as radiometals, production of radionuclides, analysis techniques and radiation safety.

Additional Information Teaching Methods

The "Flipped classroom method" is used. An active participation and full time presence of the students is expected. Some lectures will be traditional classes where the theory is presented by the teacher, some will be presented by the students where they will present on basis of the theory described in the textbook. There will be several classes devoted to problem solving and case studies, as well as written assingments.

Method of Assessment

There will be several assignments, 4 days of radiochemistry practical, an obligatory radiation safety class including an exam and cases during the course. Every presentation of the students based on the theory of the text book or a research paper will be graded, the practical work will be graded based on attitude, experimental practice and report. Finally, there will be a radiation safety exam, which needs to be passed before allowance to the practical. Upon pass on of the radiations safety exam, 15 point can be earned, when passed only at the re-exam no points can be earned. All assignments, radiation safety exam and practical will add up to 100 points, minimum amount of 55 points is needed to pass the course and the total score will be 45% of the final grade. There will be a written or oral final examination, accounting for 55 points of the final grade which must be passed > 5.5.

Literature

Radiopharmaceutical Chemistry, Lewis, Windhorst, Zeglis editors. Springer, 2019. ISBN 978-3-319-98946-4. Available as PDF via the VU library: https://link.springer.com/book/10.1007/978-3-319-98947-1

Additional Information Target Audience

mDDS, DD&S and D&I

Custom Course Registration

VUNet

Recommended background knowledge

Organic chemistry or radiopharmaceutical chemistry or medicinal chemistry or equivalent.

AI in het Onderwijs

Course Code	O_MAIIHO
Credits	3
Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. ir. F.F.J. Hermans
Examiner	dr. H.B. Westbroek
Teaching Staff	prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

Het doel van dit vak is op basis van kennis over AI en over onderwijs theorieën een helder beeld schetsen van hoe je je als docent en school wilt verhouden tot generatieve AG.

Course Content

Studenten leren in dit vak:

- Hoe generatieve AI werkt
- Wat de impact van AI op onderwijs kan zijn, vanuit een aantal verschillende theorieën en invalshoeken
- Wat de risico's van AI zijn, bijvoorbeeld mbt bias en deskilling
- Reflecteren op het gebruik van AI in de eigen onderwijspraktijk

Belangrijk: Voor dit vak is geen technische voorkennis vereist en het is voor alle studenten van de lerarenopleiding (en aanverwante studies) geschikt, zeker niet louter voor de beta's!

Additional Information Teaching Methods

Het vak bestaat uit:

- 5 verplichte hoorcolleges (10 uur)
- Huiswerkopdrachten maken (25 uur)
- Lezen literatuur (25 uur)

- Uitwerken beleidsplan (24 uur)

Method of Assessment

Studenten sluiten dit vak af door het schrijven van een beleidsplan over het gebruik van AI binnen hun vak of school.

Literature

Zie Canvas

Biomolecular Simulation in Drug Discovery

Course Code	XM_0118
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.P. Geerke
Examiner	dr. D.P. Geerke
Teaching Staff	dr. D.A. Poole, dr. D.A. Poole, dr. D.P. Geerke
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

Providing an overview of the theoretical background and application of molecular dynamics computer simulations in the context of drug discovery

Course Content

Molecular Dynamics (MD) simulations are becoming increasingly important in in silico drug design. While static structures of protein-ligand complexes can be of great help in studying protein binding and interaction, a detailed understanding of drug action upon complexation may well require to explicitly account for the conformational flexibility of the protein and the dynamics of ligand binding. MD simulations can take on this role and be used to e.g. elucidate induced fit effects upon complexation, study kinetics of protein-ligand binding, and include entropic effects in affinity prediction.

The theoretical part of this course consists of a series of lectures in which theory of and algorithms for MD simulation will be discussed, along with the models (force fields) to represent atomic interactions during simulation. In addition advanced sampling and (alchemical and end-point) methods will be presented that can be used to compute the free energy of e.g. protein-ligand binding from simulation.

A major part of the course will focus on the use of MD and its application in a drug discovery context. In a first series of computer practicals students will be made familiar with setting up, running and analyzing MD simulations. Subsequently a small project will be performed in which MD will be used to study protein-ligand binding and interactions in the context of a pharmaceutically relevant application. Assessment of the written report on this application study will substantially contribute to the final grade of the course.

Additional Information Teaching Methods

Lectures, computer lab sessions (tutorials), working independent on a small project

Method of Assessment

Written report on computer lab work, and a written or oral exam

Literature

Links to relevant literature will be provided in advance and during the course

Additional Information Target Audience

mDDS

Recommended background knowledge

BSc FAR course Molecular Modeling or MSc DDS course Principles of Drug Discovery, or equivalent

Biopharmaceuticals & Biopharma Proteomics

Course Code	XM_0108
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J. Kool
Examiner	dr. J. Kool
Teaching Staff	dr. K. Jooß MSc, dr. I. Kohler
Teaching method(s)	Seminar, Practical, Lecture

Course Objective

- Get a thorough knowledge on biopharmaceuticals (in particular monoclonal antibodies and other protein biopharma) and their importance in the pharmaceutical industry.
- Understand the processes that biopharma have to undergo in order to reach the market.
- Acquire an in-depth knowledge on bioanalytical techniques to characterize biopharmaceuticals.
- Learn about the different types of biopharma and their analysis and characterization, including vaccines, nanobodies and endogenous hormone biologicals.
- Get a thorough understanding of biopharma proteomics.
- Learn how to conduct biopharma proteomics in the laboratory, (practical research will be done on venoms-to-drugs biopharma pipelines).

Course Content

Biopharmaceuticals have become an essential class of drugs that provide new therapeutic possibilities for the treatment of diseases such as cancer, diabetes, cardiovascular diseases and autoimmune diseases. In the world's top twenty medicines around half of them are biopharmaceuticals. This course will cover a large diversity of biopharmaceuticals, with an emphasis on therapeutic protein biopharmaceuticals. Biopharmaceutical monoclonal antibodies will get a particular attention, as well as endogenous hormone biopharma (such as insulin and human growth factors) and antibody-drug-conjugates. Bioanalytical characterization of vaccines during vaccine development will also briefly be looked at.

Important aspects such as indication, drug targets, mechanisms of action, product discovery and development, production, purification and administration will be discussed. Extensive attention will be paid to the comprehensive bioanalytical characterization of protein biopharmaceuticals using analytical techniques typically used in biopharma companies. Quantitative analysis of protein drugs, as well aspects such as immunogenicity and biosimilars will also be included in this course.

Proteomics research in relation to research on protein biopharmaceuticals is a crucial aspect within bioanalytics for biopharma drug discovery & development. The pharmaceutical industry has become increasingly interested in biologics from animal venoms as a potential source for therapeutic agents. Notably, bioactive peptides in venoms, but also proteins, have raised attention of the drug discovery arena. To date, multiple drugs derived from venom peptides and proteins have been approved by the FDA and amongst others its European counterpart, and many others are currently commencing through clinical trials.

The practical part of this course will lead you through venoms-to-drugs drug discovery and teach you how to perform biopharma proteomics research. You will get hands-on experience and tutorials in studying venoms as model bioactive mixtures for finding venom-resided bioactive proteins with biopharmaceutical candidate properties towards a chosen drug target. After state-of-the art venom separations and nanofractionation, you will see how to perform HTS bioassays for a selected drug target. You will characterize the bioactive protein-based toxins found that interacted with the drug target by using nanoLC-MS and proteomics. As proteomics is an important technique for drug-target finding, the use of proteomics in target finding will also be touched upon.

Additional Information Teaching Methods

Lectures, research oriented practical days and demonstrations, tutorials, self study, assignment resulting in a presentation.

Method of Assessment

Written exam (70% of the final grade).

Lab practical performed in small groups

- Laboratory experiments related to venom proteomics (Pass/Fail)
- Short report on the results obtained during the practical (Pass/Fail)

Group presentation of an assignment on a biopharmaceutical (30%).

(The partial marks of the written exam and the presentation all should be 5.5 or higher to pass the course).

Literature

PowerPoint presentations of lectures provided as PDFs and additional study material (provided as web-links and/or as PDFs), which are made available through Canvas. Lab research manual for the practical research part of this course. Other material needed for the research practical and experimental demonstrations in the lab. Movie files of experimental procedures and data processing procedures.

Additional Information Target Audience

MSc DDS students of the Bioanalytics track, other DDS students interested in Biopharmaceuticals and/or Proteomics.

Business Management in Health and Life Sciences

Course Code	AM_470584
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K.D.S. Fernald MSc
Examiner	dr. K.D.S. Fernald MSc
Teaching Staff	dr. L.H.M. van de Burgwal, dr. E.M. Janse, dr. K.D.S. Fernald MSc
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

1. To acquire knowledge and understanding into theory of knowledge valorisation in health and life sciences
2. To acquire knowledge and insight in how to organise, protect and finance a business in health and life sciences
3. To acquire knowledge and understanding into the pharmaceutical industry’s business model and business processes
4. To acquire knowledge and understanding into the challenges that the pharmaceutical industry faces
5. To apply newly acquired knowledge and understanding by solving case examples
6. To apply newly acquired knowledge and understanding in writing a business plan
7. To reflect on and critically evaluate the role of the pharmaceutical industry in the healthcare system
8. To learn to autonomously write a (personal) business plan

Course Content

As a result of external factors (for example ageing of the population and technological advancement, leading to increased healthcare costs), our healthcare system is under pressure. As a central stakeholder in this healthcare system, the pharmaceutical industry is facing significant challenges. Business Management in the Health and Life Sciences focuses on gaining insight in the pharmaceutical industry, its business model, business processes challenges, as well as strategies and actions to overcome these challenges.

During the course, dr. Linda van de Burwal and dr. Kenneth Fernald, together with highly experienced guest lecturers from the field will teach theoretical and practical knowledge during lectures and seminars.
Tangible subjects that will be discussed during the lectures and seminars include the pharmaceutical industry’s business model and business processes, intellectual property, portfolio management, finance, risk capital, grants and subsidies, team building and people management, different legal entities, fiscal and legal aspects when starting a new company, SWOT analysis in the life sciences and clinical trials.

The newly acquired knowledge is tested via an assignment during which students will write either a personal career business plan or a ‘real’ business plan (50% of the final grade), a financial plan for the personal BP (30% of the final grade), patent seminars group assignment (15% of the final grade), and a final reflection video (5% of the final grade).

Additional Information Teaching Methods

Lectures: +-50 h
Computer seminars: 7,5 h
Work on assignment and self-study: +- 40h

Method of Assessment

- elevator pitch dream : pass/ fail
- providing feedback on pitch : pass/fail
- personal career business plan : 50 % of the final grade
- financial plan for the personal BP : 30 % of the final grade
- patent seminars group assignment : 15 % of the final grade
- final reflection video : 5 % of the final gradea
- final online exam : pass or fail

All parts have to be passed successfully.

Literature

Lectures provide key material to this course. The following book can be used as background to the lectures:

- Osterwalder, A. & Pigneur, Y. (2009). Business model generation. Self-published.

The course’s assignment is to write either a ‘real’ business plan or a 'personal' business plan. When writing a business plan, students are recommended to use the following books:

- Kubr, Marchesi & Ilar (McKinsey & company). (1998). Starting up. Achieving success with professional business planning. McKinsey & Company, Inc. The Netherlands, Amstel 344, 1017 AS Amsterdam.
- Clark (2012). Business Model You: A One-Page Method for Reinventing Your Career. Business Model You, LLC.

Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences.

Additional Information

Guest lecturers contributing to this course include*:

- Olaf Larsen, Yakult
- Bart van Wezenbeek, V.O. Patents & Trademarks
- Maurice Mooijman, RVO NL Octrooicentrum
- Marga Janse, InnovatiefLerenLeren
- Robert Al, TU/e Innovation Lab
- Loyens Loeff Belastingadviseurs
- Bart Bergstein, Forbion Capital partners
- IQVIA
- Hans Slobbe, TalentMark

Clinical Development and Clinical Trials

Course Code	AM_1180
Credits	3
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A.M. van der Geest

Examiner	dr. A.M. van der Geest
Teaching Staff	dr. A.M. van der Geest
Teaching method(s)	Study Group, Lecture

Course Objective

- To gain knowledge and insight into the function clinical trials in today's healthcare system.
- To gain knowledge and insight into the design of clinical trials.
- To gain knowledge and insight into the conduct of clinical trials, including the applying rules and regulations (including ICH-GCP)
- To gain knowledge and insight into and critically reflect on the roles, tasks and responsibilities of the stakeholders involved in clinical trials
- To gain insight into challenges in clinical development as well as in strategies to deal with these challenges
- To learn where and how to look up rules and regulations

Course Content

In today’s healthcare system, clinical trials have gained the status of golden standard to test the safety and efficacy of newly developed drugs. For new drugs to enter the market, clinical trials must be passed and as a consequence, clinical trial outcomes have major effects on our healthcare system. While our healthcare system currently is under pressure to remain affordable and available to all, at the same time, clinical trial regulations are increasingly tightened and the prominence of clinical trials in our healthcare system is being criticized. For that matter, it is of great importance to learn about and reflect on the role of clinical trials in today’s healthcare system.

The Clinical Development & Clinical Trials course will elaborate on the function, design and conduct of clinical trials, as well as the relevant stakeholders involved. The course consists of a theoretical part and an important practical part (e.g. gaining knowledge on clinical trial regulations). Classes include for example: ‘Life Cycle of a Clinical Trial’, ‘Clinical Trial Methodology’, ‘ICH-GCP Principles’, ‘The Ethics Committee’, ‘Safety Considerations in Clinical Trials’, ‘Quality Control & Quality Assurance’, ‘Compliance, Misconduct & Fraud’.

The gained knowledge and skills will be evaluated by means of a two individual assignments, one group assignment, and an exam at the end of the course.

Additional Information Teaching Methods

Lectures: +-35 h
Self study: +- 40 h

Method of Assessment

Assessment;
Two individual assignments (each 20%), one group assignment (20%), and an exam (40%) at the end of the course.
All separate assignments must be sufficient (>5,5).

Literature

- Ray, S., Fitzpatrick, S., Golubic, R. & Fisher, S. (2016). Oxford Handbook of Clinical and Healthcare Research. Oxford University Press,Oxford, UK.
- (Additional reading will be provided via Canvas and will serve as background reading for the lectures).

Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life Sciences (MPA), Societal differentiation of the Health, Life & Natural Sciences.

Additional Information

Guest lecturers, organizations/ companies:
• Eric Klaver, Trial&Eric.com/ IQVIA
* Guest lecture: SEUSS + working in the CT field.

Colloquium & Literature Thesis Bioanalytical Chemistry

Course Code	XM_0030

Credits	12

Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in bioanalytical chemistry, usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

mDDS-Bioanalytics

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Bioanalytical Chemistry (C, E, S)

Course Code	XM_0016
Credits	6
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in bioanalytical chemistry, usually in direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

see the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Only for students following the Education variant of mDDS-Bioanalytics

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Colloquium & Literature Thesis Environmental Bioanalysis

Course Code	XM_0031
Credits	12
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in environmental bioanalysis, usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and

assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

mDDS-Bioanalytics, mDDS-Molecular Toxicology (research variant only)

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Environmental Bioanalysis (C, E, S)

Course Code	XM_0017
Credits	6
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an

efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific
literature about a given topic in environmental bioanalysis, usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Only for students following the Education variant of the mDDS-Bioanalytics and mDDS-Molecular Toxicology specialization

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Medicinal Chemistry

Course Code	XM_0032
Credits	12
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in (synthetic) medicinal chemistry, drug design, receptor biochemistry or molecular pharmacology (or a combination), usually a topic of direct interest to the research group.

The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Report (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Literature

A thesis guide can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

mDDS-Synthesis of Drugs, mDDS-Molecular Pharmacology, mDDS-Computer-Aided Drug Design

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Colloquium & Literature Thesis Medicinal Chemistry (C, E, S)

Course Code	XM_0018
Credits	6
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten

Teaching Staff
Teaching method(s)

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in (synthetic) medicinal chemistry, drug design, receptor biochemistry or molecular pharmacology (or a combination), usually a topic of direct interest to the research group The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis guide can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Only for students from following the Education variant of the following tracks of the master Drug Discovery Sciences: Synthesis of Drugs, Molecular Pharmacology, Computer-Aided DrugDesign

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Molecular & Computational Toxicology

Course Code	XM_0033
Credits	12
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in molecular-, cellular toxicology or "biomolecular simulation & modeling" (or a combination), usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Research variant of mDDS-Molecular Toxicology, mDDS-Computer-Aided Drug Design (depending on the project)

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Molecular & Computational Toxicology (C, E, S)

Course Code	XM_0019
Credits	6
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows.
The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in molecular-, cellular toxicology, or biomolecular simulation & modeling (or a combination), usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher)

must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Only for students following the Education variant of the following tracks of the Master Drug Discovery Sciences: Molecular Toxicology, Computer-aided Drug Design

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Organic Chemistry

Course Code	XM_0034
Credits	12
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in synthetic organic and peptide chemistry or “Structural chemical biology” (or a combination), usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Report (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Literature

A thesis guide can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Research variant of mDDS-Synthesis of Drugs or mDDS-Molecular Pharmacology (depending on the subject)

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Colloquium & Literature Thesis Organic Chemistry (C, E, S)

Course Code	XM_0020
Credits	6
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in synthetic organic and peptide chemistry or “Structural chemical biology" (or a combination), usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Course Content

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific

objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Only for students following the Education variant of the mDDS-Synthesis of Drugs specialization

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Radiopharmaceutical Chemistry

Course Code	XM_0035
Credits	12
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given

scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in radiopharmaceutical chemistry (which may include molecular pharmacology, organic chemistry, bioanalytical chemistry), usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences".

Additional Information Target Audience

Research variant of mDDS-Synthesis of Drugs, mDDS-Molecular Pharmacology
(depending on the subject)

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Radiopharmaceutical Chemistry (C, E, S)

Course Code	XM_0021

Credits	6
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in radiopharmaceutical chemistry which may include a combination of molecular pharmacology, organic chemistry and bioanalytical chemistry, usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Only for students following the Education variant of the following master DDS specializations: Synthesis of Drugs, Molecular Pharmacology (depending on the topic)

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the

internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Theoretical Chemistry

Course Code	XM_0036
Credits	12
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in theoretical biochemistry, usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics.

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Entry Requirements

See the TER

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Research variant of mDDS-Computer-Aided Drug Design

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Colloquium & Literature Thesis Theoretical Chemistry (C, E, S)

Course Code	XM_0022
Credits	6
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The learning outcomes of a literature thesis are as follows. The student:

- is able to efficiently retrieve and critically assess relevant and recent in-depth information about a given scientific topic (research question), that deepens the knowledge gained in the master programme
- is able to conduct a literature review in an independent fashion
- is able to logically organize, describe and discuss the information in a thesis based on specific scientific objectives/questions associated with the literature review
- is able to draw conclusions on the perspectives and relevance of the subject
- is able to present and discuss the main findings in a colloquium
- is able to write a thesis report in proper academic English meeting all quality requirements as defined in the assessment criteria

Course Content

Completion of an academic master degree does not only imply practical experience and knowledge from a scientific specialization, but also that one is able to deal with substantial amounts of scientific information in an efficient way and distill this into the main points. During the literature thesis, the student will collect recent in-depth scientific literature about a given topic in theoretical biochemistry, usually a topic of direct interest to the research group. The literature information is described and discussed in a coherent form in a thesis, which is also presented orally during a colloquium.

The content of the literature thesis will depend on the involved thesis supervisor and the available topics

Additional Information Teaching Methods

Self-study, contact hours with supervisor, oral presentation, discussion

Method of Assessment

The literature project needs to be registered in the online thesis management system, where progress and

assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Thesis (37.5%), oral presentation (25%) and execution of the literature research (37.5%).

Literature

A thesis manual can be found on the Canvas community "Master Drug Discovery Sciences"

Additional Information Target Audience

Only for students following the Education variant of the following DDS specialization: Computer-Aided Drug Design

Custom Course Registration

Available positions are limited, so contact potential VU supervisors well in advance. See the Canvas community “Master Drug Discovery Sciences” for more information.

Before the start, a project proposal needs to be approved by the VU supervisor via the online thesis management system.

Consult the “Manual Literature Thesis and Colloquium” on Canvas for more information on the rules and regulations regarding literature projects. If, after reading the manual, you still have questions, please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Communication, Organization and Management

Course Code	AM_470572
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C.A.C.M. Pittens MSc
Examiner	dr. C.A.C.M. Pittens MSc
Teaching Staff	dr. S.S. Rai, dr. R.M.H. Peters, dr. C.A.C.M. Pittens MSc
Teaching method(s)	Study Group, Lecture

Course Objective

After completion of this course, you will be able to:

- identify and discuss knowledge of, and theories on organisational structures, organisational behavior, organisational change, and change management;
- describe and conceptualise organisations as embedded in a wider system, and interpret these from a transdisciplinary perspective;
- obtain a deeper understanding of communication from the perspective of sharing and influencing results, and to learn how to deal with conflicts;
- identify and discuss theories on leadership and interpersonal behaviour;
- make an informed choice between statistical techniques commonly used in health and life sciences, applying these to conduct analyses while using statistical software (e.g., SPSS) and interpreting the results of these techniques (e.g., t-tests, chi-square tests) to inform decision making;
- practise analytical and advisory skills.

Course Content

Organisations need to respond and adapt to internal and external challenges. It is increasingly recognized that addressing these challenges is not an ‘isolated process’. Organisations (need to) collaborate with multiple stakeholders in these processes of (continuous) change. In-depth understanding of the structure of an organisation and the wider system it operates in is therefore important. Different forms of knowledge can be used to gain this understanding. It requires insights gained from the analyses of qualitative and quantitative data collected within organisations and their wider context. This course offers insights into these complexities, as well as practical

(quantitative) methods to address these.

In interactive lectures, several (guest)speakers will discuss macro-aspects of organisational behavior, including designing organisations and managing them, thereby touching on the managerial skills and the strategic thinking required. In addition, concepts related to motivation, managing interpersonal behaviour, leadership, communication and developing and changing organisations will be explained. Furthermore, lecturers will discuss concepts of systems thinking, and the value of adopting multiple perspectives in analysing organisations and processes of organisational change. The (guest) speakers will illuminate concepts derived from the literature by relating them to their own experiences with informing and managing organisational change in practice. Furthermore, the role of quantitative data in managerial decision making will be explored (the role of qualitative data is covered in other courses). To that end, you will explore and discuss different statistical techniques (e.g., t-tests, chi-square tests) commonly used in health and life sciences. You will conduct analyses with these techniques in statistical software (e.g., SPSS), interpret the results and translate these to decision making processes, working towards an advisory report for a (fictive or real-world) organisation's management.

In an assignment that runs through the course, practical cases of healthcare companies will be analysed and discussed, resulting in advisory reports for management. With the other students you discuss your experiences and a coach helps you relate the experiences to theory.

Additional Information Teaching Methods

Lectures: approximately 22 hours
Response lectures: 4 hours
Training workshops: 16 hours (attendance is mandatory)
Self-study and writing project assignment: remaining hours.

Method of Assessment

Written exam (60%) and assignment (40%). Grades of both components must be sufficient.

Literature

To be announced on Canvas

Additional Information Target Audience

Compulsory course within the Master programme Management, Policy Analysis and Entrepreneurship for the Health and Life Sciences (MPA) and the Societal differentiation of Health, Life and Natural Sciences Masters programmes

Additional Information

Attendance to training workshops / working groups is indispensable

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course. It is not possible to enrol for workgroups otherwise unless announced via Canvas.

Computer-Aided Drug Design and Virtual Screening

Course Code	X_432673
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. I.J.P. de Esch
Examiner	prof. dr. I.J.P. de Esch
Teaching Staff	prof. dr. I.J.P. de Esch
Teaching method(s)	Computer lab, Lecture

Course Objective

Providing theoretical background on computer-aided drug design and virtual screening, and giving an overview of

recent developments, applications and trends.

Course Content

Introduction into most important concepts of computer-aided drug discovery and design.

- Protein homology modeling: sequence alignment methods, modeling constraints, protein-ligand interaction model refinement and validation.
- Chemoinformatics and chemogenomics and their application in drug and drug target identification: annotated ligand and protein databases, similarity searches, molecular fingerprints, machine learning, QSAR, focused library design, molecular field analysis, sequence- and structure-based comparison of binding sites.
- Structure-based virtual screening and design: molecular alignment, pharmacophore modeling, molecular docking and scoring, post-processing filters, protein-ligand interaction fingerprints, de novo design.

Students will learn to recognize the strengths and challenges of different computer-aided drug design approaches and will learn how in silico methods can be complemented with experimental studies in concrete ligand discovery and design projects.

Additional Information Teaching Methods

Lectures, case study sessions, and self-study.

Method of Assessment

Written (or oral) examination (60%) and case study (report: 20%, presentation: 15%; participation in case study sessions: 5%)

Literature

Literature that will be made available via Canvas.

Computer- Assisted Drug Design (Mason (Ed.) (references to relevant paragraphs from Mason will be included in lecture handouts and will be available as "E-book" via UBVU).

Additional Information Target Audience

mDDS-CMCT, mDDS-DD&S, mDDS-DDSA, mDDS-DDTF, mDDS-C-var, mDDS-E-var, mDDS-M-var, mCh

Recommended background knowledge

The course Computational Design & Synthesis of Drugs is recommended.

Concepts in Chemical Biology

Course Code	XM_0080
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. Drienovska
Examiner	dr. I. Drienovska
Teaching Staff	dr. S.A.K. Jongkees, dr. I. Drienovska, dr. V.U. Chukhutsina
Teaching method(s)	Written partial exam, Lecture

Course Objective

Course Objective

The ability to critically discuss published data will be trained. After the course, students are expected to know:

- the hallmarks of cancer and know the underlying principles
- core cancer signaling pathways
- inhibition strategies of signaling pathways – e.g. use of small molecules, stabilized peptides
- chemical reactions for the selective covalent modification of proteins

- bio-orthogonal and ligand directed reactions
- classical protein engineering (natural: directed evolution, disulfide…)
- Biology of the coronavirus life cycle
- SARS antiviral strategies
- (non-)natural protein modifications for improved functions
- Mechanisms of carbohydrate-active enzymes
- Chemical tools and approaches for the study of glycans

Course Content

Apart from a general introduction, four main topics are discussed, introducing different techniques central to Chemical Biology research:

- Virus Biology and Inhibition
- Chemical Glycobiology
- Targeting Hallmarks of Cancer
- Protein Engineering

Additional Information Teaching Methods

1. Lectures: Lectures will be given in English, and their content as well as the associated literature are preparation for the assignments.
2. Case Study and Assignment: The students are provided with one or more research papers. Alone or in groups of two, questions regarding these case studies are addressed in an assignment.
3. Guided Peer-to-Peer Discussions: In seminar sessions, Peer-to-Peer exchange for each topic will take place. Within these, the students are expected to exchange questions and help each other in answering these. Students are expected to participate actively. Discussions should be held in English and also a joined by one of the lecturers.

Method of Assessment

Students that have handed in all four assignments can participate in the final exam. The course grade will only be based on the grade of the exam.

Additional Information Target Audience

Recommended background knowledge: basic physico-chemical knowledge: e.g. Gibbs free energy, forces (coulomb, H-bonds, van der Waals), - basic chemical knowledge: e.g. electronegativity, bond character, acids & bases, organic synthesis - basic biochemical knowledge: e.g. general characteristics of a bacterial, mammalian and plant cell, phospholipids & biological membrane formation and function, nucleotides, DNA, RNA, double helix formation, gene regulation, amino acids and peptide bond formation, protein primary, secondary and tertiary structure, transcription, translation, post-translational modifications, signal transduction, principle of enzymatic reactions.

Additional Information

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130553

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Data Management in Drug Discovery

Course Code	XM_0117
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	T.S. Çobanoğlu MSc
Examiner	T.S. Çobanoğlu MSc
Teaching Staff	prof. dr. I.J.P. de Esch

Teaching method(s)	Seminar, Lecture

Course Objective

To understand the importance of data in drug discovery and to learn how to analyze and manage data generated and used in pharmaceutical sciences. This course will teach students how to work with different types of experimental data, curate them, retrieve information from databases, and apply practical tools for (advanced) data analysis and (basic) machine learning to solve real-world problems in drug discovery.

Course Content

In this course, you will gain an overview of diverse experimental data types and learn how to handle them in the context of drug discovery. You will explore publicly accessible databases such as ChEMBLdb, PDB, PubChem, UniProt, and DrugBank, to retrieve and analyze relevant information. You will learn about exploratory data analysis techniques (EDA) and learn how to appropriately pre-process, transform, and normalize data based on insights from EDA. The concept of statistical hypothesis testing will be introduced, with discussions on how to select the most suitable tests for different experimental datasets. Additionally, you will gain an introductory understanding of cheminformatics and machine learning models, and the course will emphasize data management best practices (implementing FAIR principles (Findability, Accessibility, Interoperability, Reusability), establishing a DMP (data management plan), to ensure effective organization, sharing and archiving of experimental data.

Throughout the course, you will work with curated datasets from various domains, including chemical datasets (structural and bioactivity data), transcriptomics, and mass spectrometry (MS)-based datasets (proteomics, metabolomics). We will demonstrate data curation and analysis methods using open-access software such as KNIME and scripting-based tools. This hands-on experience will provide an understanding of data science and AI in drug discovery as necessary in the current data era.

Additional Information Teaching Methods

Selected chapters from Open-Access books (on Introductory Biostatistics and/or Molecular Data Analysis), lectures with notes provided, practical work using tutorials, self-study, and presentations and reporting. Throughout the course, you will be using computers (bring your own device) and a variety of software packages.

Method of Assessment

Part of the course is project-based and the assignments are graded.

Additional Information Target Audience

The course is designed for MSc students in DDS and for students that are interested to develop and advance their skills in data science in the fields of drug discovery and/or molecular life sciences are highly welcome as well. Note that students should be comfortable in and have affection for using a variety of computer softwares, e.g. MS-Excel; see also below for the expected background.

Recommended background knowledge

Basic understanding of pharmaceutical sciences, biochemistry and/or molecular biology is expected as well as having followed a basic course in statistics (e.g. https://studiegids.vu.nl/nl/vakken/2024-2025/XB_43006#/ or similar). Computer programming skills are not required but it is important that the students are already experienced computer users and comfortable using a variety of computer softwares, e.g., MS-Excel. It must be emphasized that this specialized course will not be able to teach students these necessary basic computer skills.

Didactics 1

Course Code	O_MLDIDAC_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts

Teaching Staff	prof. dr. M. Meeter, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, drs. M.A. Huijskes MA, dr. H.B. Westbroek, dr. C.D.P. van Oeveren, drs. A.J.C. Monquil, dr. O.P.M. van Buuren, drs. Y.G. Meindersma, C.L. Geraedts, dr. B. de Vries, drs. R. Cornelisse MSc, M. Beukering, drs. E.D. van Noort, drs. H.R. Goudsmit, drs. M.M. Bout, prof. dr. ir. F.F.J. Hermans, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, drs. J.K.W. Riksen, drs. N.H. Ypenburg
Teaching method(s)	Seminar, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio’s vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio’s werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactics 1

Course Code	O_MLDIDAC_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, drs. J.K.W. Riksen, dr. O.P.M. van Buuren, drs. E.D. van Noort, drs. Y.G. Meindersma, dr. S.M. Calor, dr. I. Pauw, drs. A.J.C. Monquil, M. Beukering, drs. N.H. Ypenburg, dr. C.D.P. van Oeveren, dr. S. Donszelmann, drs. M.M. Bout, drs. H.R. Goudsmit, dr. H.B. Westbroek, C.L. Geraedts, N. el Abdellaoui MSc, drs. R. Cornelisse MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het handelen als leraar in simpele en overzichtelijke situaties (bijvoorbeeld op het niveau van één les met aandacht voor de klas als groep).

Course Content

De cursus Didactiek 1 is onderdeel van de eerste fase (fase I) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 1 (stage). Tijdens Didactiek 1 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

De cursus is geordend rondom zogeheten kernpraktijken die fundamenteel zijn voor het beroep van leraar. Bij Didactiek 1 staan de volgende kernpraktijken centraal: (1) Contact maken, (2) De les starten, (3) Krediet opbouwen en uitgeven, (4) De les voorbereiden, (5) Orde houden, en (6) Uitleggen, instructie geven en de les afsluiten. De reikwijdte van het didactisch denken en handelen is in deze eerste fase meestal nog beperkt tot één les. De ervaring leert dat deze zes kernpraktijken voor de meeste lio's vrijwel direct aan de orde komen in het eerste deel van de praktijkstage en daardoor een goede basis vormen.

Het fysieke onderwijs van Didactiek 1 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen.

Het onderdeel AD in de cursus Didactiek 1 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Interactieve hoorcolleges, werkcolleges en zelfstudiemodules.

Method of Assessment

Didactiek 1 wordt afgesloten met een (geschreven) startproef, bestaande uit een lesontwerp, een verantwoording op basis van theorie, en een uitwerking van eigen leerdoelen.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld via Canvas. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Didactiek 2 Semester 1

Course Code	O_MLDIDAC2
Credits	9
Period	P2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	C.L. Geraedts, prof. dr. M. Meeter, drs. M.M. Bout, drs. A.J.C. Monquil, drs. E.D. van Noort, drs. R. Cornelisse MSc, dr. H.B. Westbroek, dr. S. Donszelmann, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, N. el Abdellaoui MSc, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, drs. L.J. van Well-van Grootheest, drs. H.R. Goudsmit, drs. Y.G. Meindersma, M. Beukering, drs. M.A. Huijskes MA, drs. N.H. Ypenburg, dr. I. Pauw, dr. S.M. Calor
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Didactiek 2 is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Didactiek 2 staan de volgende kernpraktijken centraal: (1) Zien wie er in je klas zit, (2) Leerprocessen bevorderen, (3) Leerprocessen zichtbaar maken, (4) Leiding geven, (5) Een veilig leerklimaat creëren, en (6) Leerprocessen toetsen. Ten opzichte van de cursus Didactiek 1 wordt de focus verlegd van de (individuele) les naar het leerproces van de leerling. De reikwijdte van het didactisch denken en handelen wordt daarmee ook groter: er wordt een begin gemaakt met het ontwerpen en uitvoeren van reeksen van lessen.

Het fysieke onderwijs van Didactiek 2 concentreert zich op de maandag (de instituutsdag). De dag begint met een peergroupsessie (PG) waarin lio's werken aan professionele vaardigheden (reflecteren op het eigen leerproces, kritisch onderbouwen van de eigen praktijk, etc.). Vervolgens komen bij AD in periode 2 genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. In periode 3 staan, na de peergroupsessies in de ochtend, een aantal grote onderwijsthema's centraal.

Het onderdeel AD in de cursus Didactiek 2 is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

De volgende twee boeken zijn verplicht voor het onderdeel algemene didactiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd.
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast zijn wij enthousiast over de volgende boeken:

- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.
- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Deze laatste twee boeken zijn niet verplicht. Tijdens de eerste colleges zullen we kort iets over deze boeken vertellen, en is er gelegenheid om ze even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om (één van) deze boeken aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

NB: Voor studenten die vóór 1 september 2023 gestart zijn geldt in studiejaar 2023/2024 het curriculum van 2022/2023. Dit betekent dat het vak Didactiek 2 een studielast van 6 EC (ipv 9EC) heeft (en ze zich voor die variant intekenen). Studenten die de colleges van Didactiek 2 (6 EC) nog niet (geheel) gevolgd hebben sluiten aan bij colleges van het nieuwe vak Didactiek 2 (9 EC). Het is tot en met augustus 2024 nog mogelijk om Didactiek 2 (6EC) uit het oude curriculum af te ronden en het examen onder de eindtermen van het oude curriculum af te leggen.

Didactiek 2 Semester 2

Course Code	O_MLDIDAC2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. Y.G. Meindersma, drs. E.D. van Noort, prof. dr. ir. F.F.J. Hermans, drs. N.H. Ypenburg, dr. O.P.M. van Buuren, dr. H.B. Westbroek, dr. I. Pauw, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, N. el Abdellaoui MSc, dr. B. de Vries, drs. A.J.C. Monquil, M. Beukering, C.L. Geraedts, drs. R. Cornelisse MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, drs. M.A. Huijskes MA, D.S. Hoefakker-Laanstra, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. M.M. Bout, prof. dr. M. Meeter
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de basale pedagogische, algemeen-didactische en vakdidactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van (korte) lessenreeksen.

Course Content

De cursus Didactiek 2 is onderdeel van de tweede fase (fase II) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 2 (stage). Tijdens Didactiek 2 worden de onderdelen peergroup (PG), algemene didactiek en pedagogiek (AD) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

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Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Didactiek 2 wordt afgesloten met een (geschreven) basisproef, bestaande uit een ontwerp voor een reeks van drie lessen, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik.

Entry Requirements

Voorwaardelijk voor afronding van Didactiek 2: een voldoende beoordeling van Didactiek 1.

Literature

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- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.

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Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie

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Drug Target Biochemistry and Signaling

Course Code	XM_0111
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M.J. Smit
Examiner	prof. dr. M.J. Smit
Teaching Staff	prof. dr. M.J. Smit, dr. M.H. Siderius
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Deepening of the knowledge of the biochemical aspects of drug targets and their role in cellular signaling in health and disease, with the aim to understand their importance in modern pharmacology.

Course Content

This course is designed for students with a background in the field of pharmaceutical and life sciences (or have followed Principles of Pharmaceutical Sciences) and an interest in the biological and pharmacological aspects of drug discovery. The course will focus on biochemical properties, including structures, of the main families of drug targets and their involvement in cellular signaling networks in health and disease (e.g. CNS-related disorders, cancer). Different strategies to identify, validate and modulate drug targets (e.g. ‘omics’ approaches, siRNA, CRISPR/Cas9, biologicals) will be addressed. New concepts in the architecture of signaling networks and mechanisms of signal transfer will be discussed.

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Written examination and assignments

Literature

Primary literature and handouts.
'Cell signaling', second edition. Authors: Wendell Lim and Bruce Mayer ISBN: 978-0-367-27937-0

Additional Information Target Audience

1DDS, 1BMS, MSc Bioinformatics and Systems Biology with Life Science background/BSc

Recommended background knowledge

BSc Pharmaceutical Sciences or equivalent or Principles of Pharmaceutical Sciences or BSc in Life Sciences

Epidemiology

Course Code	AM_1179
Credits	3
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.S. Rai
Examiner	dr. S.S. Rai
Teaching Staff	L.L. Ruijsch, K.E. Mujica Gastelbondo MSc, L.L. Kaersenhout MSc
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

- To be able to describe the key characteristics, strengths and weaknesses of traditional epidemiological study designs and select an appropriate study design for a given research question and context;
- To be able to understand, calculate, and apply measures of occurrence and association;
- To be able to understand and assess possible bias and effect modification;
- To be able to understand and apply principles of accuracy in epidemiology;
- To gain an understanding of the principles of screening and calculate related measures;
- To be able to critically interpret and evaluate epidemiological studies (choice of study design, analyses, results, possible biases, and practical ideas for improvement)

Course Content

This is a basic epidemiology course and consists of a theoretical, contextual, and practical component. The theoretical component will focus on concepts and methodology used in epidemiological studies (e.g. study-designs, measures of occurrence and association, screening, bias, etc.). The contextual component will focus on past and current epidemiological developments, for instance, the start of the HIV/AIDS pandemic. Lastly, the practical component will focus on application of all your new skills through workgroups and journal club (practical).

Note: *This course does not include biostatistics. This course will only focus on basic epidemiological concepts, indicators/measures and related calculations, and critical appraisal and evaluation of epidemiological studies.*

Additional Information Teaching Methods

- Lectures (14 hours)
- Work groups (12 hours)
- Journal club (8 hours)
- Assignment (8 hours)
- Self-study (remaining time)

Method of Assessment

- Exam (100%)
- Assignment (insufficient/ sufficient)

Both elements need to be sufficient.

Literature

Available on Canvas

Additional Information Target Audience

This course is solely intended for students without a background in epidemiology(i.e. students who attended and completed another bachelor or master course in public-health or social sciences related methodology and applied biostatistics, epidemiology and biostatistics, or similar, are strongly discouraged from enrolling in this course).

Additional Information

For more information contact Dr. Sarju Sing Rai (s.s.rai@vu.nl)

Custom Course Registration

n/a

Ethics in Life Sciences

Course Code	AM_470707
Credits	3
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	P.R. Collard MA MSc.
Examiner	P.R. Collard MA MSc.
Teaching Staff	dr. C. Boersma MA MSc., P.R. Collard MA MSc.
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

This course:

- Provides students with knowledge of central concepts and theories in applied philosophy and professional ethics;
- Provides students with a toolbox of ethical instruments to help analyze moral problems related to (one's own) research in the life sciences and beyond.

Students who have taken and successfully finished this course:

- Are able to reflect on ethical issues related to one’s own life science specialization;
- Are capable of impartial and constructive deliberation concerning ethical issues related to one’s own life science specialization;
- Can work in teams on (hypothetical) scientific or technological developments, taking into account multiple values (amongst which safety);
- Have acquired the necessary skills to handle ethical issues in an accountable manner, as a professional academic beyond one's own inclinations and prejudices;
- Have shown a respectful and accountable attitude in dealing with group dynamics during the seminar groups and group project.

Course Content

Researchers in life sciences help generate knowledge that builds the future of our society. Therefore, professional academics should be accountable for their decisions, experimental designs and presentation of results. In this course, the principles of justification will be illustrated with cases of technology ethics, medical ethics and care ethics. An ethical dialogue on animal research is simulated through a role-play exercise. Finally, as a group training project, an ethical deliberation on a developed product that is safe by its design is prepared and executed in confrontation with another team.

Additional Information Teaching Methods

Ethics in the Life Sciences is a part-time course of four weeks (3 ECTS).

The total study time is 80 hours.

The different elements have the following study time:

- Lectures: 13 hours
- Seminar groups: 15 hours
- Group assignment: 20 hours
- Exam: 2 hour
- Group project presentation and deliberation: 3 hours
- Self study: 27 hours

Please note that attendance to the seminars is compulsory. Attendance to the lectures is highly recommended. In our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the seminars, and to pass the exam.

Method of Assessment

- Intellectual participation in the seminars and group project (10%)
- Multiple-choice exam (50%)
- Execution of the group project, the presentation and deliberation (40%)

All three elements have to be passed.

Entry Requirements

Bsc Biology, Biomedical Sciences, Psychology with profile Biological Psychology or Neuropsychology

Literature

Available on Canvas

Additional Information Target Audience

Compulsory course in all former FALW Master programmes, except Health Sciences and Neuro Sciences

Additional Information

Lectures, seminars, the exam and the final assessment are in English.

Fundamentals of Bioinformatics

Course Code	X_405052
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. H. Mouhib, dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. D. Molenaar
Teaching method(s)	Seminar, Computer lab, Written partial exam, Lecture

Course Objective

Are you interested in Bioinformatics? Do you want to find out how biology can make an exciting application domain? Or do you want to learn what more you could do with your data, and with less effort? Enter here to start!

Fundamentals of Bioinformatics (FoB) is the starting course of the Bioinformatics master. It aims to give you a broad overview of important topics relevant to the field, with a focus on current open problems. You will be made aware of these open problems during practical sessions that will cover active research in a major Bioinformatics problem (mutation impact prediction). Based on your background, you will be assigned to separate classes where you will be working to fill gaps in your background knowledge in programming and/or biology.

After completing the course successfully:

- You can explain the fundamental principles of cell biology and the genetic information flow
- You can write short code to solve simple problems and complete skeleton scripts for biological applications

- You can read algebra and calculus equations
- You can become aware of gaps in your own background knowledge.
- You can address gaps in your background knowledge (programming, mathematics and molecular biology) and skills
- You are aware of the major issues when developing and benchmarking new methodology in bioinformatics research
- You can present and communicate a joined group projects carried out with students of diverse backgrounds (Interdisciplinarity) in a respectful way
- You can apply and understand fundamental methods in bioinformatics, e.g. sequence alignment, impact prediction, and to understand and use fundamental concepts in bioinformatics, e.g. homology, evolutionary conservation, and gene function

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

Topics:

- Evolution, genomics, sequence alignment, Blast/PSI-Blast, FAIR data, benchmarking, machine learning, genome assembly, mutation impact prediction, protein stucture analysis, substitution matrices, scientific writing

Project:

- Group project to benchmark different mutation impact prediction approaches. The groups will be composed to include different BSc backgrounds.

Background knowledge (1 out of 3, see below):

- Molecular cell biology
- Programming and computational thinking in Python
- Calculus and linear algebra

Additional Information Teaching Methods

- 8 lectures (two-hour lecture in the morning, two days per week) Exercises during/in between lectures
- 12 computer practicals for the group project - compulsory attendance (online or on campus) (two hour sessions following the morning lectures, two days per week), supervised.
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Conversion classes for background knowledge:

- Molecular cell biology (7 x double lectures)
- Programming and computational thinking in Python (10 computer practicals)

Depending on your background, you will be assigned 1 of the 2 conversion classes. Students that are already well versed in both topics (molecular biology and programming) may be assigned to the following class, which is organised as part of the Introduction to Systems Biology:

- Calculus and linear algebra (10 maths classes)

Method of Assessment

- Project work [~40%] :
- Project report and/or presentation
- Digital exam to assess [~60%]:
- Lecture topics
- Concept learned during the project
- Concepts from conversion classes

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Further assessment and grading details will be posted on Canvas (resits and compensation rules)

Entry Requirements

Bachelor in any science discipline (including medicine) or strong programming background, and an interest in biological problems.

Literature

- course material (slides, scientific papers) on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics Garland Science 2008 ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will help you to set this up.

Recommended background knowledge

Programming in Python, Molecular Biology, some basics of calculus and linear algebra

Inclusieve Digitale Didactiek

Course Code	O_MIDD
Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S.M. Calor
Examiner	dr. S.M. Calor
Teaching Staff	dr. S.M. Calor
Teaching method(s)	Seminar, Lecture

Course Objective

De doordachte afstemming van complementaire face-to-face en online activiteiten moet duidelijk tot uiting komen in het ontwerp dat studenten leveren voor hun (nieuwe of bestaande cursus). Hiervoor gebruiken zij de VU Amsterdam Blended Learning (her) ontwerpsjabloon. Een belangrijk onderdeel hiervan betreft de motivatie die studenten gebruiken. Daarnaast lichten ze hun keuze voor online en face-to-face activiteiten toe en de volgorde hiervan, waarbij ze ook aandacht besteden aan constructive alignment; leerdoelen, leeractiviteiten en (formatieve en summatieve) testactiviteiten zijn op elkaar afgestemd. Inclusiviteit is een van de kernwaarden van de VU. Studenten gebruiken de VU Amsterdam Inclusiviteitskijkkader om de mate van inclusiviteit van het ontwerp in kaart te brengen. Ten slotte heeft de introductie van Generatieve Artificial Intelligence (AI) een enorm potentieel voor het onderwijs. Studenten gebruiken een VU Amsterdam doordachte didactiek bij de integratie van Generatieve AI in hun ontwerp.

Course Content

De coronaperiode heeft de inzet van digitale onderwijsvormen zoals online, blended en hybride onderwijs in een stroomversnelling gebracht. Naar aanleiding van het rapport Ruimte voor Onderwijs tijdens Corona en de motie Kwint is de wens ontstaan om meer inzicht te krijgen in de manier waarop digitale onderwijsvormen bijdragen aan het inhalen van coronavertragingen en het verbeteren van het welzijn van studenten. Daarnaast heeft de introductie van generatieve kunstmatige intelligentie (AI) een enorm potentieel voor het onderwijs. In het keuzevak “Inclusieve Digitale Didactiek” wordt aandacht besteed aan blended learning, inclusieve digitale didactiek en de integratie van generatieve AI in het onderwijs. Recente onderzoeksresultaten uit VU onderzoek worden in het keuzevak verwerkt.

De studenten van vandaag zijn opgegroeid in een wereld waar internet altijd aanwezig was, evenals mobiele telefoons en instant messaging. Ondersteunende leertechnologieën bieden een nieuwe manier om universitair

onderwijs te ontwerpen, ontwikkelen en verzorgen en vormen een aanvulling op traditionele onderwijs- en leermethoden (Orton-Johnson, 2009). Als gevolg hiervan passen universiteiten en hogescholen hun financiering aan deze veelbelovende en nieuwe technologieën aan. De mogelijkheden voor communicatie, interactie en samenwerking worden uitgebreid, waardoor het hoger onderwijs naar het kantelpunt van een belangrijke transformatie wordt geduwd, waarvoor nieuwe vaardigheden van zowel studenten als docenten nodig zijn. Leertechnologieën kunnen ongekende onderwijsmogelijkheden bieden. Het vereist echter vakmanschap om effectieve blended learning, inclusieve lessen waarin generatieve AI in is geïntegreerd, te ontwerpen en ontwikkelen. Leraren moeten kennis hebben van inhoud, didactiek en technologie, evenals het vermogen om deze kennis zodanig toe te passen dat technologie het leren verbetert.

Additional Information Teaching Methods

Dit vak is opgebouwd uit een aantal colleges. Tussentijds bestuderen studenten relevante wetenschappelijke literatuur en werken zij aan hun onderwijs ontwerp voor een vak dat zij zelf geven. Tussendoor is er gelegenheid gebruik te maken van individuele coaching. Studenten presenteren hun tussentijds ontwerp aan elkaar en ontvangen van iedereen feedback. Deze peer feedback kan gebruikt worden ter voltooiing van het onderwijsontwerp. Tijdens de presentatie besteden studenten niet alleen aandacht aan hun ontwerp, zij lichten tevens hun analyse toe alsmede hoe het blended onderwijs, inclusief onderwijs waarin generatieve AI is geïntegreerd, te ontwikkelen, te implementeren en te evalueren. Zowel presentatie als ontwerp worden beoordeeld, resulterend in één eindcijfer.

Method of Assessment

De cijfers van studenten voor deze cursus worden berekend op basis van het blended learning ontwerp en het inclusief ontwerp met daarin generatieve AI geïntegreerd en hun presentatie hiervan.

Entry Requirements

Didactiek 1, Praktijk 1 (of vergelijkbaar)

Literature

Wordt bekend gemaakt via Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Custom Course Registration

Alleen studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Major Research Project Bioanalytical Chemistry

Course Code	XM_0037
Credits	42
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of bioanalytical chemistry in one of the following settings:

- University or Academic hospital
- Research Institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

A bioanalytical chemistry research project will deal with theoretical and practical understanding of issues involved in the design, execution, analysis and interpretation of complex bioanalytical studies. The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to state-of-the-art bioanalytical techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination Regulations (TER) and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of mDDS-Bioanalytics

Additional Information

The standard length of a major internship is 42 EC (7 months). Extension up to 60 EC (10 months) can be incorporated as part of the optional part of the master program. Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences".

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Bioanalytical Chemistry (C, E, S)

Course Code	XM_0023
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of bioanalytical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

A bioanalytical chemistry research project will deal with theoretical and practical understanding of issues involved in the design, execution, analysis and interpretation of complex bioanalytical studies. The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary bioanalytical techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

E variant of mDDS-Bioanalytics

Additional Information

The standard length of a major internship is 24 EC (4 months) and can be extended to 36 EC (6 months). Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time consuming and should therefore be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Environmental Bioanalysis

Course Code	XM_0038
Credits	42
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of environmental bioanalytical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

An environmental bioanalysis research project will deal with theoretical and practical understanding of issues involved in the design, execution, analysis and interpretation of complex bioanalytical experiments in the context of environmental toxicology and chemistry (e.g. detection and quantification of drugs/toxins/metabolites in environmental and human matrices such as surface and drinking water, blood, breast milk, amniotic fluid, etc).

The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary bioanalytical techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g.

labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of mDDS-Bioanalytics, mDDS-Molecular Toxicology

Additional Information

The standard length of a major internship is 42 EC (7 months). Extension up to 60 EC (10 months) can be incorporated as part of the optional part of the master program. Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Environmental Bioanalysis (C, E, S)

Course Code	XM_0024
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of environmental bioanalytical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

An environmental bioanalysis research project will deal with theoretical and practical understanding of issues involved in the design, execution, analysis and interpretation of complex bioanalytical experiments in the context of environmental toxicology and chemistry (e.g. detection and quantification of drugs/toxins/metabolites in environmental and human matrices such as surface and drinking water, blood, breast milk, amniotic fluid, etc).

The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary bioanalytical techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination regulations (TER) and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Only for students following the Education variant of mDDS-Bioanalytics or mDDS-Molecular Toxicology

Additional Information

The standard length of this major internship is 24 EC (4 months) and can be extended to 36 EC (6 months). Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an

external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Major Research Project Medicinal Chemistry

Course Code	XM_0039
Credits	42
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

To obtain experience in doing scientific research within the fields of molecular pharmacology/receptor biochemistry, computational drug design or organic synthesis in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

In medicinal chemistry research, molecular pharmacology, biochemistry, computational chemistry and organic synthesis are important for designing and preparing small organic ligands as e.g. protein modulators as well as to interrogate (receptor) proteins and/or signaling networks in relation to health and disease. During the traineeship, the student actively participates in a research project as exerted by the host group/institute and consists of one of the following sub-disciplines:

1. Molecular pharmacology
2. Receptor biochemistry
3. Organic synthesis
4. (computational) Drug design

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary molecular pharmacological, receptor biochemistry, synthetic or computational design techniques.

Contact potential VU supervisors from the group medicinal chemistry for more information on internal internship opportunities (see Canvas community "Master Drug Discovery Sciences"

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of mDDS-Molecular Pharmacology, mDDS-Synthesis of Drugs, mDDS-Computer-Aided Drug Design

Additional Information

The standard length of a major internship is 42 EC (7 months). Extension up to 60 EC (10 months) can be incorporated as part of the optional part of the master program. Make sure you discuss your personal education plan with your master coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Medicinal Chemistry (C, E, S)

Course Code	XM_0025
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

To obtain experience in doing scientific research within the fields of molecular pharmacology/receptor biochemistry, computational drug design or organic synthesis in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

In medicinal chemistry research, molecular pharmacology, biochemistry, computational chemistry and organic synthesis are important for designing and preparing small organic ligands as e.g. protein modulators as well as to interrogate (receptor) proteins and/or signaling networks in relation to health and disease. During the traineeship, the student actively participates in a research project as exerted by the host group/institute and consists of one of the following sub-disciplines:

1. Molecular pharmacology
2. Receptor biochemistry
3. Organic synthesis
4. (computational) Drug design

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary molecular pharmacological, receptor biochemistry, synthetic or computational design techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination regulation (TER) and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Only for students following the Education variant of mDDS-Molecular Pharmacology, mDDS-Synthesis of Drugs, mDDS-Computer-Aided Drug Design

Additional Information

The standard length of this major internship is 24 EC (4 months) and can be extended to 36 EC (6 months). Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Major Research Project Molecular & Computational Toxicology

Course Code	XM_0040
Credits	42
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of molecular and computational toxicology, in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the available projects and timing of the internship, but roughly deals with one of the following areas of research:

- Biomolecular Simulation & Modeling (dr. Daan Geerke): computational methods are developed to characterize the interaction between compounds with different proteins and to predict their mode of binding and action. In close collaboration with the experimental colleagues within the division, a special focus is possible on predicting unwanted binding to drug-safety related proteins and/or the formation of metabolites by drug-metabolizing or biocatalytically active enzymes. The binding affinity and -selectivity of compounds and metabolites are calculated using efficient and accurate computational methods, and research on and development of improved biomolecular models or force fields is possible as well.

- *In vitro* cell toxicology (Prof. dr. Paul Jennings): Utilization of cultured human cells, including primary cells, renal cell lines, hepatic cell lines and induced human pluripotent stem cells for investigating mechanisms of toxicity. Assays employed include, cytotoxicity assays, stress assays, differentiation markers, mitochondrial injury (Seahorse bioanalyser), barrier function, impedance (xCELLigence), fluorescent reporter assays, transcriptomics and biokinetics.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental or computational techniques, and data analysis and interpretation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments and/or simulation, documentation (e.g. labjournal, adherence to protocols, appropriate meta tagging and documentation of workflows), data analysis, reporting on results in group meetings, a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of the following tracks of the master Drug Discovery Sciences Molecular Toxicology, Computer-Aided Drug Design (computational projects)

Additional Information

The standard length of a major internship is 42 EC (7 months). Extension up to 60 EC (10 months in total) can be incorporated as part of the optional part of the master program. Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Molecular & Computational Toxicology (C, E, S)

Course Code	XM_0026
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of molecular and computational toxicology, in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the available projects and timing of the internship, but roughly deals with one of the following areas of research:

- Biomolecular Simulation & Modeling (dr. Daan Geerke): computational methods are developed to characterize the interaction between compounds with different proteins and to predict their mode of binding and action. In close collaboration with the experimental colleagues within the division, a special focus is possible on predicting unwanted binding to drug-safety related proteins and/or the formation of metabolites by drug-metabolizing or biocatalytically active enzymes. The binding affinity and -selectivity of compounds and metabolites are calculated using efficient and accurate computational methods, and research on and development of improved biomolecular models or force fields is possible as well.
- In vitro cell toxicology (Prof. dr. Paul Jennings): Utilisation of cultured human cells, including primary cells, renal cell lines, hepatic cell lines and induced human pluripotent stem cells for investigating mechanisms of toxicity. Assays employed include, cytotoxicity assays, stress assays, differentiation markers, mitochondrial injury (Seahorse bioanalyser), barrier function, impedance (xCELLigence), fluorescent reporter assays, transcriptomics and biokinetics.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental or computational techniques, and data analysis and interpretation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments and/or simulation, documentation (e.g. lab journal, adherence to protocols, appropriate meta tagging and documentation of workflows), data analysis, reporting on results in group meetings, a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination regulation (TER) and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Only for students following the Education variant of the following specializations of the master Drug Discovery Sciences: Molecular Toxicology, Computer-Aided Drug Design (computational projects).

Additional Information

The standard length of this major internship is 24 EC (4 months) and can be extended to 36 EC (6 months). Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Organic Chemistry

Course Code	XM_0041
Credits	42
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of “Synthetic Organic and peptide chemistry “or “Structural chemical biology” in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the associated supervisor(s), available projects on host/location and the timing of the internship, but roughly deals with the following:

Development of efficient synthetic methods for making biologically relevant molecules. The synthetic toolbox that enables us to make small molecules for drug discovery is never ‘complete’, yet continuously expanding. New and efficient synthetic methods/routes are developed for making ‘drug-like’ compounds. Starting from solid theoretical knowledge of organic chemistry, both new multi-component- and cascade-reactions (in which in one step multiple covalent bonds are formed) as well as new conversions using bio-, organic or transition metal catalysis are applied. With such reactions, multiple outcomes are often possible. Using advance spectroscopic techniques, all products are characterized, and subsequently, the reaction can be optimized to one specific product. The scope of the reaction is determined after establishing the ideal conditions for the newly developed reaction and molecular diversity is the primary driver in choosing the products. Finally, to underline the utility of the newly developed reaction, interesting applications are actively pursued.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. lab journal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of mDDS-Synthesis of Drugs, mDDS-Molecular Pharmacology (depending on the subject)

Additional Information

The standard length of a major internship is 42 EC (7 months). Extension up to 60 EC (10 months) can be incorporated as part of the optional part of the master program. Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Organic Chemistry (C, E, S)

Course Code	XM_0027
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to

critically interpret and reflect on the experimental results. The project will be conducted within the field of “Synthetic Organic and peptide chemistry “or “Structural chemical biology” in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the associated supervisor(s), available projects on host/location and the timing of the internship, but deals with organic small-molecule chemistry. This includes the development of efficient synthetic methods for making biologically relevant molecules. The synthetic toolbox that enables us to make small molecules for drug discovery is never ‘complete’, yet continuously expanding. New and efficient synthetic methods/routes are developed for making ‘drug-like’ compounds.

Starting from solid theoretical knowledge of organic chemistry, both new multi-component- and cascade-reactions (in which in one step multiple covalent bonds are formed) as well as new conversions using bio-, organic or transition metal catalysis are applied. With such reactions, multiple outcomes are often possible. Using advance spectroscopic techniques, all products are characterized, and subsequently, the reaction can be optimized to one specific product. The scope of the reaction is determined after establishing the ideal conditions for the newly developed reaction and molecular diversity is the primary driver in choosing the products. Finally, to underline the utility of the newly developed reaction, interesting applications are actively pursued.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. lab journal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination regulation (TER) and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Only for students following the Education variant of the following specializations: Molecular Pharmacology (for Structural chemical biology projects), Synthesis of Drugs (organic/peptide chemistry projects)

Additional Information

The standard length of this major internship is 24 EC (4 months) and can be extended to 36 EC (6 months). Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the

application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Major Research Project Radiopharmaceutical Chemistry

Course Code	XM_0042
Credits	42
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of Radiopharmaceutical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The focus lies on the development and biological evaluation of novel radiopharmaceuticals to study and profile a disease in vitro and in vivo. Continuously new targets and drugs are being discovered and developed. However, drug development is a complex process, which is time consuming and often during late stage (phase II or III) clinical studies a drug is not further developed.
Within the radionuclide center we have the unique opportunity to synthesize PET tracers and study these radioactively labeled compounds for their in vitro and in vivo behavior as well as the disease itself. Drug lead compounds or analogues can be turned into PET tracers by labeling the molecules of interest with carbon-11 and fluorine-18 for small molecules or zirconium-89 for proteins and antibodies. Prior to any radiolabeling for PET tracer synthesis, the precursor molecules need to be synthesized, which requires state of art organic medicinal chemistry. All disease areas can be studies using PET imaging, but mainly comprise neurological or oncological disorders. Ultimately, the use of radiopharmaceutical chemistry will allow a more rapid drug development process since we can better understand both disease and the in vivo behavior of drug.

During the traineeship, the student actively participates in a research project as exerted by the host group/institute and entails one or more of the following sub-disciplines:

1. Organic synthesis
2. Radiochemistry
3. Analytical chemistry
4. Molecular biology
5. Molecular pharmacology

The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary specialist techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

In addition, an internal radiation safety course will have to be completed, including an exam, before the student is allowed to do radiochemistry experiments. This training is a part of the internship.

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of mDDS-Synthesis of Drugs, DDS-Molecular Pharmacology (Depending on the project)

Additional Information

The standard length of a major internship is 42 EC (7 months). Extension up to 60 EC (10 months) can be incorporated as part of the optional part of the master program. Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Radiopharmaceutical Chemistry (C, E, S)

Course Code	XM_0028
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of Radiopharmaceutical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The focus lies on the development and biological evaluation of novel radiopharmaceuticals to study and profile a disease in vitro and in vivo. Continuously new targets and drugs are being discovered and developed. However, drug development is a complex process, which is time consuming and often during late stage (phase II or III) clinical studies a drug is not further developed.
Within the radionuclide center we have the unique opportunity to synthesize PET tracers and study these radioactively labeled compounds for their in vitro and in vivo behavior as well as the disease itself. Drug lead compounds or analogues can be turned into PET tracers by labeling the molecules of interest with carbon-11 and fluorine-18 for small molecules or zirconium-89 for proteins and antibodies. Prior to any radiolabeling for PET tracer synthesis, the precursor molecules need to be synthesized, which requires state of art organic medicinal chemistry. All disease areas can be studies using PET imaging, but mainly comprise neurological or oncological disorders. Ultimately, the use of radiopharmaceutical chemistry will allow a more rapid drug development process since we can better understand both disease and the in vivo behavior of drug.

During the traineeship, the student actively participates in a research project as exerted by the host group/institute and entails one or more of the following sub-disciplines:

1. Organic synthesis
2. Radiochemistry
3. Analytical chemistry
4. Molecular biology
5. Molecular pharmacology

The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary specialist techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination regulation (TER) and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

In addition, an internal radiation safety course will have to be completed, including an exam, before the student is allowed to do radiochemistry experiments. This training is a part of the internship.

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Only for students following the Education variant of mDDS-Synthesis of Drugs or mDDS-Molecular Pharmacology (depending on the project)

Additional Information

The standard length of this major internship is 24 EC (4 months) and can be extended to 36 EC (6 months). Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Major Research Project Theoretical Chemistry

Course Code	XM_0043
Credits	42
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of computational experiments and to critically interpret and reflect on these results. The project will be conducted within the field of

theoretical (bio)chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the associated supervisor(s), available projects at the host/location and the timing of the internship, but generally deals with quantum chemical methods, including Density Functional Theory (DFT). DFT calculations are used to gain insights in chemical interactions, like reaction mechanisms that lead to the synthesis of drugs, or the interactions of (drug) molecules with DNA and/or proteins.

Attention will be paid to setting up new research/experiments, using state-of-the-art computational techniques, analyzing computational results and keeping adequate documentation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation, analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Reseach variant of mDDS-Computer-Aided Drug Design

Additional Information

The standard length of a major internship is 42 EC (7 months). Extension up to 60 EC (10 months) can be incorporated as part of the optional part of the master program. Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process

related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Major Research Project Theoretical Chemistry (C, E, S)

Course Code	XM_0029
Credits	24
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of computational experiments and to critically interpret and reflect on these results. The project will be conducted within the field of theoretical (bio)chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the associated supervisor(s), available projects at the host/location and the timing of the internship, but generally deals with quantum chemical methods, including Density Functional Theory (DFT). DFT calculations are used to gain insights in chemical interactions, like reaction mechanisms that lead to the synthesis of drugs, or the interactions of (drug) molecules with DNA and/or proteins.

Attention will be paid to setting up new research/experiments, using state-of-the-art computational techniques, analyzing computational results and keeping adequate documentation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation, analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Teaching and Examination regulation (TER) and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Only for students following the Education variant of mDDS-Computer-aided Drug Design

Additional Information

The standard length of this major internship is 24 EC (4 months) and can be extended to 36 EC (6 months). Make sure you discuss your personal education plan with your master (track) coordinator well in advance. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Explanation Canvas

See the Canvas community "Drug Discovery Sciences"

Managing Science and Technology in Society

Course Code	AM_470586
Credits	6
Period	P1
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.P. van Oudheusden
Examiner	dr. M.P. van Oudheusden
Teaching Staff	dr. M.P. van Oudheusden
Teaching method(s)	Study Group, Lecture

Course Objective

In this course, students:

- acquire knowledge and understanding of philosophical and social science theories about the mutual shaping of science and technology in society
- acquire knowledge and understanding of basic concepts, issues, and approaches in the field of science and technology studies
- acquire knowledge and understanding of ways in which technology can be democratized and made responsive to societal concerns and expectations
- acquire knowledge and understanding of interactive methods to direct and guide developments in science and technology
- learn to connect basic STS concepts to their own life worlds - and vice versa
- learn to communicate verbally and in scientific writing about their knowledge and understanding and to critically reflect on their own perspectives

Course Content

The ‘Managing Science and Technology in Society’ course offers an advanced introduction to the academic field of ‘Science Technology & Society Studies’ (Science and Technology Studies), as part of the second year of the master ‘Management,

Policy Analysis and Entrepreneurship for the Health and Life Sciences’.

As a MPA student you are trained to operate at the interface of your natural science discipline and society, thereby making a contribution to answering the complex problems arising in these areas. At the dawn of the 21st century, technology and science have an enormous potential for transforming life on earth. At the same time, the dimensions of our human culture shape the directions in which science and technology develop. The production of scientific knowledge and technological objects can solve some of our problems, while giving rise to new challenges. During this course you will study the interactions of science and technology with society, and the various ways in which they mutually shape one another. These interactions invoke a lot of questions. Should we embrace big data? How do new human reproductive technologies interfere with the way we deal with sexuality and social responsibilities? How can we innovate responsibly?

In this course you will be introduced to a conceptual framework to critically and reflexively assess these kinds of questions. The course aims at understanding the intertwinement of science, technology and society, and underscores the importance of a broad concern with these interactions, in order to shape collective futures that take into account societal concerns and expectations.

Additional Information Teaching Methods

'Managing Science and Technology in Society' is a fulltime course of eight weeks (6 ECTS). The course schedule is available on Canvas. The total study time is 168 hours. Tuition methods include lectures, work groups, a group project and self-study. The different elements have the following study time: lectures 22 hours; work groups 12 hours; group project 32 hours; self study (including mini-essays) 86 hours; examination (on-campus) 2.5 hours.

Method of Assessment

The examination consists of:

- Mini-essay 1 (10%)
- Mini-essay 2 (10%)
- Final essay (take-home essay exam) (50%)
- SCOB-project (30%)

Both the essay exam and the SCOB project need to be passed.

Literature

The reading material for this course can be found in Canvas.

Additional Information Target Audience

This is a compulsory course for second year students of the master Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences. The course is optional for other Master students in the health and life sciences.

Minor Research Project Bioanalytical Chemistry

Course Code	XM_0044
Credits	18
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of bioanalytical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

A bioanalytical chemistry research project will deal with theoretical and practical understanding of issues involved in the design, execution, analysis and interpretation of complex bioanalytical studies. The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary bioanalytical techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of all tracks within the master DDS

Additional Information

The length of a minor internship ranges from 18 to 36 EC (3-6 months) and can be (partly) incorporated within the optional part of the master program, although for traineeships abroad and in companies, a length of at least 30 EC is recommended. Consult the master coordinator of your track well in advance to discuss your personal education plan. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Minor Research Project Environmental Bioanalysis

Course Code	XM_0045
Credits	18
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of environmental bioanalytical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

An environmental bioanalysis research project will deal with theoretical and practical understanding of issues involved in the design, execution, analysis and interpretation of complex bioanalytical experiments in the context of environmental toxicology and chemistry (e.g. detection and quantification of drugs/toxins/metabolites in environmental and human matrices such as surface and drinking water, blood, breast milk, amniotic fluid, etc).

The specific content of the project depends on the associated supervisor(s), available projects on location and the timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary bioanalytical techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. In the case of questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of all tracks within the master DDS

Additional Information

The length of a minor internship ranges from 18 to 36 EC (3-6 months) and can be (partly) incorporated within the optional part of the master program, although for traineeships abroad and in companies a length of at least 30 EC is recommended. Consult the master coordinator of your track well in advance to discuss your personal education plan. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Minor Research Project Medicinal Chemistry

Course Code	XM_0046
Credits	18
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

To obtain experience in doing scientific research within the fields of molecular pharmacology/receptor biochemistry, computational drug design or organic synthesis in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

In medicinal chemistry research, molecular pharmacology, biochemistry, computational chemistry and organic synthesis are important for designing and preparing small organic ligands as e.g. protein modulators as well as to interrogate (receptor) proteins and/or signaling networks in relation to health and disease. During the traineeship, the student actively participates in a research project as exerted by the host group/institute and consists of one of the following sub-disciplines:

1. Molecular pharmacology
2. Receptor biochemistry
3. Organic synthesis
4. (computational) Drug design

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing

research as well as to contemporary molecular pharmacological, receptor biochemistry, synthetic or computational design techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of all tracks within the master DDS

Additional Information

The length of a minor internship ranges from 18 to 36 EC (3-6 months) and can be (partly) incorporated within the optional part of the master program, although for traineeships abroad and in companies a length of at least 30 EC is recommended. Consult the master coordinator of your track well in advance to discuss your personal education plan. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Minor Research Project Molecular & Computational Toxicology

Course Code	XM_0047
Credits	18
Period	Ac. Year (September)
Course Level	500

Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of molecular and computational toxicology, in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the available projects and timing of the internship, but roughly deals with one of the following areas of research:

1. Biomolecular Simulation & Modeling (dr. Daan Geerke): computational methods are developed to characterize the interaction between compounds with different proteins and to predict their mode of binding and action. In close collaboration with the experimental colleagues within the division, a special focus is possible on predicting unwanted binding to drug-safety related proteins and/or the formation of metabolites by drug-metabolizing or biocatalytically active enzymes. The binding affinity and -selectivity of compounds and metabolites are calculated using efficient and accurate computational methods, and research on and development of improved biomolecular models or force fields is possible as well.
2. In vitro cell toxicology (Prof. dr. Paul Jennings): Utilisation of cultured human cells, including primary cells, renal cell lines, hepatic cell lines and induced human pluripotent stem cells for investigating mechanisms of toxicity. Assays employed include, cytotoxicity assays, stress assays, differentiation markers, mitochondrial injury (Seahorse bioanalyser), barrier function, impedance (xCELLigence), fluorescent reporter assays, transcriptomics and biokinetics.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental or computational techniques, and data analysis and interpretation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments and/or simulation, documentation (e.g. labjournal, adherence to protocols, appropriate meta tagging and documentation of workflows), data analysis, reporting on results in group meetings, a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of all specializations of the master Drug Discovery & Safety

Additional Information

The length of a minor internship ranges from 18 to 36 EC (3-6 months) and can be (partly) incorporated within the optional part of the master program. For traineeships abroad and in companies a length of at least 30 EC is recommended. Consult the master coordinator of your track well in advance to discuss your personal education plan. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Minor Research Project Organic Chemistry

Course Code	XM_0048
Credits	18
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of “Synthetic Organic and peptide chemistry “or “Structural chemical biology” in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the associated supervisor(s), available projects on host/location and the timing of the internship, but roughly deals with one of the following areas of research:

- Organic small-molecule chemistry: development of efficient synthetic methods for making biologically relevant molecules. The synthetic toolbox that enables us to make small molecules for drug discovery is never ‘complete’, yet continuously expanding. New and efficient synthetic methods/routes are developed for making ‘drug-like’ compounds. Starting from solid theoretical knowledge of organic chemistry, both new multi-component- and cascade-reactions (in which in one step multiple covalent bonds are formed) as well as new conversions using bio-, organic or transition metal catalysis are applied. With such reactions, multiple outcomes are often possible. Using advance spectroscopic techniques, all products are characterized, and subsequently, the reaction can be optimized to one specific product. The scope of the reaction is determined after establishing the ideal conditions for the newly

developed reaction and molecular diversity is the primary driver in choosing the products. Finally, to underline the utility of the newly developed reaction, interesting applications are actively pursued.

- Organic and peptide chemistry: Ligands that selectively bind to biomolecular targets are a prerequisite for most strategies aiming at the elucidation or modulation of biological processes. The discovery of these ligands can be very challenging. Giving the huge amount of available structural data, known peptide binding epitopes provide a rich source of inspiration for novel ligands. Preorganization of such peptides into their bioactive conformation or the use of rigid peptidomimetic scaffolds have been applied to ensure binding affinity and bioactivity. Notably, there is a lack of approaches for the stabilization of irregular and complex peptide structures. In addition, the design of peptide-derived bioactive compounds is often hampered by their low cellular uptake and biostability. Addressing these challenges, the design of peptide-inspired ligands for an application in proximity-induced chemical reactions and as inhibitors of protein–protein interactions is pursued. The core techniques applied in corresponding projects include organic synthesis of non-natural amino acids, solid-phase peptide synthesis and/or the biophysical validation of ligand–receptor interactions (e.g. via fluorescence polarization assays).

- Structural chemical biology: The interaction networks of biomolecules such as nucleic acids or proteins are crucial for proper cell function. Once imbalanced, a variety of severe diseases can be the result. Here, we focus on key players of various cancers as well as their interaction partners. We will explore their protein-protein- as well as their protein-DNA complex formation and use small molecules as modifiers of these interactions. We will also determine their interaction mechanisms on atomic level and use this information to optimize the potency of the small molecule modulator. A subset of the following techniques may be used: sub-cloning genes of interest in suitable expression plasmids, heterologously expressing these in bacteria and purifying them via various chromatography techniques. The quality of the resulting proteins will be analyzed using SDS-PAGE and mass spectrometry. With state of the art crystallization techniques we will search for crystallizing conditions and optimize crystal formation. Subsequent X-ray analysis and structure determination will help to optimize small molecule modulators of protein-protein interactions.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. lab journal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of all tracks within the master DDS

Additional Information

The length of a minor internship ranges from 18 to 36 EC (3-6 months) and can be (partly) incorporated within the optional part of the master program, although for traineeships abroad and in companies a length of at least 30 EC is recommended. Consult the master coordinator of your track well in advance to discuss your personal education plan. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Minor Research Project Radiopharmaceutical Chemistry

Course Code	XM_0049
Credits	18
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of experiments and to critically interpret and reflect on the experimental results. The project will be conducted within the field of Radiopharmaceutical chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The focus lies on the development and biological evaluation of novel radiopharmaceuticals to study and profile a disease in vitro and in vivo. Continuously new targets and drugs are being discovered and developed. However, drug development is a complex process, which is time consuming and often during late stage (phase II or III) clinical studies a drug is not further developed.
Within the radionuclide center we have the unique opportunity to synthesize PET tracers and study these radioactively labeled compounds for their in vitro and in vivo behavior as well as the disease itself. Drug lead compounds or analogues can be turned into PET tracers by labeling the molecules of interest with carbon-11 and fluorine-18 for small molecules or zirconium-89 for proteins and antibodies. Prior to any radiolabeling for PET tracer synthesis, the precursor molecules need to be synthesized, which requires state of art organic medicinal chemistry. All disease areas can be studies using PET imaging, but mainly comprise neurological or oncological disorders. Ultimately, the use of radiopharmaceutical chemistry will allow a more rapid drug development process since we can better understand both disease and the in vivo behavior of drug.

During the traineeship, the student actively participates in a research project as exerted by the host group/institute and entails one or more of the following sub-disciplines:

1. Organic synthesis
2. Radiochemistry
3. Analytical chemistry
4. Molecular biology
5. Molecular pharmacology

The specific content of the project depends on the associated supervisor(s), available projects on location and the

timing of the internship.

Attention will be paid to setting up new research/experiments, using state-of-the-art experimental techniques, analyzing experimental results and keeping adequate documentation. In all, the student will get exposed to doing research as well as to contemporary specialist techniques.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation (e.g. labjournal), analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).
In addition, an internal radiation safety course will have to be completed, including an exam, before the student is allowed to do radiochemistry experiments. This training is a part of the internship.

Literature

Starting literature will be provided by the supervisor. The first 1-2 weeks of the traineeship will be spent on literature reading and preparation for the project.

Additional Information Target Audience

Research variant of all tracks within the master DDS

Additional Information

The length of a minor internship ranges from 18 to 36 EC (3-6 months) and can be (partly) incorporated within the optional part of the master program, although for traineeships abroad and in companies a length of at least 30 EC is recommended. Consult the master coordinator of your track well in advance to discuss your personal education plan. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Minor Research Project Theoretical Chemistry

Course Code	XM_0050

Credits	18
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	
Teaching method(s)	

Course Objective

The aim is to obtain experience in scientific research including the design and execution of computational experiments and to critically interpret and reflect on these results. The project will be conducted within the field of theoretical (bio)chemistry in one of the following settings:

- University or Academic hospital
- Research institute
- Company

The internship can be performed either within the Netherlands or abroad.

Course Content

The specific content of the project depends on the associated supervisor(s), available projects at the host/location and the timing of the internship, but generally deals with quantum chemical methods, including Density Functional Theory (DFT). DFT calculations are used to gain insights in chemical interactions, like reaction mechanisms that lead to the synthesis of drugs, or the interactions of (drug) molecules with DNA and/or proteins.

Attention will be paid to setting up new research/experiments, using state-of-the-art computational techniques, analyzing computational results and keeping adequate documentation.

Additional Information Teaching Methods

Literature review, formulating a research question, conducting experiments, keeping adequate documentation, analyzing data, reporting on results in progress/group meetings, in a written thesis and final presentation within the research group.

Method of Assessment

The internship needs to be registered in the online internship management system, where progress and assessment is recorded. The following components are assessed and a minimum passing grade (5.5 or higher) must be obtained for each component: Academic attitude (15%), execution (35%), oral presentation (15%) and written thesis (35%).

Entry Requirements

Entry requirements with respect to attained courses and number of credits apply. See the Education and Examination regulations and the internship manual Drug Discovery Sciences. For questions, consult the internship coordinator well in advance (internships_sf.beta@vu.nl).

Additional Information Target Audience

Research variant of all tracks within the master DDS

Additional Information

The length of a minor internship ranges from 18 to 36 EC (3-6 months) and can be (partly) incorporated within the optional part of the master program, although for traineeships abroad and in companies a length of at least 30 EC is recommended. Consult the master coordinator of your track well in advance to discuss your personal education plan. See the “Master Drug Discovery Sciences Internship manual” for more information (rules, regulations, tips, etc.) on the internship process. This manual can be found on the canvas community “Master Drug Discovery Sciences”.

Custom Course Registration

The number of internal (VU) internship positions is limited and there may be a selection procedure. Contact potential VU supervisors from the group for more information about internship opportunities. You can consult the canvas community “Master Drug Discovery Sciences” for more information about potential VU supervisors and the application procedure.

Be aware that a selection procedure applies for most external internship positions as well. Before the start of an external project, the UNL internship agreement has to be signed by all involved parties and the project proposal has to be approved by the VU supervisor via the online internship management system. Sometimes an additional Non Disclosure agreement (NDA) needs to be signed as well. This process can be time-consuming and should, therefore, be started several months in advance.

Consult the “Master Drug Discovery Sciences Internship manual” on Canvas for more information on the rules and regulations regarding internships. If, after reading the manual, you still have questions regarding the process related to internships, admission, approval, etc., please contact the internship coordinator via internships_sf.beta@vu.nl.

Molecular and Cellular Toxicology

Course Code	XM_0113
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. J.C. Vos
Examiner	dr. ir. J.C. Vos
Teaching Staff	dr. ir. J.C. Vos, dr. S. Moco, dr. A. Wilmes, prof. dr. P. Jennings
Teaching method(s)	Lecture

Course Objective

Students will gain knowledge on various ways cells can be compromised by exposure to exogenous chemicals, how they defend themselves against specific perturbations and the methodological aspects in determining and quantifying these processes.

Course Content

Exposure of cells to chemicals (including drugs and drug metabolites) can cause perturbations compromising the cells’ ability to function which may lead to cell death and tissue injury eventually compromising health. The course will introduce and expand on such concepts including cell injury processes, cell death, cellular defence mechanisms (e.g. p53 pathway, Nrf2 pathway, unfolded protein responses), mitochondrial toxicity and immunotoxicity.

The following aspects will be explored:

- types of adverse drug reactions,
- role of xenobiotic biotransformation (metabolism) transport in adverse drug reactions and nuclear receptors,
- reversible and irreversible interactions of toxic drugs with biological macromolecules,
- stress response pathways - cellular adaptation to exposure to chemicals and reactive oxygen species;
- cellular and molecular mechanisms leading to toxic effects,
- mitochondrial toxicity,
- immune-mediated toxicity,
- cellular in vitro models used in toxicity studies,
- cellular assays used to detect stress-responses and mitochondrial dysfunction including transcriptomics, metabolomics and proteomics.

Additional Information Teaching Methods

Lectures and self-study.

Method of Assessment

Written exam

Literature

Literature will be provided.

Additional Information Target Audience

mDDS, mBMS, mCHEM

Recommended background knowledge

Bachelor Pharmaceutical Sciences, Chemistry, Biomedical Sciences , Medical Natural, Sciences, Medical Biology or equivalent.

Molecular Modelling

Course Code	XMU_0059
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. C. Fonseca Guerra
Examiner	prof. dr. C. Fonseca Guerra
Teaching Staff	dr. T.A. Hamlin, prof. dr. C. Fonseca Guerra, prof. dr. F.M. Bickelhaupt
Teaching method(s)	Seminar, Other, Practical, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130572

Additional Information Teaching Methods

-

Method of Assessment

-

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Oncogenesis

Course Code	M_OONC03
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Medicine
Course Coordinator	prof. dr. R.H. Brakenhoff
Examiner	prof. dr. R.H. Brakenhoff
Teaching Staff	S. Biemans, dr. S.B. Baglio
Teaching method(s)	Seminar, Lecture

Course Objective

The aim of this course is to acquire state-of-the art knowledge of and deeper insights into the general concepts of oncogenesis and basic tumor biology. These include (but are not limited to) the different genetic alterations and classes of genes involved in tumor formation, their role in cellular signaling, DNA repair and cell cycle control, the (molecular) tools to study carcinogenesis including NextGen sequencing, CRISPR-Cas9 gene editing, model systems such as transgenic mouse models and high-throughput functional genomics.

In addition, hereditary cancer predisposition syndromes, exogenous factors involved in oncogenesis with emphasis on viruses will be studied. Finally, we will review markers of disease progression, genetic counselling, screening, as well as intervention options.

The students will be given an up-to-date overview of the current knowledge and opinions in this exciting field. Furthermore, the students will acquire deep insight into how to interpret literature on specific topics in oncogenesis and present their findings in a clear and concise manner.

Course Content

The course on Oncogenesis consists of three parts. The first part involves morning sessions with lectures given by experts on various topics of oncogenesis. The second part involves a literature (self-) study on a specific familial cancer syndrome followed by a group-presentation in the last week of the course. The third part involves discussion on one of the “hallmarks of cancer” for which students will summarize lectures with the aid of the handouts that will become available on CANVAS and additional literature. The latter two parts will be executed by groups of 4 students in a way that all students will be involved in a presentation.

Topics include:

1. Latest updates on the roles of cancer genes and cancer pathways
2. Modern cancer genomics and (epi)genetics including Next Gen sequencing
3. Mechanisms, models and molecular methods, including CRISPR-Cas9 genome editing
4. Viruses causing cancer
5. Clinical aspects of cancer and precancer biology
6. Cancer predisposition syndromes

Additional Information Teaching Methods

± 45 interactive lectures given by experts during morning sessions
± 8-10 hallmark discussions given by student groups
± 8-10 interactive literature presentations given by student groups

The course has approximately 56 contact hours with teachers and course coordinators.

Method of Assessment

The course will be concluded with a 2.5 hours examination (open questions) on the contents of the lectures. Handouts of the presentations, including those of the literature studies, will be available on CANVAS. There will be an examination training during the course. The hallmark discussions and presentations of the literature studies will be assessed as well.

The final mark of this course will be determined by the average of the scores for the examination (counting for 70%), the hallmark discussion (counting for 10%), and the literature presentation (counting for 20%). Students need to be present and participate actively during the course. Absence is only allowed with reason and needs to be reported on beforehand.

The (review) articles posted on CANVAS and ‘the Biology of Cancer’ (Robert A. Weinberg, second edition) are recommended as a reference work for deeper background knowledge and to facilitate studying the different subjects.

Literature

The course reader (available on CANVAS) contains (review) articles on the various topics.

Additional Information Target Audience

Only for first year Master Oncology students.

Note: Students from other programs will need to be approved by the master Oncology Admissions Board to be admitted to the course.

Additional Information

The students will have acquired insight in the latest developments in the field of oncogenesis, cancer genetics and technologies to investigate the molecular aspects of cancer. Furthermore, the students will be able to integrate knowledge and insight in the formation of opinions concerning research questions about the development of cancer, and learn how to review and present scientific work.

Coordinators:

Prof. dr. R.H. Brakenhoff
Dr. R.M.F. Wolthuis
Dr. S.R. Baglio

Custom Course Registration

All Master Oncology students are required to attend the compulsory courses of the programme. Master Oncology students must register for all courses using the registration module. All activities for which you have registered will be displayed in your personal timetable, including any changes to the timetable.
If you have not registered for a course, you will not be admitted to that course, you will not be assigned to a group, you will not be able to use CANVAS, you will not have a timetable, your grades will not be recorded, etc. In short, you will not be able to participate. Register via the registration module.

This course is open to VU students and external students from other master programmes.

Prerequisites: background knowledge of oncology and immunology. Participation is only possible if the number of students in the course does not exceed 40 and after approval by the master Oncology admission board. To apply, you must send your CV, letter of motivation and transcript of records to the programme coordinator by email: masteroncology.vu@amsterdamumc.nl

Students will be notified of their acceptance at the beginning of July.

Pedagogische en Didactische verdieping Semester 1

Course Code	O_MPEDIVER_F
Credits	9
Period	P1+2+3
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	dr. B. de Vries, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, drs. A.J.C. Monquil, dr. I. Pauw, dr. H.B. Westbroek, dr. O.P.M. van Buuren, drs. M.M. Bout, drs. J.K.W. Riksen, drs. E.D. van Noort, drs. Y.G. Meindersma, drs. N.H. Ypenburg, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. C.D.P. van Oeveren, dr. S.M. Calor, C.L. Geraedts, M. Beukering, drs. R. Cornelisse MSc, drs. L.J. van Well-van Grootheest, drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, dr. S. Donszelmann
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren,

(7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio’s ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio’s toegankelijk. Op deze manier kunnen lio’s gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio’s van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback.

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Pedagogische en Didactische Verdieping Semester 2

Course Code	O_MPEDIVER
Credits	9
Period	P4+5+6
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	C.L. Geraedts
Examiner	C.L. Geraedts
Teaching Staff	drs. M.M. Bout, drs. R. Cornelisse MSc, C.L. Geraedts, dr. H.B. Westbroek, M. Beukering, dr. O.P.M. van Buuren, D.S. Hoefakker-Laanstra, dr. B. de Vries, dr. C.D.P. van Oeveren, drs. L.J. van Well-van Grootheest, drs. E.D. van Noort, dr. S.M. Calor, J. Quartel MA, N. el Abdellaoui MSc, prof. dr. M. Meeter, dr. I. Pauw, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. S. Donszelmann, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. J.K.W. Riksen, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

Na afloop van dit vak beschikt de student over de verdiepende pedagogische en (vak)didactische kennis en vaardigheden die nodig zijn voor het zelfstandig ontwerpen en uitvoeren van doorlopende lessenreeksen, en voor het handelen als docent in complexe(re) situaties.

Course Content

De cursus Pedagogische en Didactische verdieping is onderdeel van de derde fase (fase III) van de Universitaire Lerarenopleidingen (ULO) van de VU Lerarenacademie waarin didactiek en praktijk samenkomen. Het vak loopt dan ook parallel aan de cursus Praktijk 3. Tijdens Pedagogische en Didactische verdieping worden de onderdelen algemene didactiek en pedagogiek (AD), peergroup (PG) en vakdidactiek (VD) in samenhang aangeboden. Hierbij wordt nadrukkelijk aangesloten bij de ontwikkelingsfase waarin de leraar-in-opleiding (lio) zich bevindt.

Pedagogische en Didactische verdieping is geordend rondom een aantal voor het beroep van leraar fundamentele kernpraktijken. Bij Pedagogische en Didactische verdieping staan de volgende kernpraktijken centraal: (1) Leerlijnen ontwerpen en evalueren, (2) De puber leren kennen, (3) Rekening houden met gedrags- en leerkenmerken, (4) Je onderwijs systematisch evalueren, (5) Omgaan met diversiteit in je klas, (6) Differentiëren, (7) Verantwoordelijkheid geven, (8) Mentor zijn, en (9) Je professionele ruimte benutten. De cursussen Didactiek 1 en 2 vormen samen het basisdeel van de ULO; de cursus Pedagogische en Didactische verdieping kan gezien worden als het verdiepingsdeel waarbij we nader ingaan op (het begeleiden van) individuele leerprocessen en het verzorgen van passend onderwijs, en het leren van leerlingen in een groepsdynamische setting. In Pedagogische en Didactische verdieping komen meer complexe kernpraktijken en bijbehorende thema's aan de orde. Het (vak)didactisch denken en handelen strekt zich uit over langere perioden, en er is aandacht voor het ontwerpen van leerlijnen. Ook wordt de lio nadrukkelijker uitgedaagd een eigen visie op onderwijs te geven en uit te dragen. Zo is de lesmethode niet langer leidend, maar wordt van lio’s in toenemende mate verwacht zelf invulling te geven aan de inhoud en didactiek van de lessen (waarbij zowel bestaand als eigen materiaal kan worden gebruikt). Tenslotte zullen de (vak)didactische overwegingen die ten grondslag liggen aan de eigen visie onderbouwd moeten worden met relevante literatuur en eigen praktijkervaringen.

Het fysieke onderwijs van Pedagogische en Didactische verdieping concentreert zich op de maandag (de

instituutsdag). Tijdens de (werk)colleges AD komen genoemde kernpraktijken expliciet aan de orde. In de middag komen bij VD verschillende belangrijke onderwerpen en werkwijzen aan bod die bij het leren van het eigen schoolvak van belang zijn. Tijdens deze sessies wordt zo veel mogelijk aangesloten bij de kernpraktijken. Alle inhoudscomponenten uit deze cursus worden tijdens de bijeenkomsten en in de verwerking verbonden met de lespraktijk van de lio. De lio en de werkplekbegeleider krijgen ook suggesties voor (observatie)opdrachten die kunnen bijdragen aan de ontwikkeling van de competenties die bij deze kernpraktijken horen. Tot slot zijn er peergroupbijeenkomsten o.a. gericht op intervisie en het schrijven van de meesterproef.

Het onderdeel AD in de cursus Pedagogische en Didactische verdieping is vormgegeven volgens het principe van blended learning. Tijdens het onderwijs op de maandag wordt voortgebouwd op een zelfstudiemodule die lio's ter voorbereiding online hebben doorlopen, en waarin één kernpraktijk centraal staat. Daarnaast is er nadrukkelijk ruimte voor differentiatie: alle ruim 20 kernpraktijken die samen de leerlijn AD in de gehele eerstegraads opleiding vormen, zijn in de vorm van deze zelfstudiemodules voor lio's toegankelijk. Op deze manier kunnen lio's gemakkelijk teruggrijpen naar een kernpraktijk die al eerder aan bod kwam, of vooruitwerken als de eigen lespraktijk daarom vraagt.

De colleges en workshops AD en PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Het werkcollege VD in de middag is onder begeleiding van de eigen vakdidacticus. Deze werkcolleges worden samen met lio's van hetzelfde (school)vak gevolgd.

Bij alle colleges wordt een actieve houding van de student gevraagd, zowel tijdens de bijeenkomsten als daarbuiten. Regelmatig worden verwerkingsopdrachten gegeven, waar individueel of in groepsverband aan wordt gewerkt. Deze opdrachten worden formatief geëvalueerd, onder andere door middel van (peer)feedback..

Additional Information Teaching Methods

Hoorcolleges, werkcolleges en zelfstudiemodules

Method of Assessment

Pedagogische en Didactische verdieping wordt afgesloten met een (geschreven) meesterproef, bestaande uit een ontwerp voor een lessenreeks van vier tot zes lessen, waarbij rekening wordt gehouden met een specifieke casusleerling, een verantwoording (op basis van visie, theorie, en praktijkervaringen), een videocompilatie, een evaluatie en een terugblik. Een belangrijk onderdeel is ook de systematische evaluatie waarin één aspect van het ontwerp en het effect op het leerproces van de leerling onder de loep wordt genomen.

Entry Requirements

Voorwaardelijk voor afronding van Pedagogische en Didactische verdieping: een voldoende beoordeling (of vrijstelling) van Didactiek 2.

Literature

De volgende drie boeken zijn verplicht voor het onderdeel algemene didactiek en pedagogiek:

- Blakemore, S. (2019). Inventing ourselves. The secret life of the teenage brain. London: Transworld Publishers Ltd
- Ebbens, S. & Ettekoven, S. (2020). Effectief leren. Groningen: Noordhoff Uitgevers, B.V.
- Horeweg, A. (2015). Gedragsproblemen in de klas in het voortgezet onderwijs. Amsterdam: Lannoo Campus.

Schaf deze boeken aan vóór de start van de opleiding. Twee opmerkingen daarbij: de ervaring leert dat je met een (tweedehands) exemplaar van een oudere editie van Effectief leren over het algemeen behoorlijk uit de voeten kunt. Ten tweede, het boek Inventing ourselves is ook in het Nederlands vertaald. Als je dat prettiger vindt mag je natuurlijk gebruik maken van de vertaalde editie.

Daarnaast is het volgende boek voor veel studenten een zinvolle bron:

- Teitler, P. (2017). Lessen in orde. Bussum: Coutinho.

Dit laatste boek is niet verplicht. Tijdens de eerste colleges zullen we kort iets over dit boek vertellen, en is er gelegenheid om het even door te bladeren, zodat je zelf te zijner tijd de keuze kunt maken om het boek al dan niet aan te schaffen.

Behalve van bovenstaande handboeken wordt veelvuldig gebruik gemaakt van wetenschappelijke literatuur. De desbetreffende artikelen worden tijdens de cursus ter beschikking gesteld. De literatuur die bij VD gebruikt wordt is afhankelijk van het schoolvak waarvoor wordt opgeleid.

Additional Information Target Audience

Deze cursus is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Peergroup 1 Semester 1

Course Code	O_MLPEERGR_1
Credits	0
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio’s volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef en de basisproef, en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor dit onderdeel ingeschreven te staan.

Peergroup 1 Semester 2

Course Code	O_MLPEERG_1F
Credits	0
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	N. el Abdellaoui MSc, drs. Y.G. Meindersma, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan twee leerlijnen: het onderzoekend leren kijken naar ervaringen en het leren voeren van een professionele dialoog. De bijeenkomsten maken onderdeel uit van de vakken Didactiek 1 en 2.

De student leert systematisch te reflecteren op de eigen onderwijspraktijk en hierover in gesprek te gaan. De student gebruikt hierbij feedback van derden, beeldmateriaal van de eigen praktijkuitvoering, kennis uit de school, opleiding en onderwijsonderzoek.

Subdoelen – de student kan:

- de stappen in de reflectiecirkel van Korthagen onderscheiden en doorlopen;
- leerpunten en sterke kanten van het eigen functioneren als docent benoemen;
- effectief feedback geven en ontvangen;
- tijdens de intervisie en reflectie opgedane inzichten omzetten in concrete leerpunten en organiseert de aanpak hiervan;
- verschillende typen onderwijsonderzoek onderscheiden en op kwaliteit en praktische bruikbaarheid voor zijn eigen praktijk beschouwen;
- gebruik maken van bronnen (onderzoeksliteratuur en vakbladliteratuur) voor de analyse van praktijksituaties / onderwijsontwikkelingen;
- gebruik maken van bronnen voor een onderbouwde ontwikkeling van de eigen visie en lespraktijk.

Course Content

In de peergroup staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

In werkgroepen op de maandagochtend (de instituutsdag) reflecteren studenten op hun handelen in de praktijk (stage/werkplek), leiden daaruit ontwikkelpunten af, formuleren acties en evalueren deze. Ze leren hierbij op een professionele wijze met elkaar in gesprek te gaan en ervaringen te delen.

Verschillende instrumenten en methodes worden gebruikt (logboek, reflectiecirkel, intervisie, videoreflectie, etc.) om de student in staat te stellen de complexiteit van de onderwijspraktijk te doorgronden en hiervan te leren. In de peergroups wordt literatuur bestudeerd over onderwijs en onderwijsonderzoek en er wordt onderzocht wat hiervan de relevantie is voor de eigen onderwijspraktijk.

De colleges PG worden verzorgd door een vast team van docenten (algemeen didactici, pedagogen, vakdidactici). Lio's volgen deze onderdelen met peers van verschillende schoolvakken. Bij de indeling wordt getracht rekening te houden met de voorkennis en ervaring van de verschillende studenten.

Additional Information Teaching Methods

Werkgroepen. Aanwezigheid bij de peergroup is verplicht.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de startproef (Didactiek 1) en de basisproef (Didactiek 2), en komen terug in de praktijk.

Literature

Literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Deze cursus is onderdeel van de educatieve minor, de educatieve module, de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 1 onderdeel uitmaakt van de vakken Didactiek 1 en 2 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 1

Course Code	O_MLPEERG_2F
Credits	0
Period	P1+2+3
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	drs. H.R. Goudsmit, N. el Abdellaoui MSc, drs. Y.G. Meindersma
Teaching method(s)	Study Group

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als 'professional' centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Aantal werkgroepen op maandag (instituutsdag).

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Peergroup 2 Semester 2

Course Code	O_MLPEERGR_2
Credits	0
Period	P4+5+6
Course Level	0
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. R. Cornelisse MSc, drs. H.R. Goudsmit
Teaching method(s)	Study Group, Lecture

Course Objective

De student werkt in de sessies Peergroup (PG) aan het systematisch reflecteren op de eigen onderwijspraktijk en gebruikt hierbij onder meer kennis uit de school en opleiding, feedback van derden en beeldmateriaal van de eigen praktijkuitvoering. Tijdens de beenkomsten wordt ingegaan op belangrijke zaken t.a.v. de meesterproef, zoals de casusleerling en videoreflectie. De bijeenkomsten maken onderdeel uit van het vak Didactiek 3.

Course Content

In een aantal peergroupsessies op maandag (instituutsdag) staat het handelen als ‘professional’ centraal. Studenten leren de regie te nemen over hun eigen leerproces. Ook ontwikkelen ze hun persoonlijke visie op onderwijs en een professionele identiteit, waarin ze de eisen die het beroep van docent aan ze stelt verbinden met eigen kennis, waarden en motieven.

Additional Information Teaching Methods

Werkgroepen.

Method of Assessment

Onderdelen uit de peergroup worden getoetst in de meesterproef (Didactiek 3).

Entry Requirements

Didactiek 1 en 2 (inclusief peergroup 1) afgerond (of vrijgesteld van).

Literature

Eventuele literatuur wordt nader bepaald en via de Peergroup Canvaspagina gedeeld.

Additional Information Target Audience

Peergroup 2 is onderdeel van de (eenjarige) Master LVHO en de (tweejarige) Master Educatie.

Custom Course Registration

Hoewel Peergroup 2 onderdeel uitmaakt van het vak Didactiek 3 dienen studenten apart voor het onderdeel peergroup ingeschreven te staan.

Physical Organic Chemistry

Course Code	X_435663
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. T.A. Hamlin
Examiner	prof. dr. F.M. Bickelhaupt
Teaching Staff	dr. T.A. Hamlin, prof. dr. C. Fonseca Guerra, dr. T. Hansen MSc, prof. dr. F.M. Bickelhaupt
Teaching method(s)	Seminar, Lecture

Course Objective

The student will be introduced to the fundamental principles of physical organic chemistry that dictate the reactivity of molecules.

The student will develop a deep understanding of how rates of reactions can be altered and the role of catalysts in accelerating reactions.

The student will be exposed to foundational concepts such as acid/base catalysis, carbocation/carbanion chemistry, chemical bonding, stereochemistry, isotope effects, reaction mechanisms, aromaticity, and various types of organic reactions like nucleophilic substitutions/eliminations and pericyclic reactions.

The student will be equipped with skills to explore a wide range of fields, spanning the range from fundamental organic reactions to supramolecular (biomolecular) interactions and recognition.

Course Content

This Physical Organic Chemistry course aims to cultivate critical thinking and problem-solving skills. Students will delve into the fundamental principles governing molecular reactivity, exploring topics such as catalyst-driven reaction rate modifications, acid/base catalysis, carbocation/carbanion chemistry, chemical bonding, stereochemistry, isotope effects, and reaction mechanisms. Through hands-on exercises, students will learn to formulate arrow-pushing mechanisms for experimental observations, assess existing theories, design validating experiments, and evaluate alternative explanations. The course equips students with the tools to navigate a wide range of fields, from foundational organic reactions to the complexities of supramolecular interactions and biomolecular recognition, fostering a deep understanding of the dynamic world of chemical reactions and molecular interactions.

Additional Information Teaching Methods

The course is comprised of lectures, self-study, and tutorial sessions.

Method of Assessment

Written examination.

Entry Requirements

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130522

Literature

BSc Chemistry

Additional Information Target Audience

Students interested in the topic of physical organic chemistry.

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Practice 1

Course Code	O_MLPRAK_1F
Credits	6
Period	P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. M.A. Huijskes MA, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, prof. dr. M. Meeter, N. el Abdellaoui MSc, drs. E.D. van Noort, M. Beukering, dr. H.B. Westbroek, C.L. Geraedts, dr. O.P.M. van Buuren, dr. S.M. Calor, drs. R. Cornelisse MSc, dr. S. Donszelmann, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, drs. Y.G. Meindersma, drs. M.M. Bout, dr. I. Pauw, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, drs. A.J.C. Monquil, dr. B. de Vries
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij

bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 1

Course Code	O_MLPRAK_1
Credits	6
Period	P1
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. C.D.P. van Oeveren, M. Beukering, drs. M.M. Bout, drs. A.J.C. Monquil, drs. N.H. Ypenburg, drs. H.R. Goudsmit, drs. J.K.W. Riksen, dr. H.B. Westbroek, drs. R. Cornelisse MSc, dr. S. Donszelmann, N. el Abdellaoui MSc, D.S. Hoefakker-Laanstra, dr. B. de Vries, prof. dr. ir. F.F.J. Hermans, dr. I. Pauw, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, dr. O.P.M. van Buuren, C.L. Geraedts, prof. dr. M. Meeter, drs. E.D. van Noort, dr. S.M. Calor, drs. M.A. Huijskes MA
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 1 doorloopt de lio het eerste deel van de praktijkstage onder begeleiding van een werkplekbegeleider (wpb). De stage voor Praktijk 1 omvat minimaal 48 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk gelden, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. Tijdens Praktijk 1 werkt de student o.a. toe naar het zelf geven van een hele les.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 1 voert de student tegen het eind van de eerste fase op de (opleidings-)school de opdracht ‘Presenteer jezelf’ uit. De student presenteert zichzelf als leraar in opleiding en voert daarbij bewijsmateriaal aan over zijn/haar functioneren in de praktijk gedurende de eerste fase. Vervolgens beoordeelt de werkplekbegeleider de student aan de hand van het Beoordelingsformulier Praktijk. Het functioneren van de student gedurende de stage én de inbreng tijdens de presentatie vormen input voor de beoordeling.

Additional Information Target Audience

Studenten met een baan (zij-instromers, onderwijstrainees etc) geven in dit stadium al zelfstandig les. Bij deze studenten ligt de nadruk bij de begeleiding vanuit de werkplekbegeleider op het niveau van didactisch handelen in de les.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2F
Credits	9
Period	P5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	dr. I. Pauw, prof. dr. M. Meeter, dr. H.B. Westbroek, M. Beukering, dr. C.D.P. van Oeveren, C.L. Geraedts, D.S. Hoefakker-Laanstra, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, drs. H.R. Goudsmit, dr. B. de Vries, dr. S.M. Calor, drs. A.J.C. Monquil, drs. J.K.W. Riksen, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. M.A. Huijskes MA, drs. E.D. van Noort, dr. O.P.M. van Buuren, dr. S. Donszelmann
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120

lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de module geldt:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

Voor de master geldt:

De werkplekbegeleider beoordeelt Praktijk 2 aan de hand van het Beoordelingsformulier Praktijk.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Additional Information

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 2

Course Code	O_MLPRAK_2
Credits	9
Period	P2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil

Teaching Staff	drs. A.J.C. Monquil, M. Beukering, drs. E.D. van Noort, dr. H.B. Westbroek, prof. dr. ir. F.F.J. Hermans, D.S. Hoefakker-Laanstra, C.L. Geraedts, dr. C.D.P. van Oeveren, drs. J.K.W. Riksen, dr. S. Donszelmann, drs. L.J. van Well-van Grootheest, drs. Y.G. Meindersma, drs. M.M. Bout, dr. B. de Vries, dr. I. Pauw, drs. H.R. Goudsmit, dr. S.M. Calor, dr. O.P.M. van Buuren, drs. R. Cornelisse MSc, N. el Abdellaoui MSc, prof. dr. M. Meeter, drs. M.A. Huijskes MA, drs. N.H. Ypenburg
Teaching method(s)	Study Group

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 2 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 2 van de stage omvat minimaal 72 klassencontacturen. Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. Binnen de in totaal 120 klassencontacturen die voor Praktijk 1 en Praktijk 2 gezamenlijk staan, worden ten minste 60 lesuren van 50 minuten door de student zelf verzorgd. De student werkt o.a. toe naar het zelf geven van een kleine lessenserie.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Voor studenten die de educatieve module volgen, ligt de nadruk bij Praktijk 1 en 2 in ieder geval verplicht op vmbo-tl en/of onderbouw havo/vwo (de doelgroep voor een beperkte tweedegraads bevoegdheid). Voor studenten die de gehele eenjarige master LVHO of tweejarige master Educatie volgen (en een eerstegraads bevoegdheid willen behalen) is er meer speling bij de inrichting van de stage. In hun geval geldt de eis dat binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, er minimaal 120 lesuren van 50 minuten door de student zelf worden verzorgd waarvan minimaal 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo).

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. De werkplekbegeleider wordt per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur met de wpb.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor een modulestudent geldt het volgende:

Voor de afronding van Praktijk 2 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’.

Voor een masterstudent geldt het volgende:

Praktijk 2 wordt beoordeeld door de werkplekbegeleider.

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 2. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 2 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 2: een voldoende beoordeling van Praktijk 1 en Didactiek 1.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3F
Credits	15
Period	P1+2+3
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. H.R. Goudsmit, dr. B. de Vries, D.S. Hoefakker-Laanstra, C.L. Geraedts, drs. R. Cornelisse MSc, drs. Y.G. Meindersma, R.B. van der Vos, drs. N.H. Ypenburg, M. Beukering, drs. A.J.C. Monquil, drs. E.D. van Noort, dr. S.M. Calor, drs. L.J. van Well-van Grootheest, dr. O.P.M. van Buuren, dr. I. Pauw, dr. C.D.P. van Oeveren, R.M.C. Canisius, drs. M.M. Bout, drs. J.K.W. Riksen, prof. dr. ir. F.F.J. Hermans, prof. dr. M. Meeter, dr. H.B. Westbroek, dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Deel 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op het instituut aan bod komen. Ook met onderwerpen die bij keuzevakken op het instituut ter sprake komen zal in de praktijk worden geëxperimenteerd. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Practice 3

Course Code	O_MLPRAK_3
Credits	15
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. A.J.C. Monquil
Examiner	drs. A.J.C. Monquil
Teaching Staff	drs. E.D. van Noort, drs. N.H. Ypenburg, prof. dr. ir. F.F.J. Hermans, drs. Y.G. Meindersma, dr. I. Pauw, dr. S. Donszelmann, D.S. Hoefakker-Laanstra, dr. S.M. Calor, dr. H.B. Westbroek, dr. O.P.M. van Buuren, M. Beukering, N. el Abdellaoui MSc, dr. B. de Vries, drs. M.M. Bout, drs. R. Cornelisse MSc, drs. A.J.C. Monquil, dr. C.D.P. van Oeveren, prof. dr. M. Meeter, drs. H.R. Goudsmit, drs. L.J. van Well-van Grootheest, C.L. Geraedts, drs. J.K.W. Riksen
Teaching method(s)	Lecture

Course Objective

Leservaring opdoen voor je eigen schoolvakspecialisatie in klassen onderbouw en bovenbouw vmbo-tl, havo en vwo.

Course Content

Tijdens Praktijk 3 ontwikkelen studenten zich als leraar-in-opleiding verder op pedagogisch, algemeen didactisch en vakdidactisch vlak. De kernpraktijken, die op het instituut en tijdens de stage aan bod komen, spelen daarbij een belangrijke rol. Praktijk 3 van de stage omvat minimaal 120 klassencontacturen (van 50 minuten). Tijdens de klassencontacturen observeert de student lessen, geeft de student zelf lessen/lesonderdelen, of onderneemt andere leerlinggerichte activiteiten. De student gaat meer en meer zelf(standig) lesgeven. De voorbereiding en evaluatie worden samen met de werkplekbegeleider gedaan. Ten minste 60 van de 120 lesuren worden door de student zelf verzorgd.

Het is de bedoeling dat de lio tijdens de stage een goed beeld krijgt van de verschillende jaarlagen en niveaus binnen het onderwijs. Binnen de in totaal 240 klassencontacturen die voor Praktijk 1, 2 en 3 gezamenlijk gelden, dienen minimaal 120 lesuren door de student zelf te worden verzorgd, waarvan 40 lesuren in het voorbereidend hoger onderwijs (bovenbouw havo/vwo). Het moet in ieder geval mogelijk zijn om de lessenreeks die ten tijde van Didactiek 3 wordt ontworpen (dat parallel loopt aan Praktijk 3) uit te voeren in een bovenbouwklas. Binnen Praktijk 3 zal daarom de bovenbouw veelal centraal staan.

Op school wordt de aandacht in ieder geval ook gericht op de kernpraktijken en vakdidactische onderwerpen die op

het instituut aan bod komen. De werkplekbegeleider wordt in ieder geval per mail en via de wpb-website op de hoogte gesteld van de kernpraktijken. De student bespreekt nieuwe inzichten opgedaan tijdens de instituutsdag en uit de literatuur ook met de wpb en neemt vragen, ervaringen en inzichten mee naar de instituutsdag.

Voor het in kaart brengen van de vorderingen en voor de beoordeling van lio's wordt gebruik gemaakt van praktijkbeoordelingsrubrics. Deze rubrics vormen ook een mooi formatief instrument voor wpb en student om ontwikkelpunten in kaart te brengen en te bespreken.

Additional Information Teaching Methods

Onder begeleiding van de werkplekbegeleider nemen de studenten steeds meer en actiever deel aan het lesgeven en werken in de school.

Method of Assessment

Voor de afronding van Praktijk 3 maakt de student in blok 2 of 3 een afspraak met de werkplekbegeleider en de schoolopleider voor een afrondend lesbezoek. De werkplekbegeleider en de schoolopleider beoordelen de les aan de hand van het Beoordelingsformulier Praktijk. De werkplekbegeleider en de schoolopleider komen samen tot een oordeel (= cijfervoorstel). Ieder heeft daarbij zijn eigen rol. De werkplekbegeleider betrekt bij het oordeel het complete traject dat de student doorlopen heeft en de lessen die hij gezien heeft; de schoolopleider baseert zijn oordeel op deze ‘examenles’. De VU-mentor legt ook een lesbezoek af (dit kan dezelfde ‘examenles’ zijn, maar hoeft niet).

De student dient het beoordelingsformulier met toelichting van de wpb en eigen reactie te uploaden op Canvas bij Praktijk 3. De vakdidacticus bekijkt de beoordeling. Naar aanleiding van het cijfervoorstel neemt de VU- mentor contact op met de werkplekbegeleider en stelt hierna het cijfer vast. De VU-mentor is formeel eindverantwoordelijk voor het cijfer voor de Praktijk. De student dient voor de verschillende onderdelen van de rubrics overwegend op niveau 3 te functioneren.

Entry Requirements

Voorwaardelijk voor afronding van Praktijk 3: een voldoende beoordeling van Praktijk 2 en Didactiek 2.

Explanation Canvas

Voor het onderdeel Praktijk wordt gebruik gemaakt van Canvas voor het beschikbaar stellen van het praktijkbeoordelingsformulier. De student levert het volledig door de wpb en so ingevulde formulier vervolgens ook weer in via Canvas.

Praktijkonderzoek

Course Code	O_MLPROZ
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. H.B. Westbroek
Examiner	dr. H.B. Westbroek
Teaching Staff	dr. H.B. Westbroek
Teaching method(s)	Study Group, Lecture

Course Objective

Aan het eind van deze cursus heeft de student:

1. Kennis van en inzicht in het opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek;
2. Vaardigheid in het in samenwerking met experts op het instituut en op school opzetten, uitvoeren en presenteren van kleinschalig actieonderzoek.

Concreet houdt dit in:

- Het vertalen van een praktijkprobleem naar een onderzoeksvraag;
- Het ontwerpen en uitvoeren van een kleinschalige dataverzameling;
- Het maken van een analyseplan en uitvoeren van (kwalitatieve en/of kwantitatieve) analyse;

- Het rapporteren van de gebruikte methode, de resultaten, conclusie en discussie in een presentatie.

Course Content

In het keuzevak Praktijkonderzoek staat het leren toepassen van onderzoeksvaardigheden in de onderwijspraktijk centraal. In dit keuzevak leer je een actieonderzoek opzetten en uitvoeren ten behoeve van je eigen professionalisering. Vaak wordt een professionaliseringsvraag breder gedeeld in je sectie/school. Actieonderzoek is een vorm van onderzoekend leren. Je gebruikt theorie en bevindingen uit onderwijsonderzoek om jouw praktijkprobleem te analyseren en om ontwerp/voornemens te formuleren. Je leert hoe je in je klassen data kan verzamelen over de invloed van jouw ontwerp, op de leerprocessen van je leerlingen.

Additional Information Teaching Methods

introductiecollege (1,5 uur), werkgroepen/intervisiesessies (7,5 uur), zelfwerkzaamheid en individuele begeleidingsgesprekken (157,5 uur), presentaties (1,5 uur).

Method of Assessment

Praktijkonderzoek wordt afgesloten met een presentatie. Tussentijds dient ook een onderzoeksplan te worden ingeleverd dat wordt beoordeeld (go/no go).

Entry Requirements

Vereiste voorkennis: Didactiek 2, Praktijk 1 en onderzoekservaring op bachelor/masterniveau in het eigen domeinvak.

Literature

- Literatuur uit Didactiek 1 en Didactiek 2 vormt een basis.
- Literatuur wordt aangeboden in de vorm van wetenschappelijke artikelen (CANVAS).
- Van studenten wordt verwacht dat ze aanvullend zelf literatuur zoeken en gebruiken om hun theoretisch kader en onderzoeksmethode te kunnen uitwerken en onderbouwen.

Additional Information Target Audience

Praktijkonderzoek is een van de specialisaties binnen de ULO lerarenopleiding.

Explanation Canvas

Alle informatie over de opzet van het vak, de bijeenkomsten, toetsing en literatuur wordt aangeboden via CANVAS. Communicatie gebeurt via CANVAS (discussieomgeving; mail).

Principles of Drug Discovery

Course Code	XM_0106
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. I.J.P. de Esch
Examiner	prof. dr. I.J.P. de Esch
Teaching Staff	prof. dr. A.D. Windhorst, prof. dr. I.J.P. de Esch, dr. M. Wijtmans
Teaching method(s)	Seminar, Lecture

Course Objective

To gain insight and hands-on experience in key concepts underlying drug discovery, the molecular modeling tools that enable drug design and the strategies to plan efficient synthesis routes for conceived ligands.

Course Content

- You will get acquainted with key aspects of small-molecule drug discovery, such as hit finding, hit exploration and structure-activity

relationship exploration.
- You will get acquainted with computational techniques able to generate ideas for novel compounds, such as crystal structure analysis, building homology models, predicting binding poses of ligands via molecular docking, calculating binding free energy and affinity of ligands, de novo ligand generation, and pharmacophore modeling.
- You will learn the importance of synthetic feasibility as a key and integral part of the design process and of being able to define a
synthetic pathway for the preparation of the designed compounds, because a design that cannot be synthesized is by definition a useless design. To this end, concepts covered include retrosynthesis and the incorporation of some selected moieties, such as heteroatoromatic scaffolds and known affinity-increasers.
- You will get immersed in a drug discovery case study in which you will design a potential drug and propose a feasible synthesis route for the molecule.
- You will be exposed to state-of-the-art software to guide you through your own drug discovery project, which will result in
an article report describing the research you performed.

Additional Information Teaching Methods

Lectures, tutorials, computer practicals, self-study, open offices, assignments, peer review and work on a parallel case study.

Method of Assessment

Exam

Literature

Patrick, G., An Introduction to Medicinal Chemistry 7th ed., Oxford:
Oxford University Press. 20017, ISBN: 9780198866664

Two E-Books that are accessible through UBVU at all VU computers:

- Mason: Volume 4 of Comprehensive Medicinal Chemistry II: Computer-Assisted Drug Design (Mason (Ed.).
http://www.sciencedirect.com/science/referenceworks/9780080450445
- Hoffmann: Elements of Synthesis Planning (Hoffmann (Ed))
http://www.springerlink.com/content/j81646

Additional Information Target Audience

mDDS, mCh, mBMS (biological chemistry track), mSBI (life track)

The course is not accessible for people with a BSc degree in Pharmaceutical Sciences.

Recommended background knowledge

Knowledge of basic Organic Chemistry

Principles of Drug Targets

Course Code	XM_0105
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R. Leurs
Examiner	prof. dr. R. Leurs
Teaching Staff	prof. dr. R. Leurs
Teaching method(s)	Seminar, Lecture

Course Objective

To gain insight into general steps in the process of drug discovery & development, understanding of the molecular basis of target-drug interactions and the role of pharmaco- and toxicodynamics.

- You will get acquainted with the various processes essential for drug discovery & development
- You will understand the structure and function of different drug targets
- You can explain how enzymes, receptors and DNA can be targeted by drugs
- You will understand the molecular basis of target-drug interactions and how drug-target interactions can be quantified
- You will learn the importance of drug metabolism and safety assessment

Course Content

This course is designed for students with an interest in life sciences and the biotech/pharmaceutical industry but without prior education in this field. A general introduction will be given to the process of drug discovery, drug design and synthesis, drug development and drug safety assessment. Subsequently, potential drug targets, mechanisms of drug actions (including drug- receptor/enzyme interactions and dose- response relations), drug disposition (including pharmaco-/toxicokinetics) and drug toxicity will be discussed. Various modern developments and tools will be illustrated by recent applications in the field of drug discovery.

Additional Information Teaching Methods

Lectures, tutorials, self-study, assignments.

Method of Assessment

Written exam and assignment(s)

Literature

Patrick, G., An Introduction to Medicinal Chemistry 6th ed., Oxford: Oxford University Press. 20017, ISBN: 978-0-19-874969-1, mainly Part A (Drug targets) and Part B (PDPK). Next to this textbook, some primary literature will be used to illustrate certain topics.

Additional Information Target Audience

This introductory course is primarily designed for students mDDS**without** a Bsc degree in Pharmaceutical Sciences or SBI. Accessible for students from mCh, mBMS (Biological Chemistry track) or mSBI (life track) master.

The course is **not** accessible for people with a BSc degree in Pharmaceutical Sciences or SBI.

Recommended background knowledge

Knowledge of basic biochemistry and organic chemistry

Protein Analysis

Course Code	X_435045
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Bärenfänger
Examiner	dr. J.M. Bärenfänger
Teaching Staff	dr. K. Jooß MSc, dr. J.M. Bärenfänger
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130525

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Replacement, Refinement and Reduction of animal experiments applying human cells in culture

Course Code	XM_0148
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A. Wilmes
Examiner	dr. A. Wilmes
Teaching Staff	dr. ir. J.C. Vos, dr. A. Wilmes, prof. dr. P. Jennings
Teaching method(s)	Seminar, Study Group, Other, Lecture

Course Objective

To get familiar with the principles of the 3R and cell cultures techniques including for the application in pharmacology and toxicology.

Course Content

The utilisation of whole animal studies in biological applications such as drug development has inherent ethical concerns, thus there is an impetus to develop non-animal models which may even bring advantages. Human cell culture offers a great tool in various research areas, including in the field of pharmacology and toxicology. Human cell culture systems often provide numerous advantages over animal experiments, including human relevance, reduced cost and the possibility to study mechanistic approaches. Nevertheless, choosing the best available cell model for the respective research question is critical. Furthermore, recent advances and cell culture methods and new techniques, including 3D cultures and organ on chip models evolve rapidly in this field. This course covers the basic understandings in cell culture methods as well as recent developments of advanced culturing techniques and compares different available human cell models, including (cancer) cell lines, primary cells or stem cells, such as induced pluripotent stem cells (iPSC).

In this course, the students will be familiarized with:

- The use of animals by humans: from domestication to the laboratory
- Ethical issues related to human and animal use for drug safety assessment.
- Russell and Burch and the origins of the 3Rs (Replacement, Reduction, Refinement). The principles of the humane experiment.
- Different cell culture media and coatings, including serum free media
- Physiological conditions for cell culture
- Cell proliferation, cell differentiation and respective laboratory techniques (Cell counting, cell viability assays, subculturing, cryopreservation, cell differentiation protocols)
- Primary cell culture and culture of human cell lines, including senescence, immortalization and transformation
- induced pluripotent stem cells (iPSC) culture and differentiation into target cells
- Advances cell culture techniques, including co-culture, 3D culture, organoids and microphysiological systems (MPS)/ organ on a chip
- Sterile handling and sources of contaminations, including mycoplasma
- Cell quality assurance techniques, including authentication, characterisation and validation

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Exam and assignments.

Literature

Recommended books:

1. Culture of animal cells. R. Ian Freshney. Eight Edition (March 2021). Wiley-Blackwell. ISBN: 978-1-119-51304-9
2. In vitro Toxicology Systems. Bal-Price, Jennings ISSN 1557-2153, https://link.springer.com/book/10.1007/978-1-4939-0521-8
3. The Three Rs and the Human Criterion. Micheal Balls. https://replacinganimalresearch.org.uk/the-3rs-and-the-humanity-criterion/

Additional course material will be provided via Canvas

Additional Information Target Audience

mDDS, mBMS, mCHEM, master students in life sciences

Research Methods for Analyzing Complex Problems

Course Code	AM_1182
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. D.H.J. Lynch
Examiner	drs. D.H.J. Lynch
Teaching Staff	A. el Kahal, M.S. van der Loo MSc, B.R. Chakmak, E. Blokbergen, J.R.S. Tjien Fooh MSc, A.R. de Leeuw, K.E. Mujica Gastelbondo MSc, M. Fagan, L.L. Ruijsch, H.B.A. Tahboub MSc, S.S. Mokashi, D.P. Warnaars, A.E. Wilms, D.M.E. Servania MSc, J. Putter MSc
Teaching method(s)	Computer lab, Seminar, Written partial exam, Lecture

Course Objective

The objectives of this course are:

- To understand the differences between beta- and gamma research;
- To acquire insight and understanding of a real-world research process, including knowledge of the character of complex societal issues and the needs, advantages and disadvantages of real-world research;
- Being able to create an adequate research design for the investigation of a specific complex societal problem.
- To acquire insight into relevant research methods (both quantitative and qualitative) to address complex societal problems, their underlying theoretical concepts and their relative strengths and weaknesses;
- Being able to apply these various research methods in a specific societal context;
- To analyze and interpret qualitative findings;
- To acquire insight into reflexivity on the process of doing research in the real-world

Course Content

Contemporary societies increasingly face complex social problems, such as climate change, HIV/ AIDS or ethnic and religious diversity. These complex problems involve a variety of social actors: policy-makers, professionals, NGOs, industries, science and, of course, the public at large. Addressing these complex issues demands an approach that investigates, analyzes and integrates the positions and knowledge of different actors.

This course offers an (advanced) introduction to various research methods used in real world research, including questionnaires, surveys, semi-structured interviews, and focus groups. These methods are commonly used in research into complex problem contexts, communication and opportunities for intervention. Strengths and weaknesses of each research method and technique will be discussed, as well as its possibility to be applied in different societal contexts.

Additional Information Teaching Methods

Research Methods for Analyzing Complex Problems is a parttime course of eight weeks (6 ECTS). The total study time is 160 hours. Tuition methods include lectures, workgroups, workshops, group project work and self-study. The different elements have the following study time: - lectures 20 hours - workgroups and training 36 hours - examination 2 x 2 hours - project work & reading (self-study) Remaining hours Please note that attendance to the

workgroup sessions is compulsory. If you miss one workgroup, with a good reason, you will receive an additional assignment. If you miss more than one workgroup session it is no longer possible to pass the project part of the course. Attendance to the lectures is highly recommended. In our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

Method of Assessment

The course grade is based on the group assignment 'research design' and two individual exams.
Individual Exams 60% of final grade
Group assignment 'research design' 40% of final grade

Literature

The literature of this course consists of selected scientific articles that are provided on Canvas, and the books:

- Verschuren, D.E. and Doorewaard, H. (2010). Designing a Research Project (2nd edition)Eleven International Publishing, the Hague. ISBN 978-90-5931-572-3.
- Gray, D.E. Doing Research in the Real World (3rd or 4th edition) Publications Ltd, United Kingdom. ISBN 978-1-4462-6019-7

An overview of additional literature per lecture will be provided on Canvas.

Additional Information Target Audience

The course ‘Research Methods for Analyzing Complex Problems’ is a compulsory course for first year master students ‘Management, Policy Analysis and Entrepreneurship in Health and Life Sciences’. This course is also a compulsory course within the Majors 'Science Communication' and 'Science in Society' of Health, Life and Natural Sciences Master programmes.

It can be followed as an optional course for other Life Sciences Master program students at the VU University, but only when taken in parallel with Analysis of Governmental Policy (AM_470571). Please contact the course coordinator prior to enrollment if you belong to this group.

Additional Information

The entire course is in English.

Please note that attendance to all workgroup sessions is compulsory.

Attendance to the lectures is highly recommended. From our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

Contact: Durwin Lynch (d.lynch@vu.nl)

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course, based on the actual number of students that start with the course. It is not possible to enrol for workgroups otherwise - unless announced via Canvas.

Research Skills and Career Perspectives

Course Code	XM_0002
Credits	0
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Scholten
Examiner	dr. D.J. Scholten
Teaching Staff	prof. dr. L.M. Bouter, dr. H.F. Vischer, I.T. Braadbaart MA, dr. A. Wilmes, dr. J.M. Bärenfänger, dr. M. Wijtmans, dr. D.J. Scholten, dr. D.P. Geerke
Teaching method(s)	Seminar, Lecture

Course Objective

The goal of this course is to provide information on various topics related to research skills and career perspectives.

By finishing the course, students:

1. understand the preconditions for successfully arranging an internship (including rules, and finding/arranging an internship)
2. are able to write an appropriate CV and motivation letter that satisfies professional standards to apply for a job or internship
3. understand what are the main issues and challenges in scientific research with respect to integrity and potential solutions to these
4. understand how the scientific peer review process is organized and how to engage in a professional feedback process
5. can explain what career opportunities exist after finishing the master's program, and identify concrete actions to prepare for their career

Course Content

Within this course, various topics will be covered, including:

- How to arrange an internship and write a good motivation letter
- Scientific integrity
- Professional peer review
- Career opportunities (Career event)
- Other related topics

See Canvas for the most up-to-date overview of the course

Additional Information Teaching Methods

Lectures, symposium, assignments

Method of Assessment

Students will only pass the course when all the different parts are successfully completed, which includes both active participation and successful completion of assignments (see Canvas syllabus).

Additional Information Target Audience

All first-year students Drug Discovery Sciences (mDDS)

Additional Information

This course is mandatory for all master students Drug Discovery Sciences

Contact the course coordinator if you have any questions regarding this course

Science and Communication

Course Code	AM_470587
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	dr. C. Boersma MA MSc., T.F.L. Roedema MSc, L.B. Clauss MSc, K.E. De Long MSc
Teaching method(s)	Study Group, Lecture

Course Objective

- Gain theoretical insight in the nature of science,
- Gain theoretical insight in the nature of communication,
- Gain theoretical insight in the relationship between science and society,

- Gain insight in the role of science communication in this relationship,
- Acquire knowledge of different theories and models of science communication,
- Acquire knowledge of different strategies, media and activities for science communication,
- Learn how to practically apply theoretical concepts from the field of science communication in communicating science,
- Develop practical skills for science communication (especially writing and giving oral presentations).
- Reflect on your own knowledge and competencies pertinent to your projected (ideal) role as science communicator.

Course Content

Science is all around us and shapes our lives in many different ways. From the vaccines you need to get when traveling abroad to the smartphone you use on a daily basis, and from the public transportation you use to get to the university to the ingredients of your toothpaste: scientific knowledge is elemental to all of these. Simultaneously, society shapes the ways in which science and technology develop too. Science, technology and society influence each other continuously—or, to put it differently, they ‘communicate’.

Students of the Science Communication specialization are expected to become experts in understanding and designing interactions between science and society. In order to make this interaction fruitful and valuable for both science and society, it is first of all important to gain theoretical knowledge about science, about communication and about science communication. Science and Communication provides students with the theoretical and conceptual foundations of the discipline of science communication. Thus, you will develop an in-depth understanding of communication processes at the core of several interfaces, including those between scientists from different disciplines, between different sciences and their stakeholders, and between science and the public.

Additional Information Teaching Methods

Expected contact hours:
Lectures (20 h)
Workgroups (25 h)
Home-study for group assignments (15 h)
Home-study for individual assignments/exam (100 h)

Method of Assessment

a) Participation. (pass/fail)
This consists of the following:
- the 'CARQ' presentations and summaries;
- Active participation in the work groups.
All these are assessed as pass or fail. For each one you fail, you have to do an alternative assignment.
Nota bene: if you fail your participation, this cannot be compensated with an alternative assignment!

b) An interview with a science communicator of your own choosing, including a reflection report. (20%)
d) "TED-talk" in which you present the research you did (e.g. for your Bsc thesis or (first) Msc internship) and reflect on it. (20%)
e) Exam. (60%)

To pass, your grades for the assignments and exam all need to be 5.5 or higher.

Resit: In case your grade is not sufficient, you have to take a resit. This can either consist of a second attempt at (b), (c) or (d), or a re-exam.

Literature

Academic articles. Direct links to articles will be provided on Canvas.

Additional Information Target Audience

The course Science and Communication is a compulsory course for students of the Master specialisation Science Communication (Wetenschapscommunicatie) and is a prerequisite for the internship. Science and Communication is an optional course for students from other master programs in the health and life sciences.

Science in Dialogue

Course Code	AM_1002
Credits	6

Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E. Gommeh
Examiner	dr. E. Gommeh
Teaching Staff	mr. D.T. van der Lugt MSc, P.E. van der Aar, M.C. Vos MA MSc., J. Putter MSc, A.M. Aris, A.R. de Leeuw
Teaching method(s)	Seminar, Study Group, Written partial exam, Lecture

Course Objective

To gain knowledge of and insight into:

- the basic concepts and issues in understanding science-society interactions and interfaces, both from a science and technology studies and communication science perspective
- the nature and course of interpersonal and group communication processes relevant to the formal and informal dialogue between science and society
- the nature and form of dialogical science communication, aimed at reflective learning and mutual understanding

To acquire or improve:

- individual skills for effective interpersonal communication
- individual skills for the design and facilitation of science-society dialogue

Course Content

This course invites you to explore the democracy of science and technology with a particular focus on the communicative aspects of a fruitful science - society dialogue. Science, technology and innovation hold a great promise for the progress of our societies. Take for example the opportunities of nanomedicine for targeted drug delivery or synthetic biology for renewably energy or the recycling of plastic waste. At the same time these developments are potentially controversial. They lead to a variety of questions and concerns related to risks, benefits and wider moral issues. For example, nanotechnology may contain risks for health and environment. Synthetic biology may radically change the nature and meaning of life.

Clearly, advances in science do not always match the needs, desires and expectations of stakeholders and citizens in society. On the other hand, parts of society might not always appreciate the nature of scientific findings and what they have to offer. For a fruitful relationship between science and society – especially regarding topics that are potentially controversial – a constructive science-society dialogue is necessary.

This course offers advanced lectures on the basic concepts and theories about the dynamic relationship of science and society, the nature of societal controversies surrounding science, technology and innovation, and the basic concepts and issues of dialogical science communication: communication, learning, dialogue, understanding, controversy, and democracy. A series of seminars and group assignments present communicative tools and interaction spaces that can be used to design and facilitate science-society interactions. Through training workshops you will focus on improving your individual communication and facilitation skills.

Additional Information Teaching Methods

Lectures (14h), Workgroups (28h), Training workshops (24h), Dialogue presentations (12h), Selfstudy (remaining hours)

Method of Assessment

Group assignment (50%), Two digital exams (30%), Reflection report (20%).

All assignments must be passed (grade > 6).

Literature

Is announced on Canvas one month before start of the course

Additional Information Target Audience

Optional course in the MSc specialization Science Communication

Additional Information

Independence and a cooperative attitude is expected. Attendance to workgroup/ seminars and training sessions is mandatory.

Science Journalism

Course Code	AM_471014
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	A.M. Aris, K.E. De Long MSc, dr. C. Boersma MA MSc., T.F.L. Roedema MSc
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

After following this course, you are able to:

- Review the key changes in science journalism due to changes in the science-society relationship and digitalization of the media
- Examine contemporary issues of science journalism including the emergence of issues and publics, framing, values and emotions, expertise and evidence, uncertainty and ignorance.
- Apply an interdisciplinary toolkit to think about the interactions between science, media and the public and the role of science journalism
- Useyour skills in different science communication media, such as writing, video, and documentary to generate meaningful exchange between science and society
- Describe what the professional practice of science journalism entails

Course Content

Science journalism is needed now more than ever. Science heavily influences our society, not only through the application of its knowledge but also through its worldview. Although science and technology have shown the potential to improve our lives and living conditions significantly, they also contribute to new problems, risks and concerns. This awareness has changed the science and society relationship and put science and technology under increasing public scrutiny. In the context of the complex challenges of today, such as climate change or food security, ‘the facts’ are regularly called into question. Public trust in the solutions presented by science is under pressure. There is clear demand for the scientific world to communicate effectively with the rest of society. But how to shape the public conversation about science in society? And how do the media shape our perception and appreciation of science? What is the role of science journalism? Especially with respect to the larger transformations that are taking place in the relationship between science, media and public. Digitalization has revolutionized the media landscape. Everyone now has the means to either find or generate any information themselves. This provides new opportunities, but may also spread low-quality, sensationalized and sometimes intentionally misleading information. The challenge for science journalism today is to reinvent its practice and help shape a meaningful exchange between science and society.

In this course we review the key changes in science journalism. We examine how changes in both the science-society relationship and digitalization of the media landscape have changed the science journalism practice. We look at different roles of journalism such as reporters, storytellers, watchdogs and educators and discuss what new roles might need to be developed in light of the abovementioned changes. We analyze the communication and interaction dynamics in the current media landscape according to a set of core questions: How do issues and publics emerge? What is the role of framing? What is the role of values and emotions? What is the role of expertise, evidence? What is the role of uncertainty and ignorance? Finally, we train specific skills that you need as a science journalist, such as popular writing, making popular science videos, interviewing, and program design. Throughout the course, we work together closely with professionals from the field. Our guest speakers work as freelancer, editor or producer at diverse science media, such as newspapers, broadcasting media, online magazines and documentary film. They contribute to our course via the lectures, assignments and discussions.

Additional Information Teaching Methods

Lectures and seminars on theory and practice of science journalism and skill training (36h). Considerable time is

set aside for performing science journalism in assignments (108h). All sessions indicated as training sessions in the schedule are mandatory. The assignments are assessed by lecturers and fellow students (peer-review process). Self-study (remaining hours).

Method of Assessment

Several individual assignments (60%), several small group assignments (40%).
All assignments must be passed (grade > 6).

Literature

Announced on Canvas one month before start of the course

Additional Information Target Audience

All Master students with a Beta-Bachelor degree. Students taking this course as part of their VU Science Communication Specialisation or UvA Major Science Communication will have precedence over other students. Students from other faculties and or universities need to get formal consent from the course coördinator (Frank Kupper) before enrolment.

Additional Information

The course is taught in English. Because language proficiency is crucial, there is the option to hand-in assigments in Dutch. More information: t.f.l.roedema@vu.nl.

Scientific Writing in English

Course Code	X_400592
Credits	3
Period	P3, P6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	B.I.M. Schouten
Examiner	B.I.M. Schouten
Teaching Staff	
Teaching method(s)	Study Group

Course Objective

The aim of this course is to provide Master's students with the essential linguistic know-how for writing a scientific article in English that is well organized, idiomatically and stylistically appropriate and grammatically correct.
At the end of the course students:

- know how to structure a scientific article;
- know what the information elements are in parts of their scientific article;
- know how to produce clear and well-structured texts on complex subjects;
- know how to cite sources effectively;
- know how to write well-structured and coherent paragraphs;
- know how to construct effective sentences;
- know what collocations are and how to use them appropriately;
- know how to adopt the right style (formal style, cohesive style, conciseness, hedging)
- know how to avoid the pitfalls of English grammar;
- know how to use punctuation marks correctly;
- know what their own strengths and weaknesses are in writing;
- know how to give effective peer feedback.

Final texts may contain occasional spelling, grammatical or word choice errors, but these will not distract from the general effectiveness of the text.

Course Content

The course will start with a general introduction to scientific writing in English. Taking a top-down approach, we will then analyse the structure of a scientific article in more detail. As we examine each section of an article, we will peel back the layers and discover how paragraphs are structured, what tools are available to ensure coherence

within and among paragraphs, how to write effective and grammatically correct sentences and how to choose words carefully and use them effectively.

Topics addressed during the course include the following:

- Structuring a scientific article
- Considering reading strategies: who is your readership? How do they read your text? What do they expect? How does that affect your writing?
- Writing well-structured and coherent paragraphs
- Composing effective sentences (sophisticated word order, information distribution).
- Academic tone and style: hedging – why, how, where?
- Using the passive effectively
- Understanding grammar (tenses, word order, etc.)
- Understanding punctuation
- Referring to sources: summarising, paraphrasing, quoting (how and when?)
- Avoiding plagiarism
- Vocabulary development: using appropriate vocabulary and collocations

Additional Information Teaching Methods

Scientific Writing in English is an eight-week course and consists of 2 contact hours a week. Students are required to spend at least 6 to 8 hours of homework per week. They will work through a phased series of exercises that conclude with the requirement to write several text parts (Introduction, Methods, Discussion and Abstract). Feedback on the writing assignments is given by the course teacher and by peers.

Method of Assessment

Students will receive the three course credits when they meet the following requirements:

- Students hand in three writing assignments (Introduction, Methods, Discussion)
- Students get a pass mark for all writing assignments;
- Students provide elaborate peer feedback (Introduction, Methods, Discussion, Abstract);
- Students attend at least 7 out of 8 sessions;
- Students are well prepared for each session (i.e. study the theory for each session and participate in online discussions about the theory);
- Students participate actively in class and respect deadlines;
- Students do not plagiarise or self-plagiarise.

Writing assignments:

- If students have a BSc thesis in a traditional thesis form (e.g., 20+ pages) and written in English, they may use this for the writing assignments.
- If students have a BSc thesis in a traditional form (e.g., 20+ pages) written in another language than English, they may use this for the writing assignments.
- If students have written a paper or report in English that’s not already in article form, they may use this for the writing assignment.
- If students are working on their MSc thesis or internship report when taking Scientific Writing in English, they may use this for the writing assignments. They will have to notify their supervisor to make sure that they won’t be accused of self-plagiarism.
- If students cannot or do not wish to use any of the above-mentioned texts for the writing assignments (1-4), they are expected to do a limited Literature Review on a topic in their field of research, using at least 5 articles.

Students are not allowed to use the following texts for the writing assignments:

- A BSc thesis written in English that’s already in article form.
- A MSc thesis written in English that’s already in article form (and that has already been marked).
- An internship report written in English that’s already in article form (and that has already been marked).
- A paper or report written in English that’s already in article form.

Literature

Effective Scientific Writing: An Advanced Learner's guide to Better English, 5th edition (May 2017) (A. Bolt & W. Bruins, ISBN 978 90 8659 617 1). VU bookstore: €34.50

Additional Information Target Audience

This course is only open to students of the two-year Master's programmes of the Faculty of Science. These students are only eligible to the course if they have already conducted scientific research (e.g. for their Bachelor’s

thesis) or if they will be working on a research project when taking Scientific Writing in English.

Additional Information

To do well, students are expected to attend all lessons. Group schedules are to be found at rooster.vu.nl and on Canvas.

A VUnet registration for this course automatically gives access to the corresponding Canvas site. Group registration only takes place via Canvas.

Make sure Scientific Writing in English does not overlap with another course.

If you have registered for a group in Canvas, you are expected to attend all sessions (eight). If you decide to withdraw from the course, do so in time in VUnet. This will avoid a 'fail' on your grade list for not taking part in this course and allows other students to fill in a possible very wanted group spot.

Full time students with their main registration at VU will be given preferential treatment for placement in this course. For secondary students proof of enrollment is not a guarantee of placement.

Custom Course Registration

Important: each group has a minimum of 18 and maximum of 24 participants, so students should register on time through VUnet to ensure a place in one of the (designated) groups. If you have registered for a group in VUnet, you are expected to attend all sessions (eight). If you decide to withdraw from the course, please do so in time. This allows other students to fill in a possible very wanted group spot.

Societal Entrepreneurship in Health and Life Sciences

Course Code	AM_470575
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K.D.S. Fernald MSc
Examiner	dr. K.D.S. Fernald MSc
Teaching Staff	dr. L.H.M. van de Burgwal, dr. K.D.S. Fernald MSc
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

This course consists of both a theoretical and a practical component.

Both components run simultaneously so that the theoretical knowledge can be applied to the practical component. In the theoretical component, you learn about different aspects of societal entrepreneurship, both from a theoretical and from a practical perspective.

This knowledge can be applied in the practical component which revolves around a group assignment based on a real-life business case. Using the Business Model Canvas (Osterwalder & Pigneur, 2010) you will develop a business plan covering aspects such as value propositions, key activities, key partners, customer segments, cost structure, and revenue streams. In setting up this business plan, commercial, organisational and societal aspects of entrepreneurship play a key role.

Learning objectives

- Understand the relevance of entrepreneurship and innovation for science disciplines.
- Explain the importance of valorization of findings from the health and life sciences and business ideas for a knowledge-based economy.
- Apply different models and strategies for societal entrepreneurship
- Outline the financial, social, and ecological aspects (societal entrepreneurship) of value-adding opportunities.
- Recognize and design opportunities that create economic and social value.
- Recognize and understand different entrepreneurial processes and configurations.
- Critically reflect on the topic of societal entrepreneurship within the health and life sciences
- Construct a fundable business plan (investment memo) for a socially responsible, innovation-driven enterprise

Course Content

This course focuses on societal aspects of entrepreneurship. Although the shape and form of Corporate Social Responsibility (CSR) has changed over the years, it is broadly accepted that corporations need to act socially responsibly. In the health and life sciences, Access to Medicine, sustainable innovation, and precautionary principles are important concepts when it comes to social responsibility. During the course, you will study the meaning of societal and responsible entrepreneurship in the context of health and life sciences.

In the course, theoretical insights are combined with practical experiences. Both types of knowledge will culminate in the development of a thorough and competitive business plan that conveys an understanding of societally relevant, financially viable, and commercially interesting aspects.

Lecturers from Athena and experts from the field discuss various relevant topics, such as in-depth insight into the elements of a business plan, different business model configurations and the role of societal impact. The course is relevant for a wide range of business cases in the health and life sciences, ranging from NGO-like organizations to strong business-driven life sciences enterprises.

You will work in small teams, each team working on their own case. You will be in charge of this assignment; showing leadership and taking responsibility for the academic quality of your final product.

Real cases from the life sciences will be made available for students to work on. In addition, students that have truly considered becoming entrepreneurs themselves are encouraged to formulate a health and life sciences related business case (as a group of 3-5 students) before registering for this course.

Additional Information Teaching Methods

Lectures and workshops are key elements of this course. Lectures provide key knowledge for the assignments. Additionally, each week students have compulsory workshops in which specific parts of the assignment are further developed. Schedule and study time: The total study time is 160 hours. The following hours are contact hours: - Lectures: 34 hours - Workshops: 15 hours - Exam: 3 hours - Practical assignment: 66 hours - Self-study for remaining 50 hours

Method of Assessment

- 50% group assignment (business plan and pitch)
- 50% individual assignments (peer feedback, reflection and presentation)

Entry Requirements

Proven knowledge of business aspects in the Health and Life Sciences is required (e.g. by having passed the Business Management course).

Literature

- Business Model Generation (Osteralder & Pigneur, 2010)
- Additional literature will consist of management articles, scientific management literature and management tools.

Additional Information Target Audience

Optional course for Master students Management, Policy Analysis and Entrepreneurship in Health and Life sciences (MPA), M-differentiation of the Health, Life & Natural Sciences, Biology, Biomedical Sciences.

Additional Information

Active participation of individuals in work group meetings and group assignments is considered a prerequisite to pass the course.

Custom Course Registration

VU-net registration will close 4 weeks before the start of the course.

Groups for the assignment will be made via Canvas.

Spectroscopic Approaches in Medicinal Chemistry

Course Code	XM_0109

Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	A.C. Janssen
Examiner	A.C. Janssen
Teaching Staff	dr. O.P.J. van Linden, prof. dr. I.J.P. de Esch, A.C. Janssen
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

The overall objective of this course is to obtain a thorough understanding of the role of advanced and less common NMR experiments in medicinal chemistry research.

Specifically:

1. You will be introduced to a wide variety of advanced NMR techniques and other spectroscopic measurements to unambiguously confirm the structure of drug-like molecules.

- What kind of experiments are available?
- What experiment or technique suits what situation?

2. You will learn how to interpret various spectra and deduce structural information from these experiments.

- What information does one get from the different NMR experiments?
- How does one interpret NMR spectra in a structured manner?

3. You will get acquainted to the broader role of NMR spectroscopy in structural elucidation of small molecules in drug research.

- What other applications of NMR besides structure elucidation are there?
- How can they be applied in medicinal chemistry?

4. You will get familiar with how to properly handle spectroscopic data so it can be used in scientific reporting and publications.

- How does one report NMR data for publications and reports?
- What software is available for NMR data processing?
- Collaborate as a team while working on case studies.

Course Content

Among the many analytical tools available to medicinal and organic chemists, Nuclear Magnetic Resonance (NMR) spectroscopy stands out as the most powerful and widely used technique in the laboratory.

This course will begin with a brief review of fundamental NMR theory, followed by an in-depth exploration of advanced NMR experiments. Emphasis will be placed on methods used to elucidate and confirm the two- and three-dimensional structures and conformations of drug-like molecules, including NSAIDs, terpenes, steroids, saccharides, antibiotics, and peptides, typically ranging from 150 to 1500 Da in molecular weight.

When basic NMR experiments, such as 1H, 13C, COSY, HSQC, and HMBC, fall short to provide all the necessary information for the full elucidation of a synthesized compound, additional techniques can be employed to gather the missing details. To further enhance structure elucidation, we will explore advanced NMR techniques like 1,1-ADEQUATE, INADEQUATE, and selective 1D/ 2D experiments. Mass Spectrometry (MS) alongside NMR will also be addressed to complete the characterization process.

The course includes an introduction to NMR processing software (MestReNova) and will feature hands-on practice with real drug-like molecules synthesized by research groups in the O|2 research building, allowing you to apply theoretical knowledge to actual data.

Lastly, we will cover best practices for reporting spectroscopic data in scientific publications, ensuring students understand how to clearly communicate their findings in a professional and reproducible manner.

Additional Information Teaching Methods

Lectures, tutorials, case studies and self-study

Method of Assessment

Method of Assessment

Two partial exams (50 % each)

Two case studies (Pass/fail)

Literature

- NMR for Chemists and Biologists, Rodrigo J. Carbajo and José Neira, Springer, 2013 (ebook): https://link-springer-com.vu-nl.idm.oclc.org/book/10.1007%2F978-94-007-6
- Handouts
- Selected articles

Additional Information Target Audience

Medicinal (DDS) and Organic Chemistry (Molecular Sciences) students

Recommended background knowledge

Basic knowledge of Nuclear Magnetic Resonance
Basic knowledge of organic chemistry

Synthetic Approaches in Medicinal Chemistry

Course Code	XM_0110
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Wijtmans
Examiner	dr. M. Wijtmans
Teaching Staff	prof. dr. I.J.P. de Esch, dr. M. Wijtmans
Teaching method(s)	Seminar, Lecture

Course Objective

To obtain detailed knowledge and understanding of vital organic chemistry reactions and aspects used in medicinal chemistry.

Course Content

- You will master the synthetic toolbox that medicinal chemists use to make exciting molecules.
- You will immerse yourself in the most important and generally used synthetic reactions, including reactions of nucleophilic carbon intermediates, nucleophilic substitutions, electrophilic additions to carbon-carbon multiple bonds, reductions, oxidations, cycloadditions, aromatic substitution reactions, rearrangements, and reactions of transition metals as well as of Group I and II metals.
- You will learn how to critically inspect chemo-, regio- and stereoselectivity questions.
- You will be exposed to contemporary articles and emerging concepts in organic synthesis (such as photoredox chemistry and C-H activation).
- You will learn how to discuss organic chemistry topics in a scientific research community.

Additional Information Teaching Methods

The Flipped classroom method is used. This means that all lectures are offered as slidecasts for self study, while all contact hours are used for interactive sessions.

Method of Assessment

Two written examinations.

Literature

Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry, Part B, 5th edition.

Additional Information Target Audience

mDDS, Synthesis of Drugs specialisation.

Recommended background knowledge

Knowledge of organic chemistry

Teaching Assistant

Course Code	XM_432741
Credits	3
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. J.E. van Muijlwijk-Koezen
Examiner	prof. dr. J.E. van Muijlwijk-Koezen
Teaching Staff	
Teaching method(s)	

Course Objective

The main goal is to improve your skills in supervision and teaching through work in a (practical) course, including familiarising yourself with the specific STEM didactics.

Course Content

Teaching skills are practised and developed during a (practical) course and assessed through both a performance evaluation and a written reflection. Students are recommended to follow the course 'Tutoring Students' prior for a stronger theoretical basis. There is also additional required material on Canvas for students who work in wet labs.

Additional Information Teaching Methods

Hands on course:

You will become a member of the team of supervisors for practical courses or working classes for undergraduates and assist in the lab or classroom. With the aid of feedback, you will improve via learning by creating goals, taking (teaching) actions to develop these goals, and reflecting on the success of these actions in a final report.

Method of Assessment

Execution during the course (50%) and concluding reflective assignment (50%). Both must be successfully completed.

Additional Information Target Audience

mDDS, mChem, mPhys, mMath, mMNS, mSBI

Additional Information

Limited spots available. This course can be done in any period during the year. Make sure you arrange everything well in advance (see also Canvas page).

Find a coordinator/teacher of a (practical) course that needs a student assistant for guiding a practical course or tutorials. You are responsible for finding a Student-as-Teacher position yourself. The duration and the number of hours required to supervise in this assistantship will determine whether you are eligible for the 3 EC or 6 EC variant of this course.

Custom Course Registration

Contact your master coordinator IN TIME. The earlier the better to ensure that this fits in your Personal Education Plan.

Explanation Canvas

All information is on Canvas and it is essential that students take responsibility for taking initiative, following the provided instructions, and contacting the relevant teachers on time.

Recommended background knowledge

X_432625, period 3, Tutoring Students or equivalent via Center for Teaching & Learning

Teaching Assistant

Course Code	XM_432742
Credits	6
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. J.E. van Muijlwijk-Koezen
Examiner	prof. dr. J.E. van Muijlwijk-Koezen
Teaching Staff	
Teaching method(s)	

Course Objective

The main goal is to improve your teaching skills and get familiar with the specific pharmaceutical sciences/chemistry didactics (Science students as teachers)

Course Content

Practice during a (practical) course supervised by the course lecturer and a written reflection. For assistance during practical work the course 'tutoring students' is a requirement.

Additional Information Teaching Methods

hands on course:
You will become a member of the team of supervisors for practical courses or working classes for undergraduates and assist in the lab or classroom. With the aid of feedback and intervision, you will improve via learning by doing. The theoretical background has been taught in the course 'tutoring students'

Method of Assessment

Execution during the course and concluding reflective session

Additional Information Target Audience

mDDS, mChem, mPhys, mMath, mMNS, mSBI

Additional Information

Limited seats available, fixed periods

Find a coordinator/teacher of a (practical) course that needs a student assistant for guiding a practical course or tutorials. You are responsible for finding a student assistant position yourself. The duration and the number of hours required to supervise in this assistantship will determine whether you are eligible for the 3 EC or 6 EC variant of the Teaching assistant course. Make sure this is agreed upon beforehand! The teaching assistant course can be done in any period during the year. Make sure you arrange everything well in advance (see also Canvas page).

Custom Course Registration

Contact your master coordinator IN TIME. The earlier the better.

Recommended background knowledge

X_432625, period 2, tutoring students or equivalent via Center for Teaching & Learning

Translational Radiopharmaceutical Sciences

Course Code	XM_0067
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. D.J. Vugts
Examiner	prof. dr. D.J. Vugts
Teaching Staff	prof. dr. A.D. Windhorst, dr. W. Beaino, prof. dr. D.J. Vugts, dr. S.V.S. Golla
Teaching method(s)	Practical, Lecture

Course Objective

To obtain detailed knowledge of translational radiopharmaceutical sciences

Course Content

Radiopharmaceutical sciences concern the development and evaluation of radiolabeled compounds, so-called radiopharmaceuticals or tracers. These radiopharmaceuticals can be either imaging agents for Positron Emission Tomography (PET) and Single Photon Emission Computed Tomography (SPECT) or therapeutic agents. The stunning clinical successes of nuclear imaging and targeted radionuclide therapy have resulted in a rapid growth of the field of radiopharmaceutical sciences, a truly translational scientific discipline in which design, synthesis and application of radiopharmaceuticals is key.
This course focusses on radiopharmaceutical design, including target validation with molecular biology techniques, chemistry and vector molecules, lead finding and optimization for radiopharmaceuticals, in vitro investigations, preclinical research including imaging and metabolism, radiopharmacy aspects like Good Manufacturing Practice, regulatory matters (eg Eudralex 4) and European Pharmacopea and finally an introduction to PET data analysis.

Additional Information Teaching Methods

The "Flipped classroom method" is used. An active participation and full time presence of the students is expected. Some lectures will be traditional classes where the theory is presented by the teacher, some will be presented by the students where they will present on basis of the theory described in the textbook. There will be several classes devoted to problem solving sessions and case studies.

Method of Assessment

There will be several assignments, 2 practicals, and cases during the course which will account for 30% of the final grade. There will be a written or oral final examination, accounting for 70 % of the final grade.

Literature

Radiopharmaceutical Chemistry, Lewis, Windhorst, Zeglis editors. Springer, 2019. ISBN 978-3-319-98946-4. Available as PDF via the VU library: https://link.springer.com/book/10.1007/978-3-319-98947-1

Additional Information Target Audience

mDDS, DD&S and D&I

Custom Course Registration

VUNet

Recommended background knowledge

Basic knowledge about organic chemistry, eg organic chemistry I bachelor pharmaceutical sciences, chemistry or alike

Tutoring Students

Course Code	X_432625
Credits	3
Period	P3

Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	I.T. Braadbaart MA
Examiner	prof. dr. J.E. van Muijlwijk-Koezen
Teaching Staff	dr. M. Wijtmans, prof. dr. J.E. van Muijlwijk-Koezen, I.T. Braadbaart MA, dr. D.J. Scholten
Teaching method(s)	Lecture

Course Objective

This course aims to prepare students for coaching tasks in tutorials and practical courses. Students will explore different aspects of teacher-student interaction, and by the end of the course will be able to reflect critically on their beliefs and practices about teaching, guide pre- and post-experiment discussions, support students in problem-solving, plan for safety in practical work, and give written feedback.

Course Content

The course asks students to engage in literature, discussion and practical simulations to support students' understanding of appropriate teaching and coaching practices in tutorials and practical courses. Students will analyse various learning aims within a laboratory setting and prepare, conduct and reflect on a presentation of a pre- and post-experiment discussion in tutorials and practical courses. They will observe and interpret the application of problem solving and coaching models in tutorials and practical courses. Attention will be paid to strengths and weaknesses in models of teacher-student interaction, with a focus on questioning. An important constituent is the student's analysis of and reflection on his/her own teaching approach. Topics on safety and lab journal procedures in practical courses as well as on the grading of lab reports are also included.

Additional Information Teaching Methods

3,5 consecutive hours per session (2x per week over four weeks):

- Lectures and whole class discussions
- Hands-on practice moments
- Self-study
- Group work

Method of Assessment

A collection of written group assignments completed throughout the course, including:

- a summary of good guiding practices
- two lesson plan designs
- a plan for pre- and post-experiment discussion
- a rubric

A collection of individual assignments completed throughout the course, including:

- a reflection on previous learning experiences
- a reflection on pre- and post-experiment discussion
- a reflection on attitudes towards guided- and open-inquiry labs
- an analysis of various strategies to repair student misconceptions in the sciences
- a reflection and analysis of problem-solving approaches
- a final critical reflection on your development throughout the course, accompanied by a formative presentation and feedback session

Literature

Will be provided.

Additional Information Target Audience

mDDS, mChem, mPhys, mMath, mMNS, mSBI

Additional Information

This course is recommendable for any MSc student who has a general interest in educational coaching strategies

and models. Specific attention is given to guiding experimental / laboratory work and problem-solving skills.

Number of participants is limited to 24 (first-come, first-serve basis).

Custom Course Registration

VUnet

Explanation Canvas

Canvas will be used to share literature, submit assignments, and for other relevant course communications.

Tweetalig Onderwijs

Course Code	O_MTTO
Credits	3
Period	P2
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S. Donszelmann
Examiner	dr. S. Donszelmann
Teaching Staff	dr. S. Donszelmann
Teaching method(s)	Seminar

Course Objective

Na afloop van het vak heb je een beeld van het Tweetalig Onderwijs (TTO) binnen het Nederlandse voortgezet onderwijs en beschik je over basale vaardigheden om de Engelse taal op een zinvolle manier als instructietaal én als leertaal in te zetten.

Course Content

Op hoe langer hoe meer scholen wordt tweetalig onderwijs aangeboden (TTO). Het doel van dit type onderwijs is dat leerlingen tegelijk met vakinhoud ook Engels leren. Soms pakt dit goed uit, maar het kan ook zijn doel voorbij schieten. Hoe zit dit?

Het gebruik van de doeltaal (Engels in dit geval) leidt niet bij voorbaat tot het leren van Engels. Leerzaam inzetten van de doeltaal is gebonden aan allerlei didactische voorwaarden waarvan de docent zich vaak maar beperkt bewust is. Het gevolg kan heel goed zijn dat leerlingen bij veel vakken nauwelijks Engels bijleren ten opzichte van het Engels dat zij al beheersten, en bovendien minder vakinhoud meekrijgen dan in het Nederlands het geval zou zijn omdat de docent zich omslachtig in een vreemde taal uitdrukt. Doeltaal-Leertaal onderzoek (Dönszelmann, 2019) geeft inzicht in de vraag welk didactisch instrumentarium er nodig is om leerlingen op leerzame manier onder te dompelen in de vreemde taal, terwijl er tegelijkertijd ook vakinhoud wordt overgebracht.

Binnen het keuzevak onderzoeken we wat dit betekent voor de TTO-praktijk binnen Nederlandse VO-scholen, en hoe goed TTO-onderwijs eruit moet zien.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Kort literatuurverslag, of onderzoek binnen de eigen praktijk (als hier TTO-onderwijs voorkomt).

Entry Requirements

Didactiek 2, Praktijk 1

Literature

Wordt via Canvas bekend gemaakt.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten

Custom Course Registration

Er geldt een maximum van 5 voor bijvakstudenten van andere lerarenopleidingen.

Vernieuwende onderwijsconcepten

Course Code	O_MLVNON
Credits	3
Period	P2, P5
Course Level	500
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. M. Meeter
Examiner	prof. dr. M. Meeter
Teaching Staff	prof. dr. M. Meeter, dr. H.B. Westbroek
Teaching method(s)	Seminar, Lecture

Course Objective

Na het volgen van dit keuzevak, kun je:

1. beschrijven hoe het onderwijs in Nederland de afgelopen eeuwen vorm gekregen heeft en hoe met het onderwijs antwoord gegeven is op de vragen: wat willen we onze kinderen leren; waartoe dient onderwijs en hoe geven we onderwijs vorm?
2. een aantal vernieuwende vormen van VO-onderwijs opsommen (traditionele vernieuwende onderwijsconcepten, zoals Montessori, Dalton, Vrije School en hedendaagse vernieuwende onderwijsconcepten zoals gepersonaliseerd leren/Kunskapsskolan, Agora) en beschrijven wat de ideeën achter de vernieuwingen zijn.
3. een eigen leervraag over vernieuwende onderwijsconcepten en/of processen die daarbij horen formuleren en vragen bedenken die je tijdens het schoolbezoek aan docenten en leerlingen wilt stellen om de leervraag te kunnen beantwoorden.
4. de eigen leervraag beantwoorden met gebruikmaking van theorie, gesprekken met docenten en leerlingen, de bijeenkomsten en het schoolbezoek en een onderbouwd antwoord geven op de vraag: Wat vind ik belangrijk bij het onderwijs dat ik verzorg? Wat spreekt mij aan?
5. een begrippenkader hanteren waarmee je kunt nadenken en praten over vernieuwingsprocessen en vernieuwende onderwijsconcepten.
6. een beredeneerd oordeel vormen en een professionele dialoog voeren over voor- en nadelen van vernieuwingen en ongeteste aannames en kritische implementatiedetails benoemen.

Course Content

In het Nederlandse onderwijs, zeker in competitieve markten als de Amsterdamse, wordt continu vernieuwd. Scholen sturen bijvoorbeeld aan op het implementeren van gepersonaliseerd leren of projectmatig werken of onderwijs in gemengde stamgroepen. Bij dit vak bezoeken we een aantal vernieuwingsscholen en analyseren we de beweegredenen en de proceskant van onderwijsvernieuwing (de implementatie).

Additional Information Teaching Methods

Werkcolleges (waarin de dialoog centraal staat en waarin je collega's uit het de scholen interviewt) en schoolbezoeken.

Method of Assessment

Analyse van een schoolvernieuwing in de vorm van een academisch essay of in de vorm van een podcast of educlip (keuze) aan de hand van wetenschappelijke- en vakliteratuur, interviews, observaties en schoolbezoeken.

Entry Requirements

Didactiek 1 en 2, Praktijk 1

Literature

Informatie over te lezen artikelen wordt gedeeld op Canvas.

Additional Information Target Audience

Dit keuzevak is bedoeld voor studenten van de VU lerarenopleidingen (VU Lerarenacademie). Ook studenten van andere eerstegraads lerarenopleidingen (Utrecht, UvA, Delft) kunnen deelnemen als bijvakstudent. Er geldt een maximum van 5 bijvakstudenten.

Wereldburgerschap deel 1

Course Code	O_MWRLDBS_1
Credits	3
Period	P1, P4
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	D.S. Hoefakker-Laanstra, dr. I. Pauw, drs. H.R. Goudsmit
Teaching method(s)	Study Group

Course Objective

Na het volgen van dit keuzevak:

- kan je het belang van wereldburgerschap voor leerlingen, docenten en scholen onderbouwd over het voetlicht brengen.
- ken je de verschillende definities van wereldburgerschap en kan je op grond daarvan een onderbouwde eigen formulering geven.
- kan je de verschillende vormen van wereldburgerschap noemen en kan je bij de verschillende vormen leerlijnen en concrete activiteiten noemen.
- kan je aan de hand van de gekozen definitie en de methodologie van wereldburgerschap een onderbouwde leerlijn wereldburgerschap voor een school ontwerpen
- kan je in de klas in het kader van wereldburgerschap een maatschappelijke controverse les ontwerpen en uitvoeren .
- kan je in je klas met leerlingen met behulp van de dialoog maatschappelijke thema’s bespreken

Course Content

Leerlingen van nu groeien op in een gepolariseerde multiculturele omgeving in een globaliserende wereld, waarin belangrijke thema’s als klimaat, de energietransitie en ook oorlog en vrede de landsgrenzen overstijgen. De samenleving van nu vraagt nieuwe en andere competenties van leerlingen. Zo is het belangrijk, dat leerlingen in dialoog met elkaar kunnen gaan in plaats van in debat. En moeten leerlingen zich een aanpak eigen maken om de grote maatschappelijke thema’s voor zichzelf behapbaar te maken. Ook vraagt het van leerlingen om een open en onderzoekende houding ten opzichte van andere meningen en ideeën. Anders gezegd, de maatschappij vraagt om leerlingen, die zich zowel nu in hun eigen omgeving als later in de maatschappij kunnen ontwikkelen tot wereldburgers . En het is de taak van het onderwijs hen handvatten mee te geven om wereldburgers te worden. Hoe je als docent en als school dat kan doen is het onderwerp van deze specialisatie.

Additional Information Teaching Methods

Werkcolleges

Method of Assessment

Verschillende toetsvormen: leerlijn Wereldburgerschap ontwerpen, controverse les ontwerpen en uitvoeren

Entry Requirements

Didactiek 1, Praktijk 1

Literature

- Wereldgericht Onderwijzen.Biesta in praktijk, B.Eidhof e.a, 2021
- Artikelen via Canvas

Wereldburgerschap Deel 2

Course Code	O_MWRLDBS_2
Credits	3
Period	P2, P5
Course Level	400
Language Of Tuition	Dutch
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	drs. H.R. Goudsmit
Examiner	drs. H.R. Goudsmit
Teaching Staff	dr. I. Pauw, drs. H.R. Goudsmit, D.S. Hoefakker-Laanstra
Teaching method(s)	Seminar, Lecture

Course Objective

Doelen.

- Studenten kennen de verschillende internationale onderzoeken naar de diverse schoolsystemen, kunnen de indicatoren van de verschillende onderzoeken aangeven en zijn in staat tot het geven van een onderbouwde kijk op deze onderzoeken.
- Studenten kennen de kritische momenten in een schoolsysteem en zijn in staat deze momenten te waarderen
- Studenten zijn in staat andere onderwijssystemen dan het Nederlandse in kaart te brengen en deze te relateren aan het eigen onderwijsstelsel
- Studenten kunnen het begrip Internationalisering definieren, de mate van Internationalisering op hun eigen school in kaart brengen en een leerlijn Internationalisering voor een school ontwerpen
- Studenten kunnen hun visie op goed onderwijs verdiepen naar aanleiding van analyseren van andere schoolsystemen. Daarnaast zijn ze ook in staat een meer onderbouwde kijk op het Nederlands onderwijssysteem te geven

Course Content

Onlangs sprak een bekende Nederlandse publicist naar aanleiding van hevige discussies rond de klimaatcrisis de gevleugelde woorden: "Velen hebben een mening, slechts weinigen komt een oordeel toe." Nederlanders staan er om bekend overal een mening over te hebben en die te pas en te onpas te ventileren.

Je kan je afvragen of dat ook niet voor het onderwijs geldt. Er wordt immers veel gezegd en geschreven over ons onderwijs zonder dat het altijd duidelijk waarop het geventileerde is gebaseerd.

En hoewel wij er dagelijks deel van uit maken is het maar de vraag hoe onderbouwd onze eigen visie is op het Nederlands onderwijssysteem. Wordt onze eigen kijk ook niet teveel ingegeven door een beperkte hoeveelheid aan eigen ervaringen en missen wij niet het overzicht of de helikopter view om te komen tot een gefundeerd oordeel? Missen wij ook niet voldoende vergelijkingsmateriaal om tot een werkelijk gefundeerd oordeel te komen? Deze keuzemodule wil dan ook in de eerste plaats een stimulans zijn om tot een dergelijk verdiepte visie te komen.

Overal in de wereld tonen regeringen en politici interesse in de uitkomsten van de benchmarks, die internationaal worden georganiseerd door instellingen als de OESO (PISA; Eduaction at Glance). De uitkomsten van deze rapporten zorgen in de deelnemende landen vaak voor onderwijshervormingen. Maar het doen van internationaal vergelijkend onderzoek roept ook veel discussie op. Kan je onderwijssystemen wel vergelijken op grond van cijfers, zijn de onderzoeken wel betrouwbaar en valide, kun je onderwijssystemen uberhaupt wel met elkaar vergelijken? In dit keuzevak zullen we dan ook ingaan op de verschillende onderzoeken, onderzoeksmethodes en kritieken. Maar ook gaan we zelf (een klein) onderzoek doen naar de verschillende onderwijssystemen om te kijken of we ons een beeld kunnen vormen van andere schoolsystemen.

Dit keuzevak is enerzijds een aanvulling op deel 1 van Wereldburgerschap maar is ook als "los"keuzevak te volgen

Additional Information Teaching Methods

Werkcollege

Method of Assessment

Het schrijven van een essay

Entry Requirements

Didactiek 1 en praktijk 1

Literature

Artikelen uitgereikt op college

Recommended background knowledge

Didactiek 1 en Praktijk 1